MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR MODULE, KEY CODE MHA

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1 1 Scope This document establishes the requirements and test procedures for the procurement of an electronic microprocessor module

1 2 Classification This specification covers two classes of modules Class I and Class II as specified in MIL-STD-1389

2 APPLICABLE DOCUMENTS

2 1 Issues of documents The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATION

MILITARY

MIL-M-28787	-Modules, Standard Electronic, General Specification for

STANDARDS

MILITARY

MIL-STD-1378	-Requirements for Employing Standard Electronic Modules
MIL-STD-1389	-Design Requirements for Standard Electronic Modules
MIL-STD-1665	-Test Equipment for the Standard Electronic Modules Program

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Space and Naval Warfare Systems Command, ATTN SPAWAR 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5963

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3 REQUIREMENTS

3 1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3 2 Correlation requirements Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

3 3 Failure rate requirements All modules procured to this specification shall have a predicted failure rate no greater than _____ failures per million operating hours for class I or _____ failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389 When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 6 3)

3 4 Performance requirements The module shall conform to the electrical requirements specified in table I Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6 3)

3 5 Module configuration requirements The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein

3 5 1 Key Code The module key code shall be MHA

3 5 2 Size The module size shall be 2A Format A

3 5 3 Weight The module shall have a maximum weight of 70 grams

3 5 4 Contact pins The module shall have 80 contact pins positioned 1A1 through 1A40 and 2A1 through 2A40

3 5 5 Fin position The module shall have an insertion and extraction and heat dissipation fin on increment A

3 5 6 Depth The module shall have no additional depth requirements

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
A	ISOLATION	IV								
	Initial		S	10	INF	10	INF	10	INF	Mohm
	End-of-life		S	1	INF	1	INF	1	INF	Mohm
B1-B4	CONTINUITY	V	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
C	Vcc STRESS	VI	S	---------- MUST PASS ALL REMAINING TESTS ----------						
D1-D16	Ibd	VII	S	0.0	1.0	0.0	0.1	0.0	0.1	mA
D17	Ibd	VII	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	VII	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1-E40	Ios	VIII	S	-9.1	-12.4	-9.1	-12.4	-9.1	-12.4	mA
E41,E43-E45	Ios	VIII	S	-20	-55	-20	-55	-20	-55	mA
E42,E46,E47	Ios	VIII	S	-20	-100	-20	-100	-20	-100	mA
F1-F40	Vclamp	IX	S	0.0	-1.5	0.0	-1.5	0.0	-1.5	V
G1	Vinj	X	X	0.88	1.63	0.88	1.63	0.88	1.63	V
G2	Vinj	X	X	0.91	1.68	.91	1.68	.91	1.68	V
G3	Icc	X	X	20	195	20	195	20	195	mA
H1-H6,H16, H8-H14,H18	Iih	XI	X	0	80	0	80	0	80	uA
H7,H15,H17	Iih	XI	X	0	40	0	40	0	40	uA

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
H19	Iih	XI	X	-2.6	-3.4	-2.6	-3.4	-2.6	-3.4	mA
H20-H23,H25	Iih	XI	X	0	300	0	300	0	300	uA
H24	Iih	XI	X	0	1000	0	1000	0	1000	uA
J1-J6,J16, J8-J14	Iil	XII	X	0.0	-3.50	0.0	-3.50	0.0	-3.50	mA
J7,J15,J18	Iil	XII	X	0.0	-1.75	0.0	-1.75	0.0	-1.75	mA
J17	Iil	XII	X	0.0	-0.88	0.0	-0.88	0.0	-0.88	mA
J19	Iil	XII	X	-5.0	-10.8	-5.0	-10.8	-5.0	-10.8	mA
J20-J23,J25	Iil	XII	X	0	-10	0	-10	0	-10	mA
J24	Iil	XII	X	0	-20	0	-20	0	-20	mA
K1-K24	Voh	XIII	X	5.22	5.55	5.22	5.55	5.22	5.55	V
K25-K40	Voh	XIII	X	5.05	5.55	5.05	5.55	5.05	5.55	V
K41-K47	Voh	XIII	X	2.4	4.5	2.4	4.5	2.4	4.5	V
L1-L47	Vol	XIV	X	0.0	0.4	0.0	0.4	0.0	0.4	V
M1,M2	FUNCTIONAL, priority interrupt and clock shaping	XV	X	---------- SEE TABLE XV ----------						
N1-N4	DYNAMIC, min and max injector current	XVI	X	---------- SEE TABLES XVI THRU XIX ----------						

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
P1-P40	tTHL C1= 100 pF	XX	Q	5	40	5	40	5	45	nS
P41, P43-P45	tTHL C1= 50 pF	XX	Q	4	50	4	50	4	55	nS
P42,P46,P47	tTHL C1= 50 pF	XX	Q	4	20	4	20	4	25	nS
Q1,Q3-Q5	tTLH C1= 50 pF	XXI	Q	10	80	10	80	10	90	nS
Q2,Q6,Q7	tTLH C1= 50 pF	XXI	Q	3	45	3	45	3	55	nS
R1	tPLH C1= 50 pF	XXII	Q	10	58	10	60	10	60	nS
R2	tPLH C1= 50 pF	XXII	Q	10	43	10	45	10	45	nS
R3-R18,R28, R42,R60	tPLH C1= 50 pF	XXII	Q	10	61	10	63	10	63	nS

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
R19,R34-R37 R47,R49,R52 R53,R64,R67 R70,R72	tPLH Cl= 50 pF	XXII	Q	10	56	10	58	10	58	nS
R20-R27	tPLH Cl= 50 pF	XXII	Q	10	51	10	53	10	53	nS
R29-R33, R38-R41, R43-R45,R50 R51,R54,R55 R58,R59,R61 R62,R65,R68 R71,R73,R75 R77,R79	tPLH Cl= 50 pF	XXII	Q	10	46	10	48	10	48	nS
R46,R48,R56 R57,R63,R66 R69,R74,R76 R78	tPLH Cl= 50 pF	XXII	Q	10	50	10	52	10	52	nS
R80-R83	tPLH Cl= 50 pF	XXII	Q	5	29	5	31	5	31	nS
S1,S28,S42 S60	tPHL Cl= 50 pF	XXIII	Q	10	68	10	71	10	71	nS

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
S2,S29,S43 S61	tPHL Cl= 50 pF	XXIII	Q	10	38	10	41	10	41	nS
S3-S18	tPHL Cl= 50 pF	XXIII	Q	10	48	10	51	10	51	nS
S19	tPHL Cl= 50 pF	XXIII	Q	10	41	10	44	10	44	nS
S20-S27	tPHL Cl= 50 pF	XXIII	Q	10	36	10	39	10	39	nS
S30-S33, S38-S41, S44,S45,S50 S51,S54,S55 S58,S59,S62 S65,S68,S71 S73,S75,S77 S79	tPHL Cl= 50 pF	XXIII	Q	10	37	10	40	10	40	nS
S34-S37,S47 S49,S52,S53 S64,S67,S70 S72	tPHL Cl= 50 pF	XXIII	Q	10	63	10	66	10	66	nS
S46,S48,S56 S57,S63,S66 S69,S74,S76 S78	tPHL Cl= 50 pF	XXIII	Q	10	42	10	45	10	45	nS

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
S80-S83	tPHL Cl= 50 pF	XXIII	Q	5	44	5	47	5	47	nS
T1,T2	Dynamic functional and prop delay time check	XXIV	Q	---------- SEE TABLES XXV AND XXVI and 6.3----------						
U1,U3	Max frequency check, clock shaping (PRP2)	XXVII	Q	201	207	201	207	201	207	nS
U2,U4	Max frequency check, clock shaping (PWA2)	XXVII	Q	52	69	52	69	52	69	nS

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TABLE I. Module electrical requirements. (CLASS I)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		0 ± 5 deg C 4/		60 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
V1	Power Supply Capacitance, Vcc	XXVIII	S	3.22	5.98	3.22	5.98	3.22	5.98	uF
V2	Power Supply Capacitance, INJ	XXVIII	S	200	1500	200	1500	200	1500	pF
W1-W16	Cin	XXIX	Q	2	20	2	20	2	20	pF
W17-W41	Cin	XXIX	Q	2	40	2	40	2	40	pF

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
A	ISOLATION	IV								
	Initial		S	10	INF	10	INF	10	INF	Mohm
	End-of-life		S	1	INF	1	INF	1	INF	Mohm
B1-B4	CONTINUITY	V	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
C	Vcc STRESS	VI	S	---------- MUST PASS ALL REMAINING TESTS ----------						
D1-D16	Ibd	VII	S	0.0	1.0	0.0	0.1	0.0	0.1	mA
D17	Ibd	VII	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	VII	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
E1-E40	Ios	VIII	S	-9.1	-12.4	-9.1	-12.4	-9.1	-12.4	mA
E41,E43-E45	Ios	VIII	S	-20	-55	-20	-55	-20	-55	mA
E42,E46,E47	Ios	VIII	S	-20	-100	-20	-100	-20	-100	mA
F1-F40	Vclamp	IX	S	0.0	-1.5	0.0	-1.5	0.0	-1.5	V
G1	Vinj	X	X	0.88	1.63	0.88	1.63	0.88	1.63	V
G2	Vinj	X	X	0.91	1.68	.91	1.68	.91	1.68	V
G3	Icc	X	X	20	195	20	195	20	195	mA
H1-H6,H16, H8-H14,H18	Iih	XI	X	0	80	0	80	0	80	uA
H7,H15,H17	Iih	XI	X	0	40	0	40	0	40	uA

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
H19	Iih	XI	X	-2.6	-3.4	-2.6	-3.4	-2.6	-3.4	mA
H20-H23,H25	Iih	XI	X	0	300	0	300	0	300	uA
H24	Iih	XI	X	0	1000	0	1000	0	1000	uA
J1-J6,J16, J8-J14	Iil	XII	X	0.0	-3.50	0.0	-3.50	0.0	-3.50	mA
J7,J15,J18	Iil	XII	X	0.0	-1.75	0.0	-1.75	0.0	-1.75	mA
J17	Iil	XII	X	0.0	-0.88	0.0	-0.88	0.0	-0.88	mA
J19	Iil	XII	X	-5.0	-10.8	-5.0	-10.8	-5.0	-10.8	mA
J20-J23,J25	Iil	XII	X	0	-10	0	-10	0	-10	mA
J24	Iil	XII	X	0	-20	0	-20	0	-20	mA
K1-K24	Voh	XIII	X	5.22	5.55	5.22	5.55	5.22	5.55	V
K25-K40	Voh	XIII	X	5.05	5.55	5.05	5.55	5.05	5.55	V
K41-K47	Voh	XIII	X	2.4	4.5	2.4	4.5	2.4	4.5	V
L1-L47	Vol	XIV	X	0.0	0.4	0.0	0.4	0.0	0.4	V
M1,M2	FUNCTIONAL, priority interrupt and clock shaping	XV	X	---------- SEE TABLE XV ----------						
N1-N4	DYNAMIC, min and max injector current	XVI	X	---------- SEE TABLES XVI THRU XIX ----------						

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
P1-P40	tTHL Cl= 100 pF	XX	Q	5	40	5	40	5	47	nS
P41, P43-P45	tTHL Cl= 50 pF	XX	Q	4	50	4	50	4	57	nS
P42,P46,P47	tTHL Cl= 50 pF	XX	Q	4	20	4	20	4	27	nS
Q1,Q3-Q5	tTLH Cl= 50 pF	XXI	Q	10	80	10	80	10	95	nS
Q2,Q6,Q7	tTLH Cl= 50 pF	XXI	Q	3	45	3	45	3	60	nS
R1	tPLH Cl= 50 pF	XXII	Q	10	58	10	60	10	61	nS
R2	tPLH Cl= 50 pF	XXII	Q	10	43	10	45	10	46	nS
R3-R18,R28, R42,R60	tPLH Cl= 50 pF	XXII	Q	10	61	10	63	10	64	nS

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
R19,R34-R37 R47,R49,R52 R53,R64,R67 R70,R72	tPLH Cl= 50 pF	XXII	Q	10	56	10	58	10	59	nS
R20-R27	tPLH Cl= 50 pF	XXII	Q	10	51	10	53	10	54	nS
R29-R33, R38-R41, R43-R45,R50 R51,R54,R55 R58,R59,R61 R62,R65,R68 R71,R73,R75 R77,R79	tPLH Cl= 50 pF	XXII	Q	10	46	10	48	10	49	nS
R46,R48,R56 R57,R63,R66 R69,R74,R76 R78	tPLH Cl= 50 pF	XXII	Q	10	50	10	52	10	53	nS
R80-R83	tPLH Cl= 50 pF	XXII	Q	5	29	5	31	5	32	nS
S1,S28,S42 S60	tPHL Cl= 50 pF	XXIII	Q	10	68	10	71	10	72	nS

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
S2,S29,S43 S61	tPHL Cl= 50 pF	XXIII	Q	10	38	10	41	10	42	nS
S3-S18	tPHL Cl= 50 pF	XXIII	Q	10	48	10	51	10	52	nS
S19	tPHL Cl= 50 pF	XXIII	Q	10	41	10	44	10	45	nS
S20-S27	tPHL Cl= 50 pF	XXIII	Q	10	36	10	39	10	40	nS
S30-S33, S38-S41, S44,S45,S50 S51,S54,S55 S58,S59,S62 S65,S68,S71 S73,S75,S77 S79	tPHL Cl= 50 pF	XXIII	Q	10	37	10	40	10	41	nS
S34-S37,S47 S49,S52,S53 S64,S67,S70 S72	tPHL Cl= 50 pF	XXIII	Q	10	63	10	66	10	67	nS
S46,S48,S56 S57,S63,S66 S69,S74,S76 S78	tPHL Cl= 50 pF	XXIII	Q	10	42	10	45	10	46	nS

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ± 5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
S80-S83	tPHL Cl= 50 pF	XXIII	Q	5	44	5	47	5	48	nS
T1,T2	Dynamic functional and prop delay time check	XXIV	Q	---------- SEE TABLES XXV AND XXVI and 6.3 ----------						
U1,U3	Max frequency check, clock shaping (PRP2)	XXVII	Q	201	207	201	207	201	207	nS
U2,U4	Max frequency check, clock shaping (PWA2)	XXVII	Q	52	69	52	69	52	69	nS

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TABLE I. Module electrical requirements. (CLASS II)

TEST COND	PARAMETER	REF TEST TABLE	TEST TYPE 1/-3/	25 ± 5 deg C 4/		-55 ± 5 deg C 4/		85 ±5 deg C 4/		UNITS
				MIN	MAX	MIN	MAX	MIN	MAX	
V1	Power Supply Capacitance, Vcc	XXVIII	S	3.22	5.98	3.22	5.98	3.22	5.98	uF
V2	Power Supply Capacitance, INJ	XXVIII	S	200	1500	200	1500	200	1500	pF
W1-W16	Cin	XXIX	Q	2	20	2	20	2	20	pF
W17-W41	Cin	XXIX	Q	2	40	2	40	2	40	pF

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Notes:

1/ "X" parameters shall be 100 percent tested during the module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1% or two modules from each inspection lot, whichever is greater) at 25 deg C
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life

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3 5 7 Holddown holes The module shall have no additional requirements for location or quantity of holddown holes

3 5 8 Marking Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/394-1 for class I and M28787/394-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may act as a heat barrier)

3 6 Circuit pin configuration The functional block diagram shall conform to figure 1 The pin assignments shall conform to figure 2

3 7 Acceptance test All production modules shall be tested in accordance with footnotes 1 and 2 of table I

3 8 End-of-life limits Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25 deg C end-of-life limits of table I (see 6 2)

3 9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (Vcc) of 5 0±0 5Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5 5 Vdc ± 1 percent

4 QUALITY ASSURANCE PROVISIONS

4 1 Quality assurance The quality assurance provisions of MIL-M-28787 shall apply

4 2 Acceptance tests Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787

4 3 Test equipment The following test equipment is necessary to conduct acceptance tests for this module Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 Designator	No required
Megohm Bridge	R06	1
Ohmmeter	R10	1
Oscilloscope, High Frequency	D06	1
Capacitance Meter	C05	1
Capacitance Bridge	C01	1
DC Power Supply	P04	1
Pulse Generator	J01	2

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Resistors (1/4 watt unless specified otherwise)

	No Required
536 Ohms ± 1%, 1/8W	40
1 KOhms ± 5%, 1/8W	46
51.1 Ohms ± 5%, 1/2W	1
4.02 Ohms ± 1%, 2W	1
392 Ohms ± 1%,	1
6 34 KOhms ± 1%	2
17 8 KOhms ± 1%	1
4 32 KOhms ± 1%	4

Note All resistors shall be metal film and all capacitors shall be ceramic, unless otherwise specified

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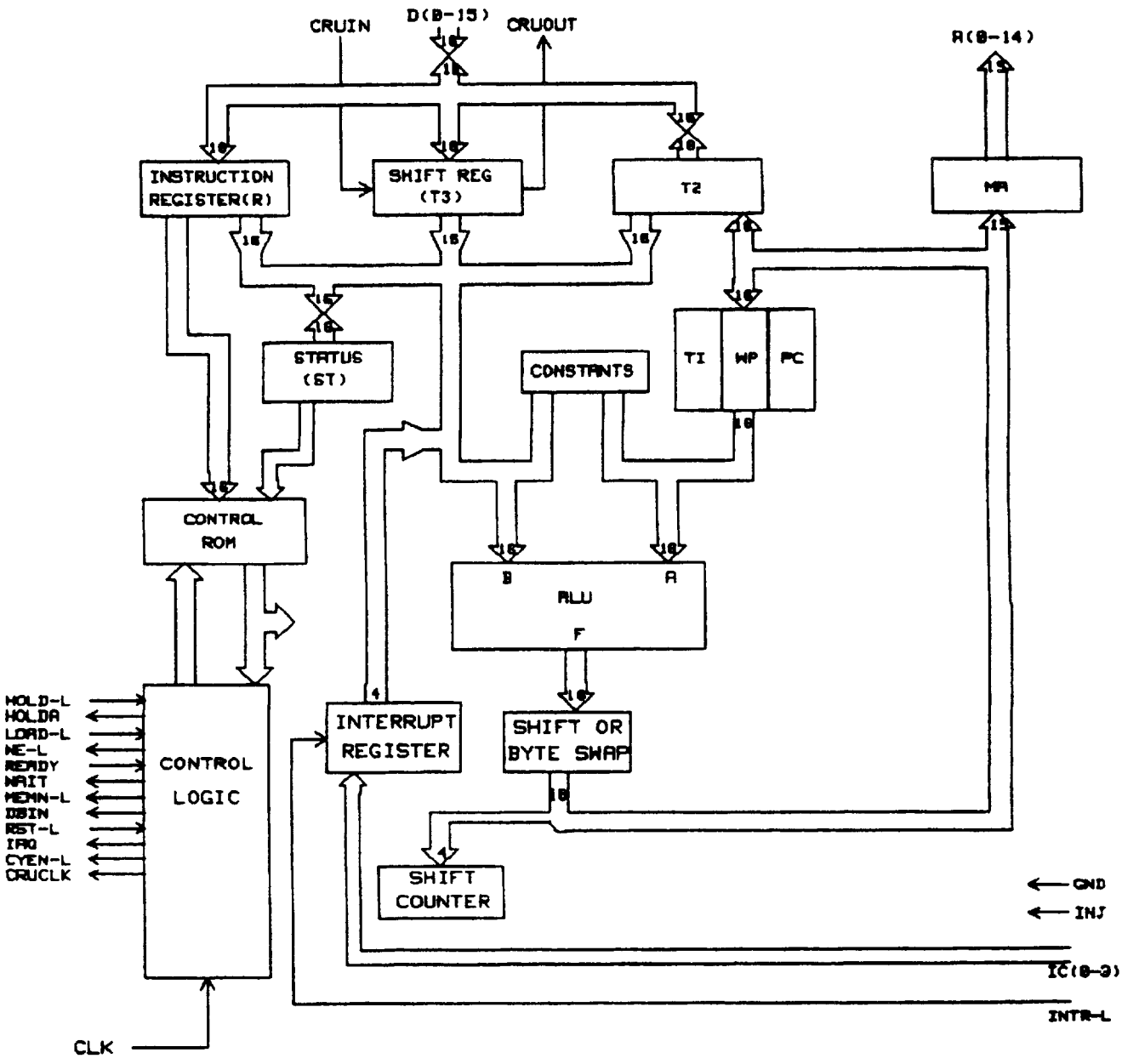


FIGURE 1 MHA Functional Block Diagram (CPU)

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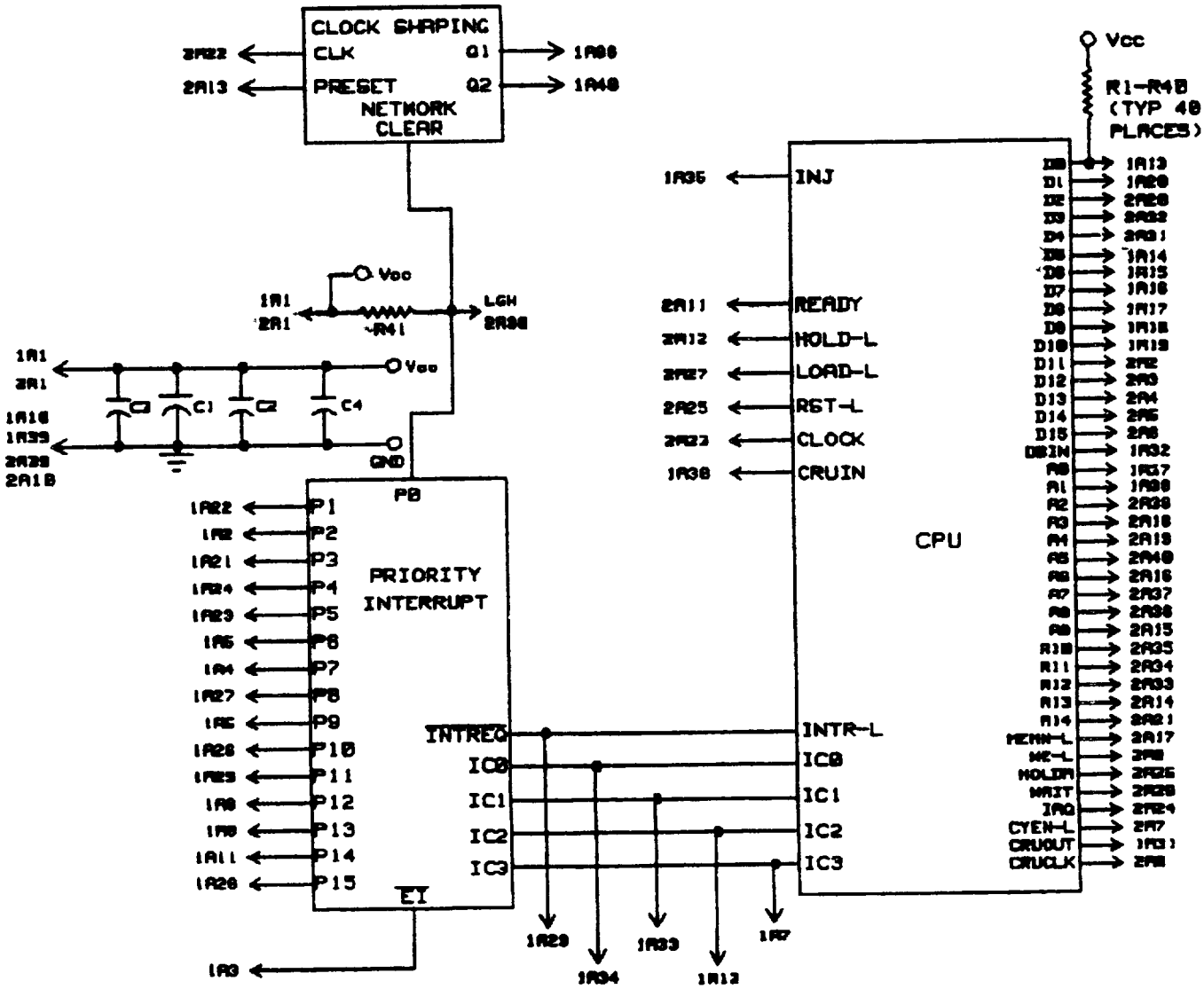


FIGURE 2 MHA Pin Assignment.

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4 4 Test procedures The module shall be tested as specified in table I

4 4 1 Alternate test methods Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787

4 4 2 Order of testing Test conditions A through W shall be performed in any order with the following exceptions Test conditions A,C, D,E and F are stress tests and will be performed only at 25 ±5 deg C If test conditions A,C,D,E or F, are part of a sequence of tests, the order of testing shall be as follows A,C,D,E,F where applicable, followed by the remaining tests in the sequence

4 5 Qualification testing Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions

4 5 1 Operating temperature tests The operating temperature tests shall be conducted in accordance with MIL-M-28787 Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4 4 2 using the initial limits at 25 deg C ambient When electrical testing is called for at the end of each temperature stabilization period, test conditions B and G through V shall be performed in the order specified in 4 4 2 using the initial limits for applicable operating temperature At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3

4 5 1 1 Component temperature test The component temperature test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 5

4 5 2 Vibration tests The vibration tests shall be conducted in accordance with MIL-M-28787 During the vibration tests, the module shall be operated in the test circuit shown on figure 7 Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25 deg C ± 5 deg C ambient

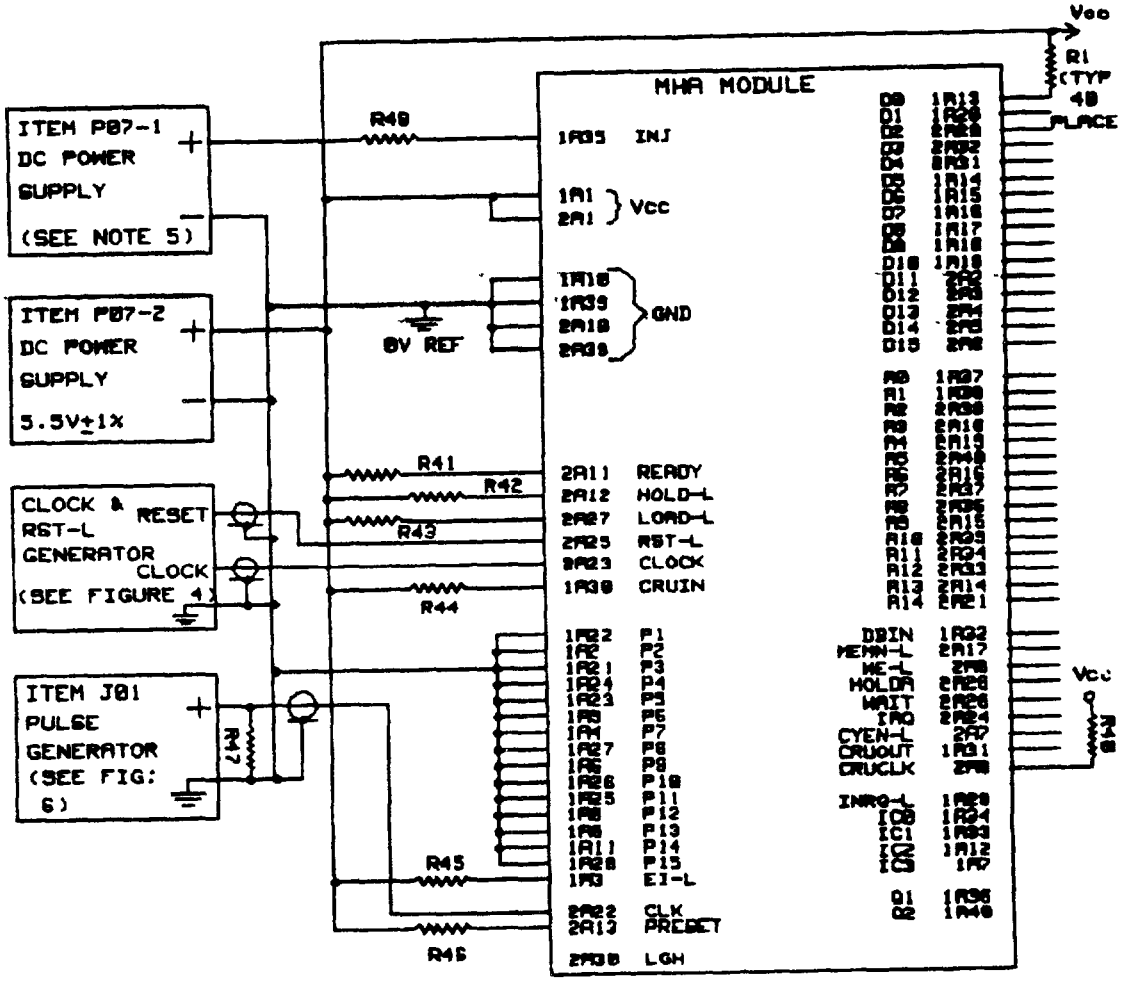
4 5 3 Life test The life test shall be conducted in accordance with MIL-M-28787 The module shall be operated as shown on figure 3 Following the life test and while still at the maximum class temperature, electrical tests B and G through V shall be performed in the order specified in 4 4 2 using the operating temperature end-of-life requirements of table I Following the return of the module to 25 deg ± 5 deg C, electrical tests B and G through V shall be performed in the order specified in 4 4 2 using the 25 deg ± 5 deg C ambient end-of-life requirements of table I

4 5 4 Thermal shock test The thermal shock test shall be conducted in accordance with MIL-M-28787 The module shall be unpowered during this test

5 PACKAGING

5 1 Packaging requirements The requirements for packaging shall be in accordance with MIL-M-28787

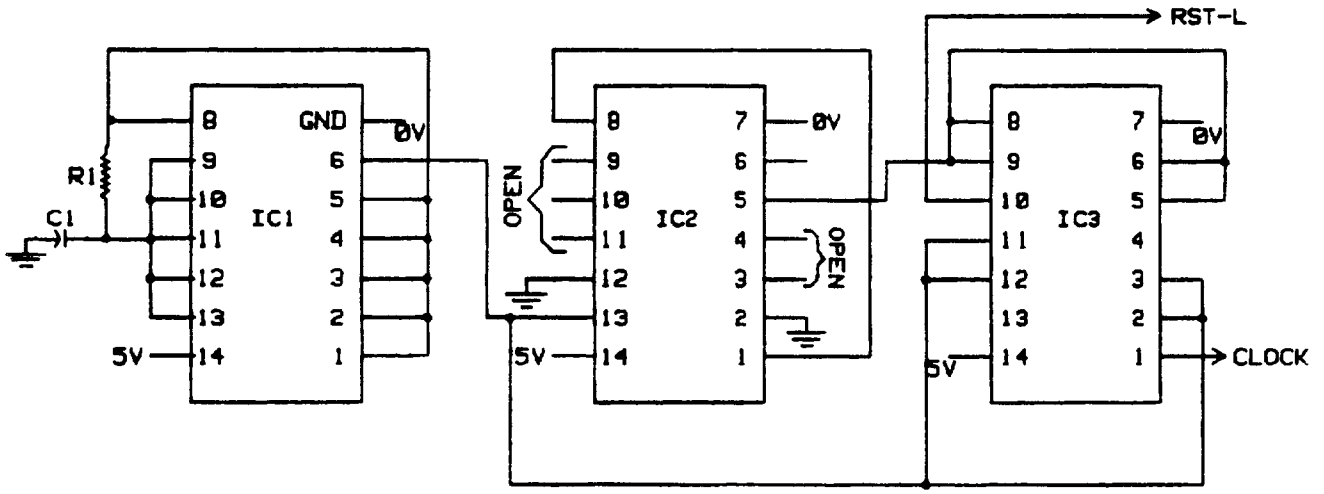
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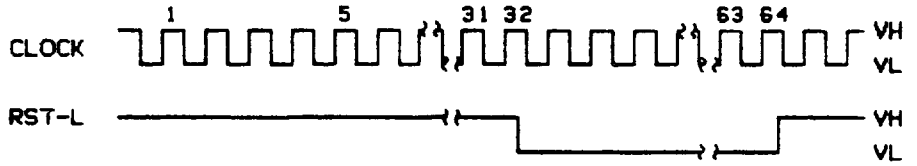


Notes
R1-R40= 536 Ohms ±1%, 1/8 W
R41-R46= 1 0 KOhms ± 5%, 1/8W
R47= 51 Ohms ±5%, 1/2 W
R48= 4 Ohms ± 1%, 2 W
Adjust item P07-1 voltage to limit pin 1A35 current to 520 mA ±1%

FIGURE 3 Operating temperature and life test circuit

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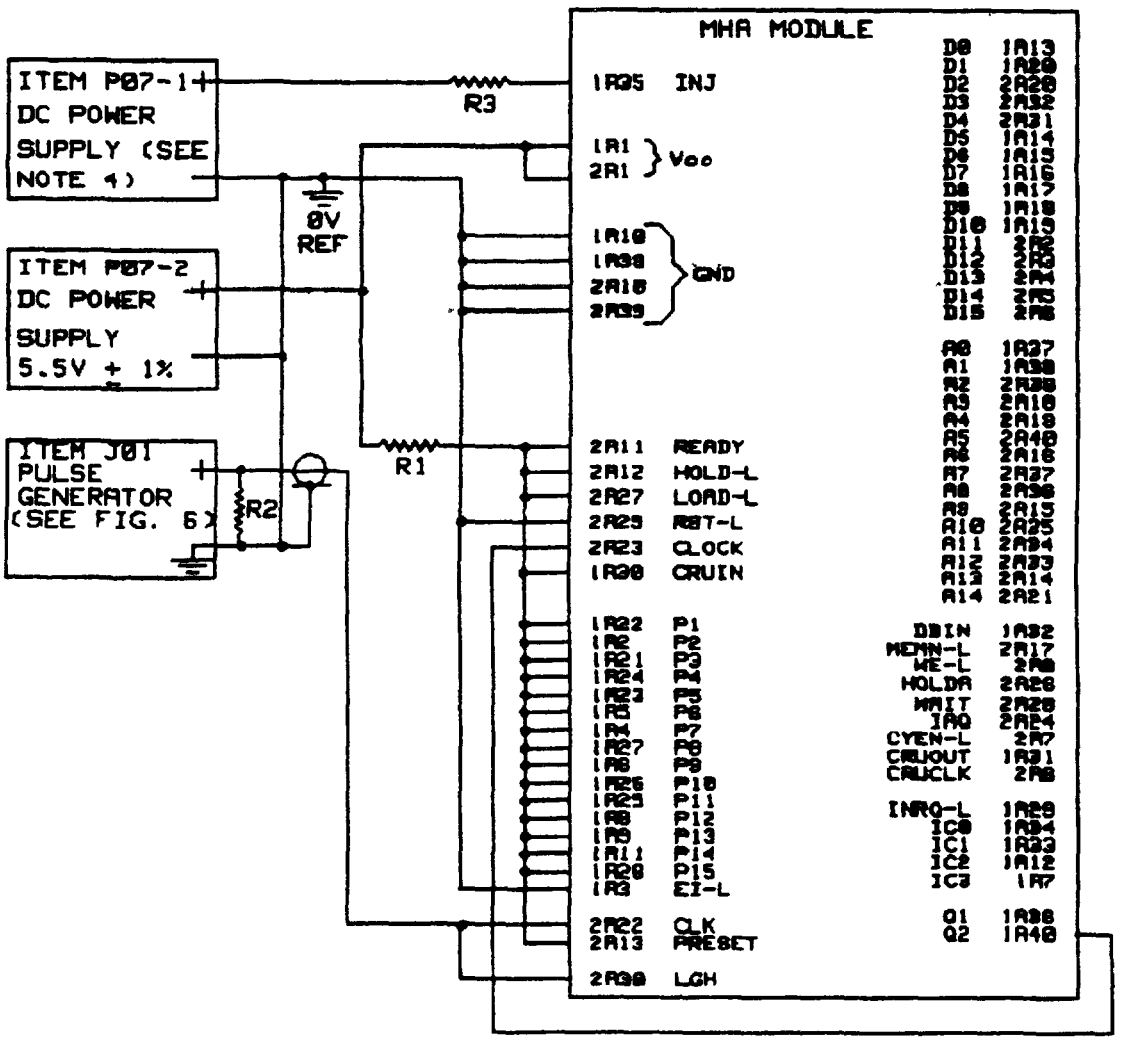




NOTES
1. R1= 390 Ohms ±5%, 1/4 W
2. C1= 001 uF + 20%, -10%
3. IC1= 5413 or equivalent
4. IC2= 54393 or equivalent
5. IC3= 54128 or equivalent

FIGURE 4 CLOCK and RST-L generator circuit

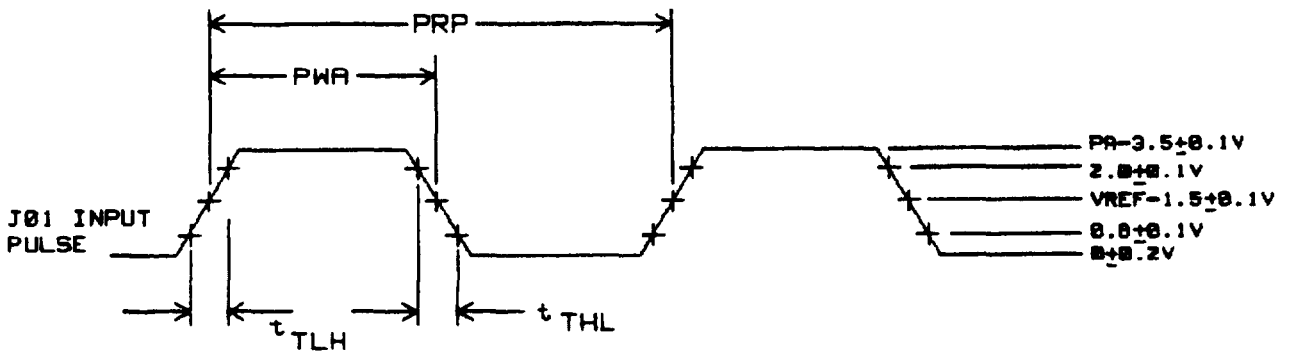
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NOTES
R1= 1 0 KOhms ±5%, 1/8 W
R2= 51 Ohms ±5%, 1/2 W
R3= 4 Ohms ±1%, 2W
Adjust item P07-1 voltage to limit pin 1A35 current to 520 mA ±1%

FIGURE 5 Component temperature test circuit

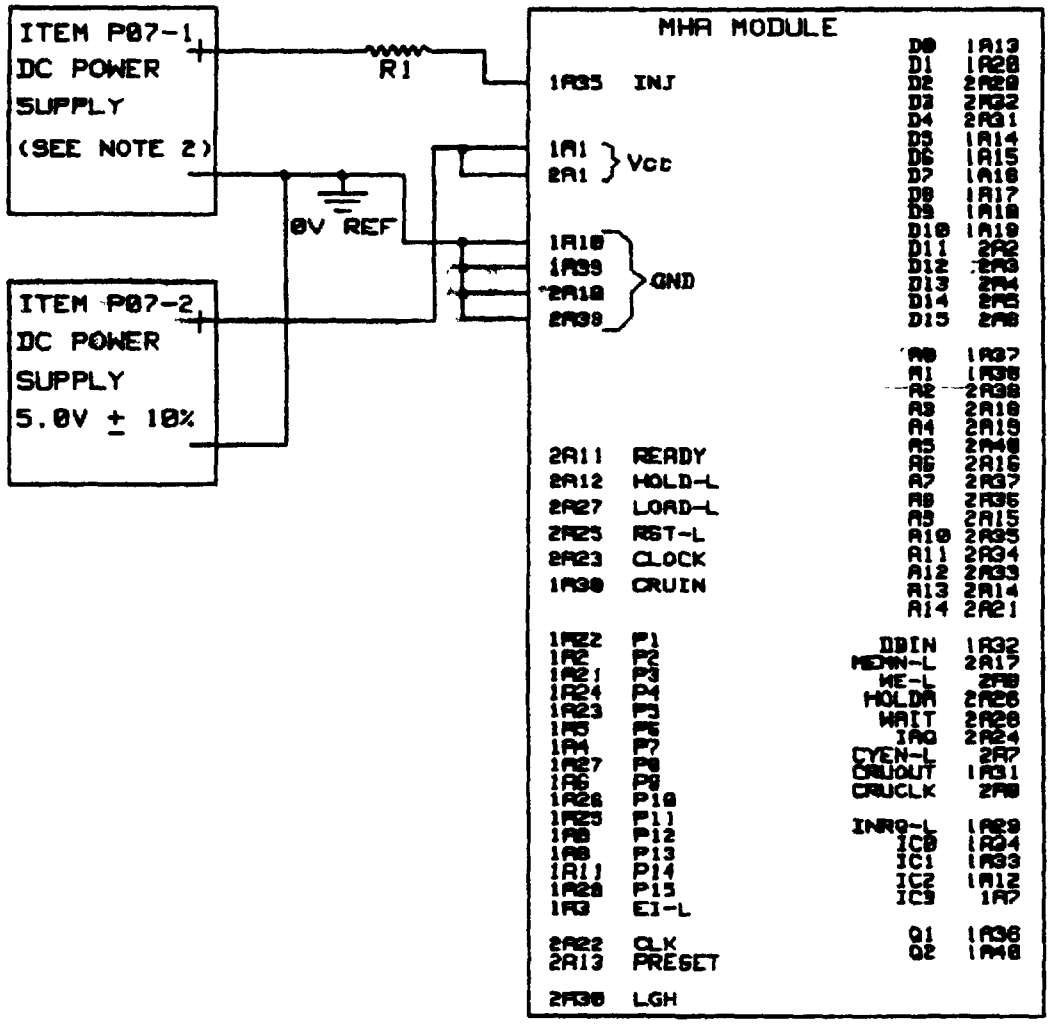
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NOTE Input pulse parameters
tTLH= tTHL= 15±5 ns
PWA= 45±2 5 ns
PRP= 111 ±5 6 ns

FIGURE 6 Operating and component temperature and life test circuit waveform for figures 3 and 5

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NOTES.
1 R1= 4 Ohms ±1%, 2 W
2 Adjust item P07-1 voltage to limit pin 1A35 current to 500 mA ±1%

FIGURE 7. Vibration test circuit

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6 NOTES

6 1 Definitions Definitions are as specified in MIL-STD-1331

6 2 End-of-life When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3 8)

6 3 Substitutability Class II modules are substitutable for class I modules When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3 3 and 3 4)

6 4 CPU FUNCTIONAL DESCRIPTION

6 4 1 CPU BLOCK DIAGRAM The CPU block diagram is shown in figure 1 Addresses are supplied to the Address Bus (A0-A14) by the Memory Address Register (MA) Instructions read from memory are loaded into the Instruction Register (IR) via the Data Bus (D0-D15)

Bit-oriented input/output operations are provided by the Communication Register Unit (CRU) interface whereby 1 to 16 bits may be transferred by a single instruction

6 4 1 1 Arithmetic Logic Unit(ALU) The arithmetic logic unit (ALU) is the computational component of the device It performs all arithmetic and logic functions required to execute instructions The functions include addition, subtraction, AND, OR, exclusive OR, and complement A seperate comparison circuit performs the logical and arithmetic comparisons to control bits 0 through 2 of the status register

The ALU is arranged in two 8-bit halves to accommodate byte operations Each half of the ALU operates on one byte of the operand During word operand operations, both halves of the ALU function in conjunction with each other However, during byte operand processing, the least significant half of the ALU operates in a passive mode, performing no operation on the data that it handles The most signigicant half of the ALU performs all operations on byte operands so that the status circuitry used in word operations is also used in byte operations

6 4 1 2 Internal registers The following three internal registers are accessible to the user

a Program Counter (PC)
b Status Register (ST)
c Workspace Pointer (WP)

Other internal registers which are utilized during instruction acquisition or execution are inaccessible to the user

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6 4 1 2 1 Program Counter (PC) The Program Counter is a 15-bit register (left justified with 16th bit hardwired to 0) that contains the word address of the instruction currently executing The microprocessor increments this address to fetch the next instruction from memory

If the current instruction in the microprocessor alters the contents of the PC, then a program branch occurs to the location specified by the altered contents of PC All context switching operations plus simple branch and jump instructions affect the contents of the PC

6 4 1 2 2 Status Register (ST) The Status Register is a fully implemented 16-bit register that reports the results of program comparisons, indicates program status conditions, supplies the arithmetic overflow enable and interrupt mask level to the interrupt priority circuits, and provides the external bit for memory expansion to 64K words and beyond Each bit position in the register signifies a particular function or condition that exists in the microprocessor Figure 9 illustrates the bit position assignments Some instructions use the Status Register to check for a prerequisite condition, others affect the values of the bits in the register, and others load the entire Status Register with a new set of parameters

6 4 1 2 3 Workspace The device uses blocks of memory words, called workspaces, for instruction operand manipulation instead of internal hardware registers A workspace occupies 16 contiguous memory words in any part of memory The individual workspace registers may contain data or addresses, and function as operand registers, accumulators, address registers, or index registers Some workspace registers take on special significance during execution of certain instructions Figure 10 illustrates the workspace map A user defined number of workspaces may exist in memory which provides a high degree of program flexibility

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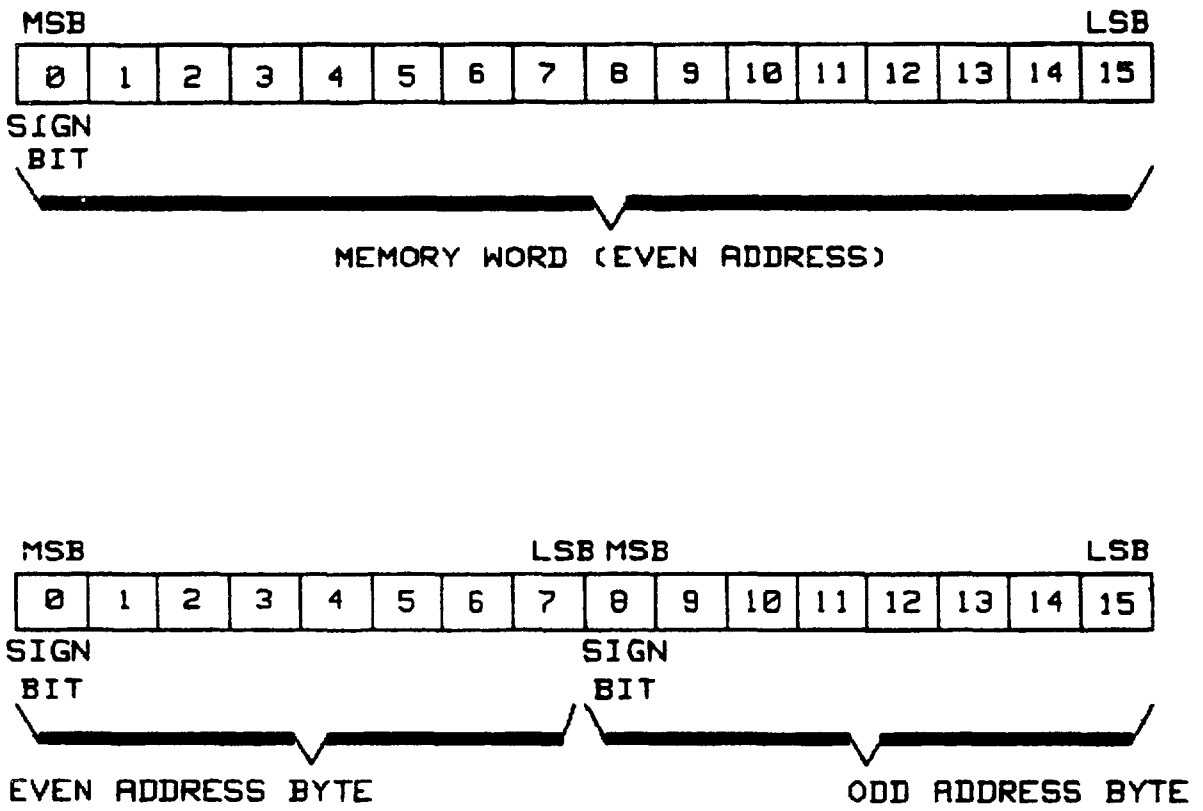


FIGURE 8 Word and byte formats

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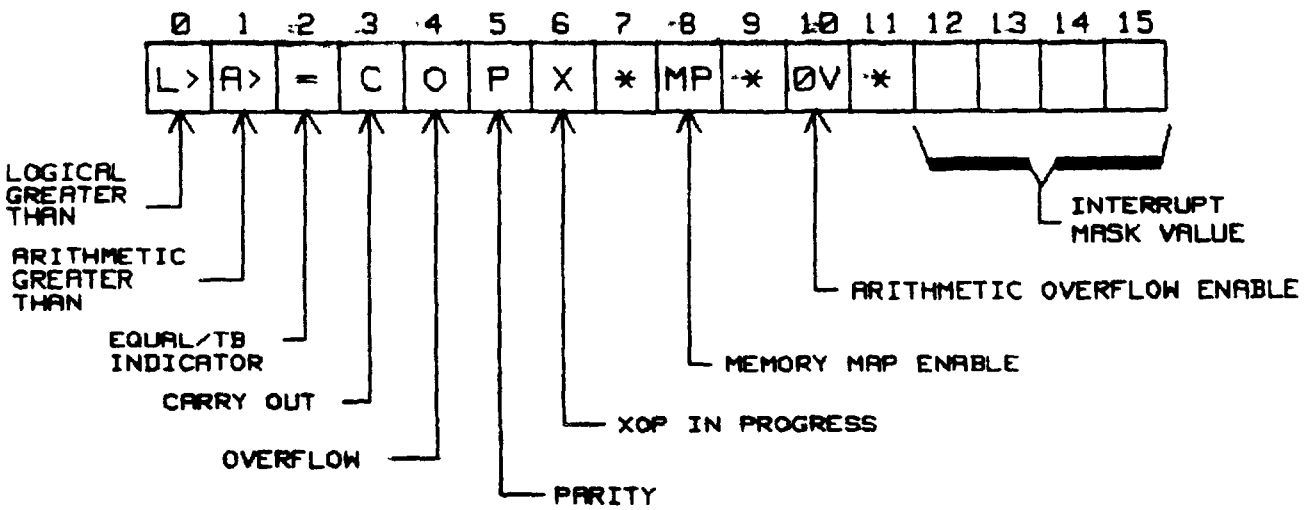
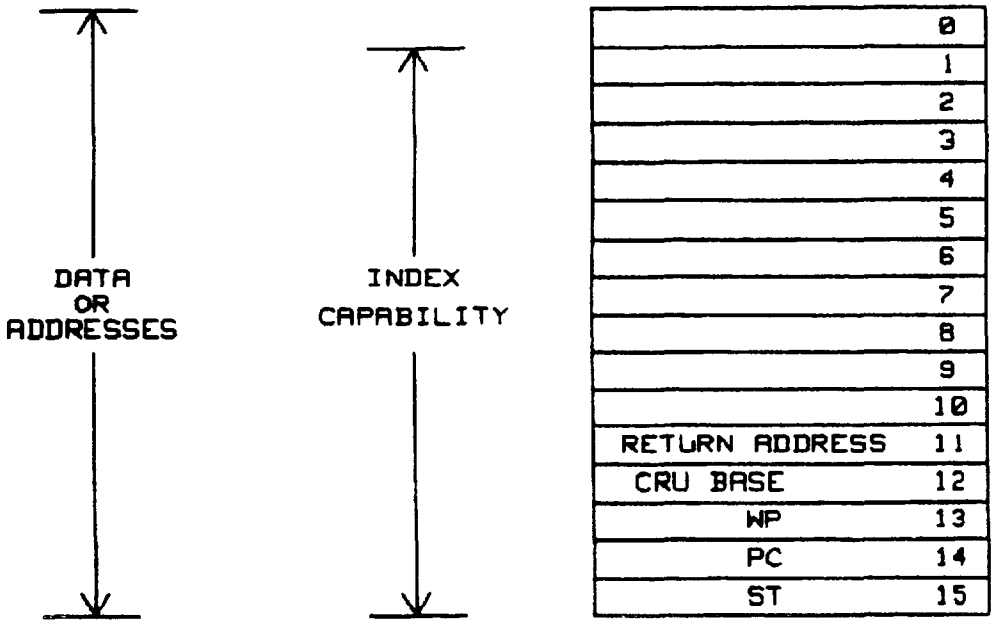


FIGURE 9 Status register bit assignments

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NOTE
The WP register contains the address of workspace register 0

FIGURE 10 Workspace map

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6 4 1 2 4 Workspace Pointer (WP) To locate the workspace in memory, one hardware register called the Workspace Pointer (WP) is used The Workspace Pointer is a 15-bit register (left justified with the 16th bit hardwired to 0) that contains the memory address of the first word in the workspace The device can then access any register in the workspace by adding two times the register number to the contents of the Workspace Pointer and initiating a memory request for that word Figure 11 illustrates the relationship between the Workspace Pointer and its corresponding workspace in memory

6 4 1 2 5 Context Switching The workspace concept is particularly valuable during operations that require a context switch, which is a change from one program environment to another, or to a subroutine, as in the case of an interrupt Such an operation using a conventional multi-register arrangement requires that at least part of the contents of the register file be stored and reloaded using a memory cycle to store or fetch each word This operation is accomplished by changing the Workspace Pointer A complete context switch requires only three store cycles and three fetch cycles, exchanging the Program Counter, Status Register and Workspace Pointer After the switch, the Workspace Pointer contains the starting address of a new 16-word workspace in memory for use in the new routine A corresponding time saving occurs when the original context is restored Instructions that result in a context switch include Branch and Load Workspace Pointer (BLWP) and Return From Subroutine (RTWP) Device interrupts, the arithmetic overflow interrupt, illegal opcode detection trap, RESET and LOAD also cause a context switch by forcing a trap to a service subroutine

6 4 2 INTERFACES

6 4 2 1 Memory Interface The Memory Bus provides the interface between the devices and the memory system It consists of a Data Bus (DB), an Address Bus (AB), and control signals Figure 12 illustrates the signals contained in the memory interface Figures 13 and 14 illustrate the timing relationships of these signals

During each memory-read or memory-write cycle, MEMN-L becomes active (logic level LOW) along with valid memory-address data appearing on the Address Bus (A0 through A14)

In the case of a memory-read cycle, DBIN becomes active (pulled to logic level HIGH) at the same time memory-address data becomes valid, the memory write strobe, WE-L, remains inactive (pulled to logic level HIGH) If the memory-read cycle is initiated for acquisition of an instruction, IAQ becomes active (pulled to a logic level HIGH) at the same time MEMN-L becomes active At the end of a memory-read cycle, MEMN-L and DBIN together become inactive At that time, though the address may change, the Data Bus remains in the input mode until terminated by the next low-to-high transition of the clock

In the case of a memory-write cycle, WE-L becomes active (logic level LOW) with the first high-to-low transition of the clock after MEMN-L becomes active, DBIN remains inactive (LOW) At the end of a memory-write cycle, WE-L and MEMN-L become inactive (MEMN-L on the low-to-high transition of clock, WE-L on the preceding high-to-low transition of clock)

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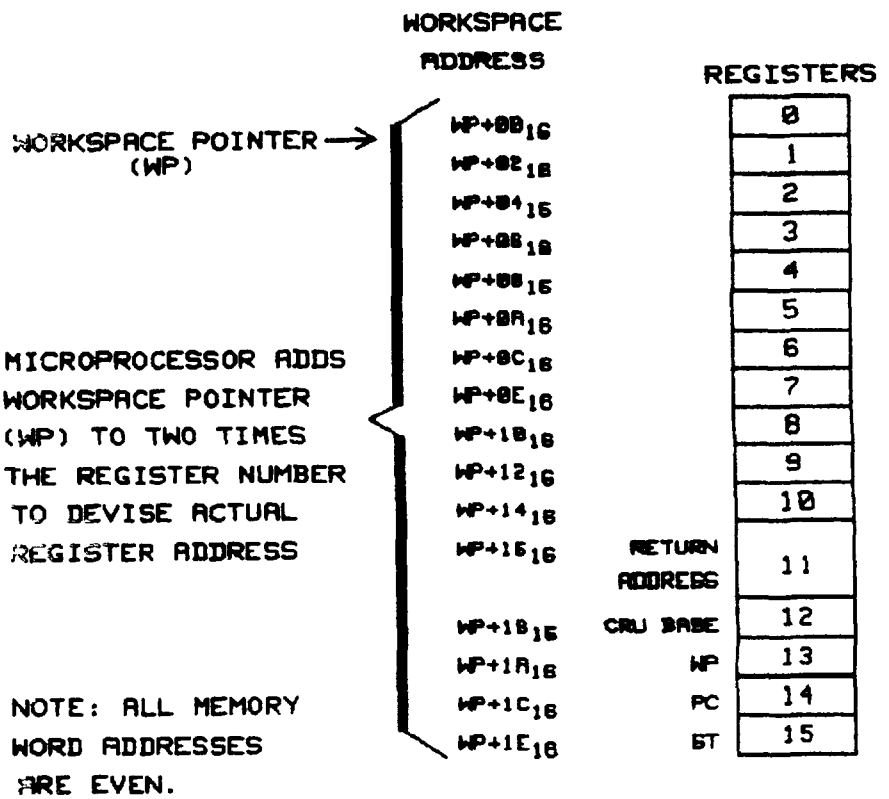


FIGURE 11 Workspace pointer and registers

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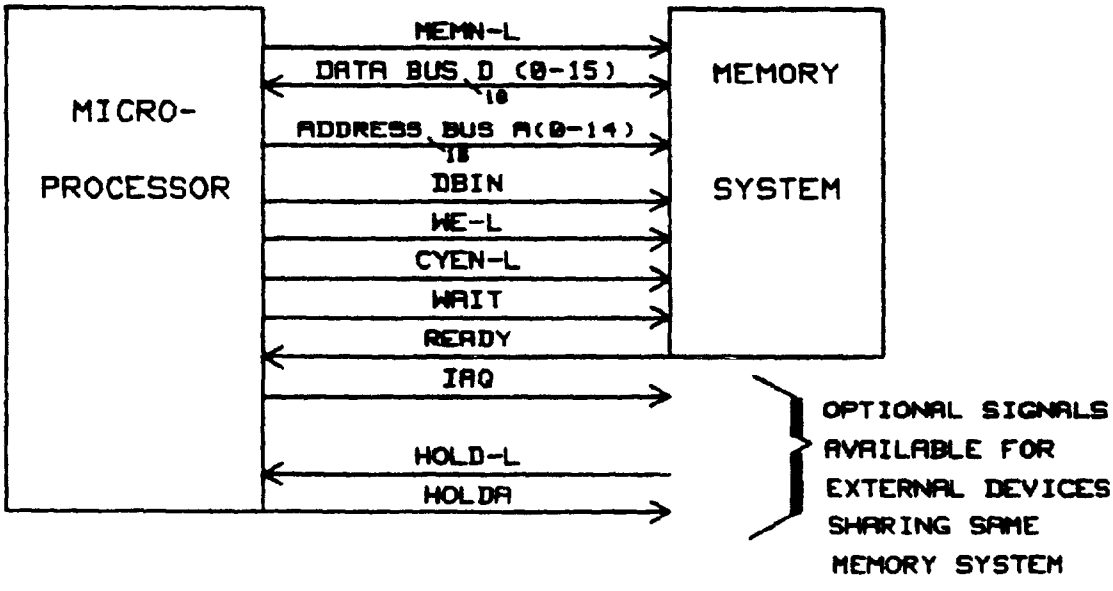
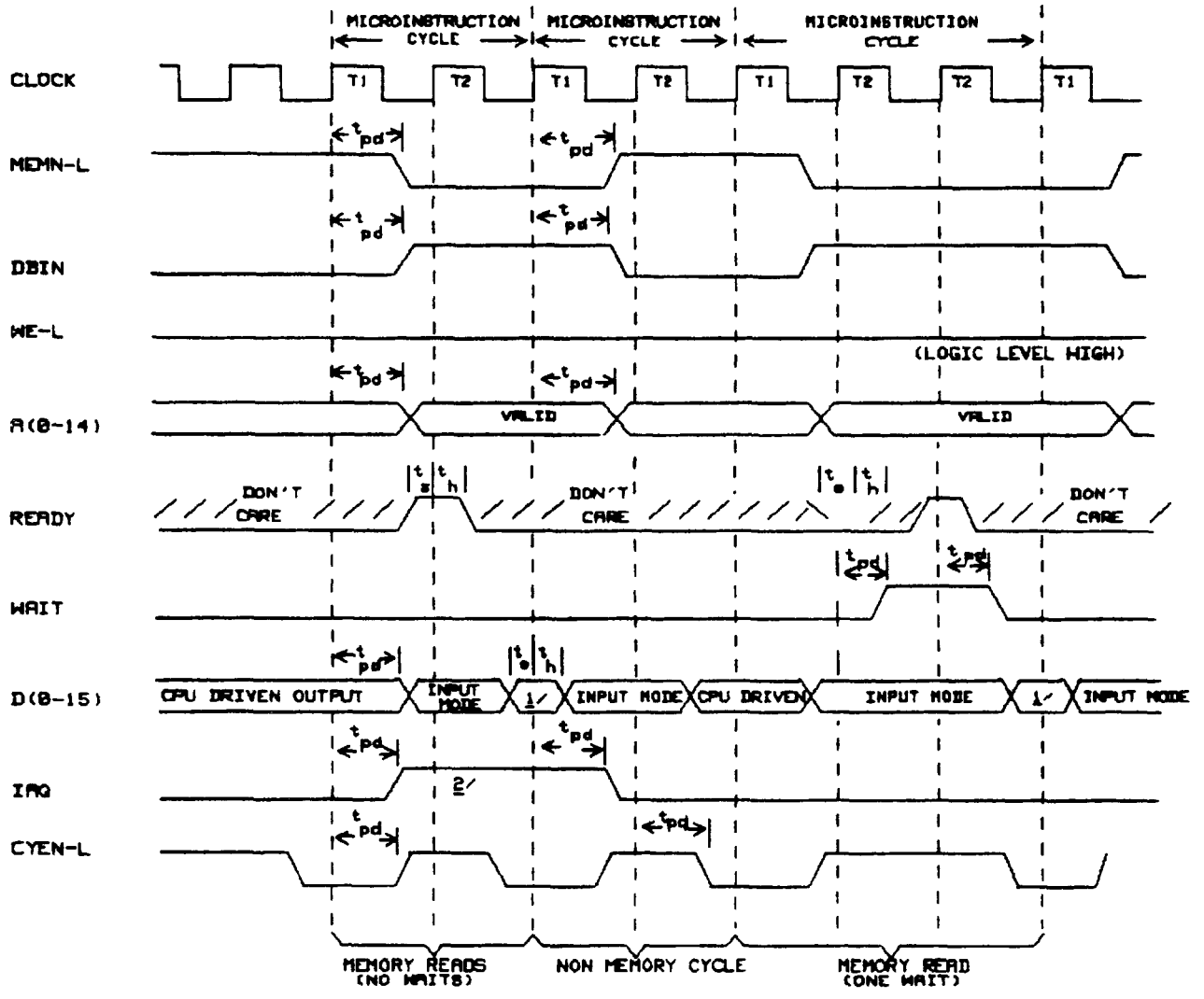


FIGURE 12 Memory interface

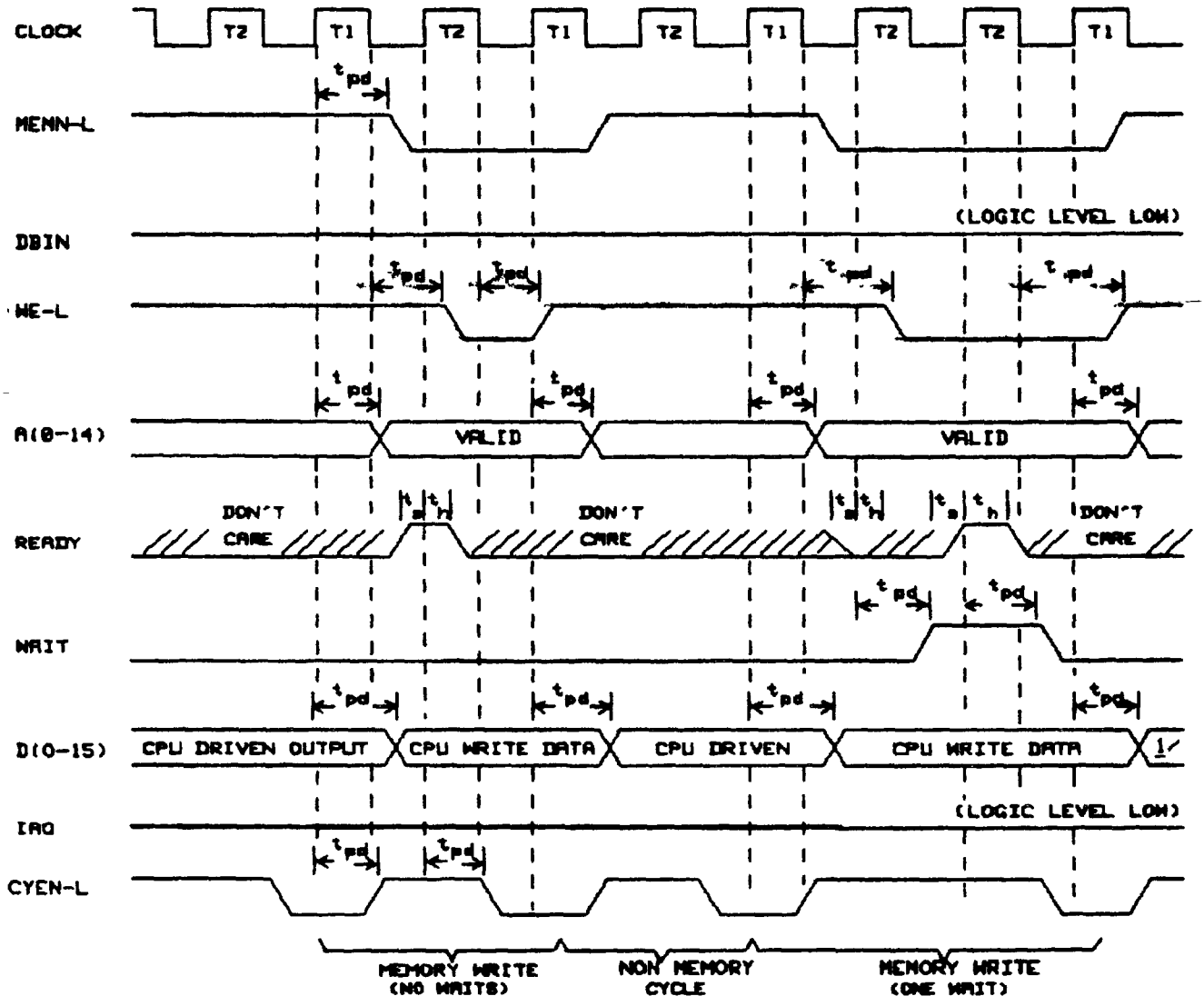
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1/ Valid memory read data
2/ Shown assuming IAQ cycle

FIGURE 13 Memory bus timing read operation

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1/ CPU driven

FIGURE 14 Memory bus timing write operation

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During either a memory-read or a memory-write operation, READY may be used to extend the duration of the associated memory cycle such that the speed of the memory system may be coordinated with the speed of the device If READY is inactive (logic level LOW) during the first low-to-high transition of the clock after MEMN-L becomes active, the device will enter a wait state suspending further progress of the memory cycle The first low-to-high transition of the clock after READY becomes active terminates the wait state and allows normal completion of the memory cycle The cycle end signal (CYEN-L) will be activated (logic level LOW) for one clock period during each microinstruction cycle (i e , memory operation, non-memory internal operation, etc) CYEN-L will be activated on the low-to-high transition of clock which initiates the last clock period of a microinstruction cycle and will be deactivated on the next low-to-high clock transition

The device hold facilities allow both the device and external devices to share a common memory To gain memory bus control, an external device requiring direct memory access (DMA) sends a hold request (HOLD-L) to the device When the next cycle (memory or non-memory) occurs, the device enters a hold state and signals its surrender of the memory bus to the external device via a hold acknowledge (HOLDA) signal

Receiving the hold acknowledgement, the external device can proceed to utilize the common memory After its memory requirements have been satisfied, the external device returns memory bus control by releasing HOLD-L

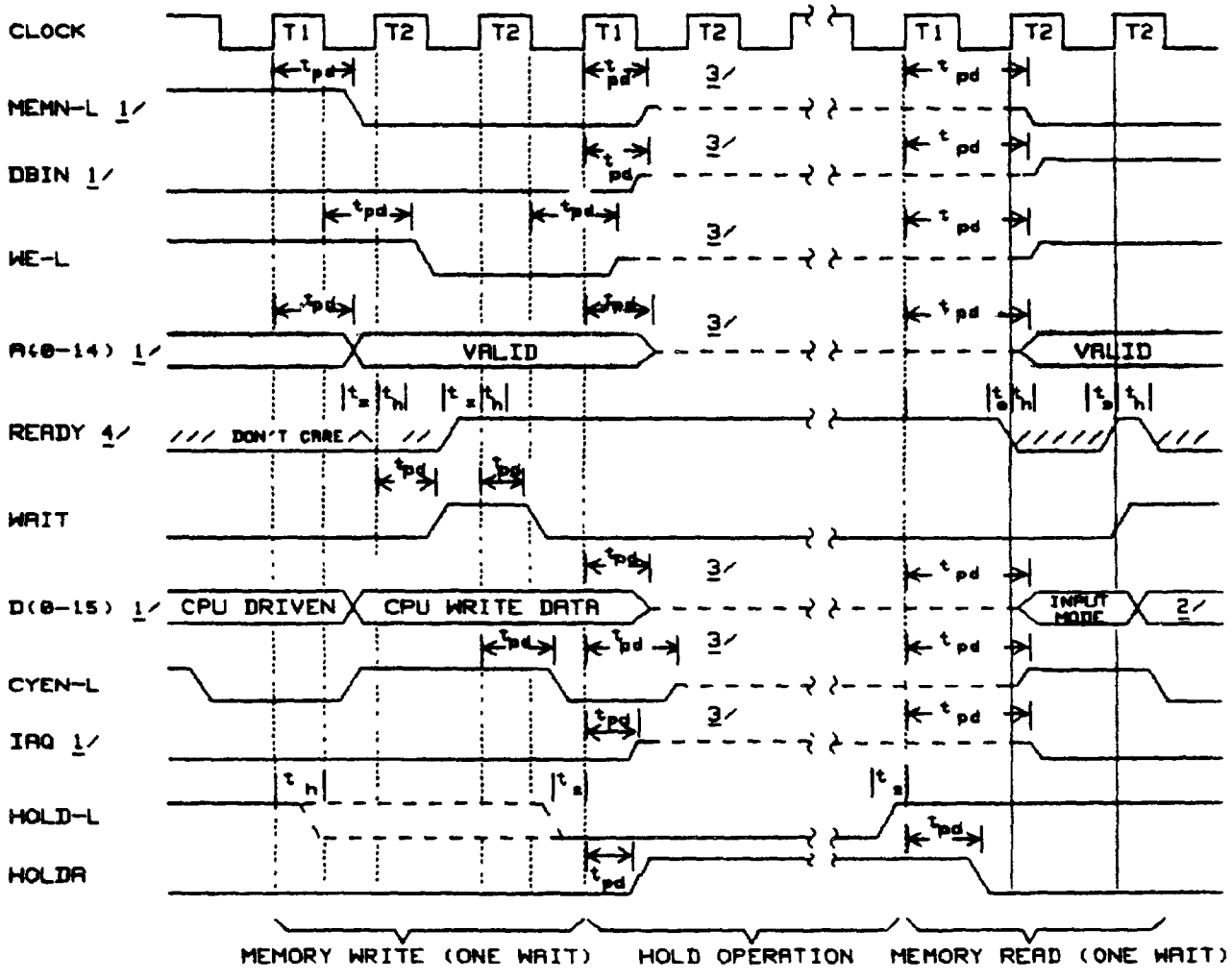
When HOLD-L becomes active (logic level LOW), the device enters a hold state at the beginning of the next available cycle as shown in figure 15 Upon entering a hold state, HOLDA becomes active (pulled to logic level HIGH) with the following signals pulled to a HIGH logic level by individual pullup resistors tied to each respective open-collector output DBIN, MEMN-L, IAQ, MPEN-L, WE-L, CYEN-L, A0 through A14, and D0 through D15 When HOLD-L becomes inactive, the device exits the hold state and regains memory bus control

6 4 3 INTERRUPTS

The device employs 16 interrupt levels with the highest priority level 0 and lowest level 15 Level 0 is reserved for the RST-L function Level 2 is reserved at the user's option for the arithmetic overflow interrupt and/or an illegal op-code trap Interrupt levels 1 through 15 may be used for external device interrupts

External device interrupts are input via the interrupt request (INTR-L) signal line and the four interrupt code lines (IC0-IC3) Figure 16 shows the timing for the sampling of these inputs and their effect on device operation Activation of the INTR-L input causes a comparison with the interrupt code (IC0-IC3) with the interrupt mask contained in status register bits 12 through 15 When the level of the pending interrupt is less than or equal to the enabling mask level (higher or equal priority interrupt), the device recognizes the interrupt and initiates a context switch following completion of the currently executing instruction The new context (WP and PC) is fetched from the interrupt vector locations Then, the previous context (WP, PC and ST) is stored in workspace registers 13, 14, and 15, respectively, of the new workspace Bits 7 through 11 of the status register are forced to zero and set the interrupt mask to a value that is one less than the level of the interrupt being serviced, except for level-zero interrupt, which loads zero into the mask This allows only interrupts of higher priority to interrupt a service routine All interrupt requests should remain active until recognized in the device service routine The individual service routines must reset the interrupt requests before the routine is completed

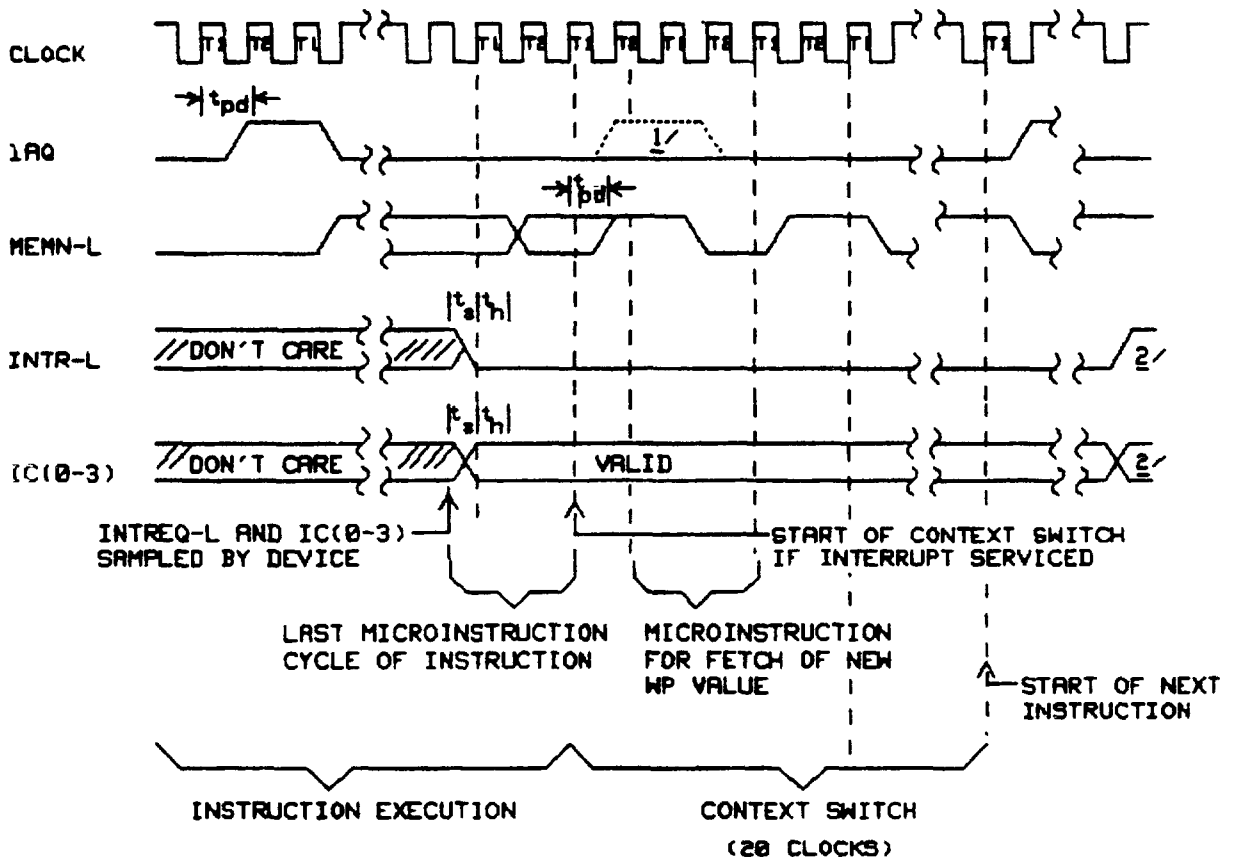
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1/ There is maximum delay of 135 nS for these signals to return to valid levels on leaving a hold state
2/ Valid memory read data
3/ Driven by external device
4/ "Ready" should be held high during a hold operation

FIGURE 15 Memory bus timing hold operation

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1/ Next IAQ if interrupt not serviced
2/ INTR-L and IC(0-3) changed by interrupt service routine
3/ This diagram assumes valid IC is of higher priority than code in status register

FIGURE 16 Interrupt timing

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If a higher priority interrupt occurs, a second context switch occurs to service the higher priority interrupt When that routine is complete, a return instruction (RTWP) restores the routine parameters to complete processsing of the lower priority interrupt All interrupt subroutines should terminate with the return instruction to restore original program parameters The interrupt vector locations, device assignments, enabling mask value, and the interrupt codes are shown in Table II

During a device hold state, resulting from the activation of HOLD-L, the device will sample the interrupt code lines Upon activation of INTR-L, if the code is less than or equal to the device interrupt mask level, the device will continue to sample the interrupt code lines until HOLD-L is released

6 4 3 1 Interrupt Level 0- RESET-L Interrupt Level 0 is reserved for the RST-L input to the device When asserted (logic level LOW), RST-L causes the device to reset itself and inhibit WE-L and CRUCLK

When RST-L is released, a level zero interrupt sequence is initiated, acquiring the WP and PC trap vectors from memory locations 0000(16) and 0002(16), sets all status register bits to logic level LOW, and then fetches the first instruction of the reset program if LOAD-L is not active If LOAD-L is active, the LOAD-L trap occurs after the RST-L function is completed The device continuously samples RST-L on low to high clock transitions as shown in figure 17 To be recognized, RST-L must be active for one low-to-high transition of the clock and satisfy the setup and hold time requirements Interrupt level 2 has two additional capabilities associated with its usage Arithmetic overflow conditions, as indicated by status register bit 4 (ST4) being set to a 1, can cause a level 2 interrupt to occur at the end of the instruction which generated the overflow condition Servicing of this overflow interrupt can be enabled/disabled by status register bit 10, the Arithmetic Overflow Enable Bit (i e , ST10 = 1 enables overflow interrupt, ST10= 0 disables overflow interrupt) The overflow interrupt can also be inhibited by the interrupt mask (ST12-ST15) or overridden by a pending level 0 or level 1 interrupt If servicing the overflow interrupt is overridden by the servicing of a higher priority (level 0 or 1) interrupt, the occurrence of the overflow condition will be retained in the contents of the status register which are saved by the higher priority interrupt context switch Returning from the higher priority subroutine via an RTWP instruction causes the overflow condition to be reloaded into status register bit 4 and the overflow interrupt to occur (upon completion of the RTWP instruction) Servicing of a level 2 arithmetic overflow interrupt forces the interrupt mask to 0001(2) The arithmetic overflow interrupt service routine must reset ST4 to zero before the routine is complete It should be noted that only those instructions specifically denoted as affecting ST4 will change ST4 (either to 0 or 1) The acquisition and execution of "illegal" device (undefined) op-codes cause a trap operation to occur using the information stored in the level 2 vectors The op-codes which cause the trap are 0000-007F, 00A0-017F, 0320-033F, 0780-07FF, and 0C00-0FFF The occurrence of the trap is non-maskable (i e , not controlled by the interrupt mask value) and the trap will override any level interrupt Interrupts are inhibited until the first instruction of the trap subroutine is executed The occurrence of the "illegal" op-code trap does not change the interrupt mask

6 4 4 Communication Register Unit (CRU) Interface The communications register unit (CRU) is a direct command driven bit-oriented I/O interface The CRU may directly address, in bit-fields of one to sixteen bits, up to 4096 peripheral input bits, and up to 4096 peripheral output bits The device executes three single-bit and two multiple-bit CRU instructions The single-bit instructions include TEST BIT (TB), SET BIT TO ONE (SBO), and SET BIT TO ZERO (SBZ), the multiple-bit instructions include LOAD CRU (LDCR) and STORE CRU (STCR)

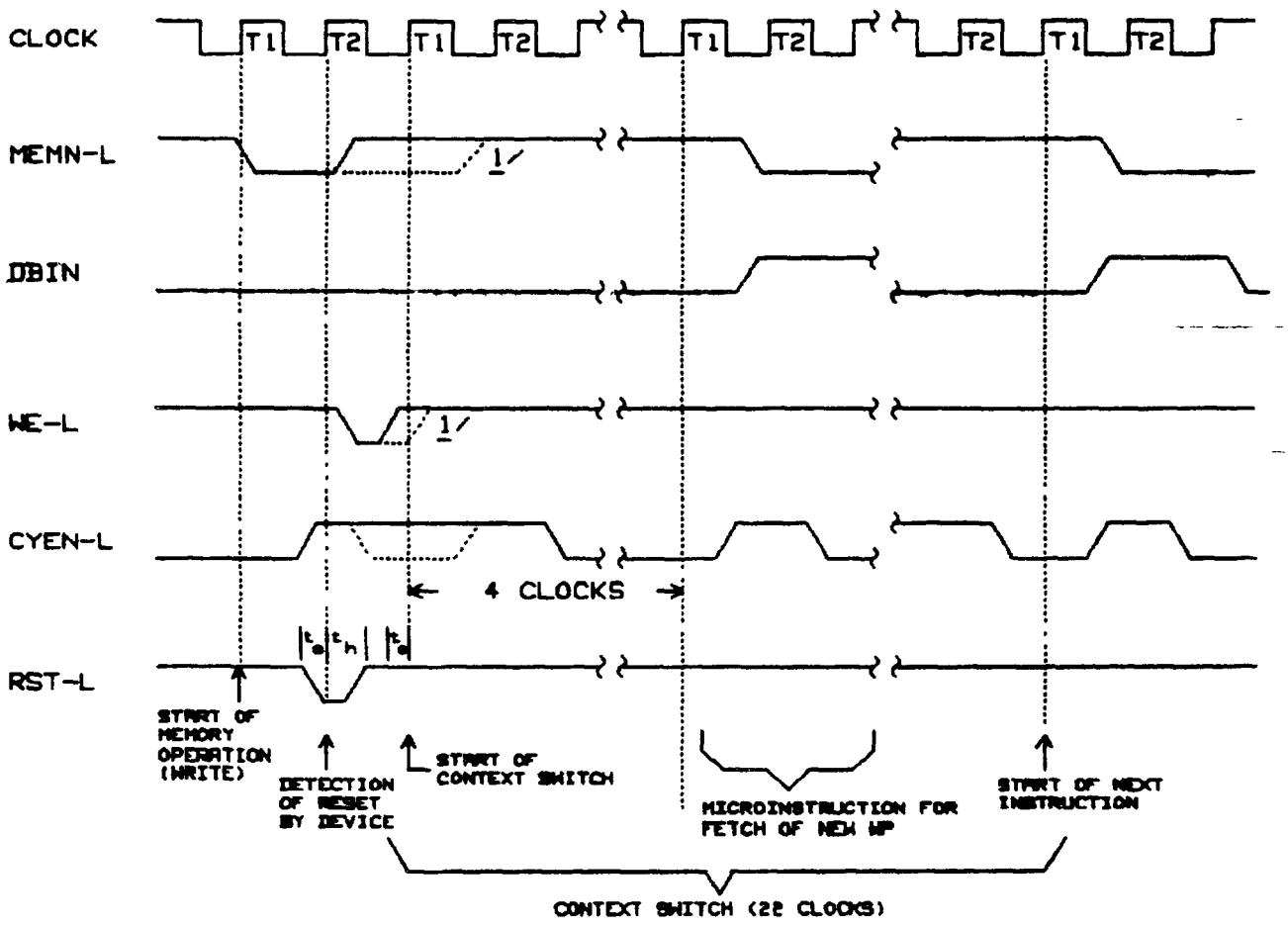
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TABLE II Interrupt level data

INTERRUPT LEVEL		VECTOR LOCATION (MEMORY ADDRESS IN HEX)	DEVICE ASSIGNMENT	ENABLING MASK VALUES (ST12 THRU ST15)	INTERRUPT CODES (IC0 THRU IC3)
(Highest priority)	0	00	Reset	0 through F (see Note 1)	0000
	1	04	External Device	1 through F	0001
	2	08	Arithmetic Overflow or	2 through F (see Note 2)	See Note 2
	2	08	Illegal op-code or	0 through F (see Note 3)	See Note 3
	2	08	External Device	2 through F	0010
	3	0C	•	3 through F	0011
	4	10	•	4 through F	0100
	5	14	•	5 through F	0101
	6	18	•	6 through F	0110
	7	1C	•	7 through F	0111
	8	20	•	8 through F	1000
	9	24	•	9 through F	1001
	10	28	•	A through F	1010
	11	2C	•	B through F	1011
	12	30	•	C through F	1100
	13	34	•	D through F	1101
	14	38	•	E and F	1110
(Lowest priority)	15	3C	External Device	F only	1111

NOTES
1 Level 0 cannot be disabled
2 Arithmetic overflow interrupt is generated internal to the device and is enabled/disabled by bit 10 of the status
3 Illegal op-code trap is operated internal to the device and it cannot be disabled by the interrupt mask

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1/ Dashed lines indicate signal waveforms if RST-L had not occurred
2/ Case shown assumes RST-L is detected during memory operation
3/ Synchronize RST-L with IAQ on T1 to prevent loss of instruction in progress
4/ RST-L may occur on any low-to-high clock transition
5/ Context switch will start on low-to-high clock edge following inactivation of RST-L

FIGURE 17 Reset timing.

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As shown in figure 18, the device utilizes three dedicated I/O signals CRUIN, CRUOUT, CRUCLK, the least significant twelve bits of the address bus, DBIN, READY and WAIT to support the CRU interface CRU interface timing is shown in figures 19 and 20

To transfer a data bit to a CRU device, the device outputs the corresponding CRU-bit address on ADDRESS BUS bits A3 through A14, a CRUCLK pulse and the respective data bit on CRUOUT (ADDRESS BUS bits A0 through A2 are set to zero during CRU transfers) This process is repeated until transfer of the entire field of data bits specified by the CRU instruction has been accomplished To transfer a data bit from a CRU device, the microprocessor outputs the corresponding CRU-bit-address on ADDRESS BUS bits A3 through A14 and receives the respective data bit on CRUIN No CRUCLK pulses occur during a CRU input operation

During all CRU input and output operations, the device will activate the DBIN output (logic level HIGH) and place the 16-bit Data Bus in the input mode Activation of DBIN will allow users to detect the occurrence of a CRU cycle and facilitate cycle stealing by a DMA device since all uses of the DATA and ADDRESS busses will be indicated by the activation of DBIN or MEMN-L

During either a CRU input or output operation, READY may be used to extend the duration of the associated CRU operation such that the speed of the CRU device may be coordinated with the speed of the device If READY is inactive (logic level LOW) during the first low to high transition of clock after the initiation of the CRU operation, the device will enter a wait state (i e , WAIT becomes active) suspending further progress of the CRU cycle The first low-to-high transition of clock after READY becomes active terminates the wait state and allows normal completion of the CRU operation

6 4 4 1 Single Bit CRU Operations The device performs three single-bit CRU functions Test bit (TB), set bit to one (SBO), and set bit to zero (SBZ) To identify the bit to be operated upon, a CRU-bit address is developed and placed on the ADDRESS BUS, A3 to A14 For the two output operations (SBO and SBZ), the device also activates DBIN, generates a CRUCLK pulse, indicating an output operation to the CRU device, and places bit 7 of the instruction word on the CRUOUT line to accomplish the specified operation (Bit 7 is a one for SBO and a zero for SBZ) A test bit (TB) instruction transfers the addressed CRU bit from the CRUIN input line to bit 2 of the status register (ST2)

The device develops a CRU-bit address for the single-bit operations from the CRU base address contained in workspace register 12 and the signed displacement count contained in bits 8 through 15 of the instruction The displacement allows two's complement addressing from base minus 128 bits through base plus 127 bits The base address from workspace register 12 is added to the signed displacement specified in the instruction and the result is loaded onto the ADDRESS BUS Figure 21 illustrates the development of a single-bit CRU address

6 4 4 2 Multiple Bit CRU Operations The device performs two multiple-bit CRU operations store communication register (STCR) and load communications register (LDCR) Both operations perform a data transfer from the CRU-to-memory or from memory-to-CRU as illustrated in figure 22 Although the figure illustrates a full 16-bit transfer operation, any number of bits from 1 through 16 may be involved The LDCR instruction fetches a word from memory and right shifts it to serially transfer it to CRU output bits The LDCR involves eight or fewer bits, those bits come from the right-justified field within the addressed byte of the memory word If the LDCR involves nine or more bits, those bits come from the right justified field within the whole memory word When transferred to the CRU interface, each successive bit

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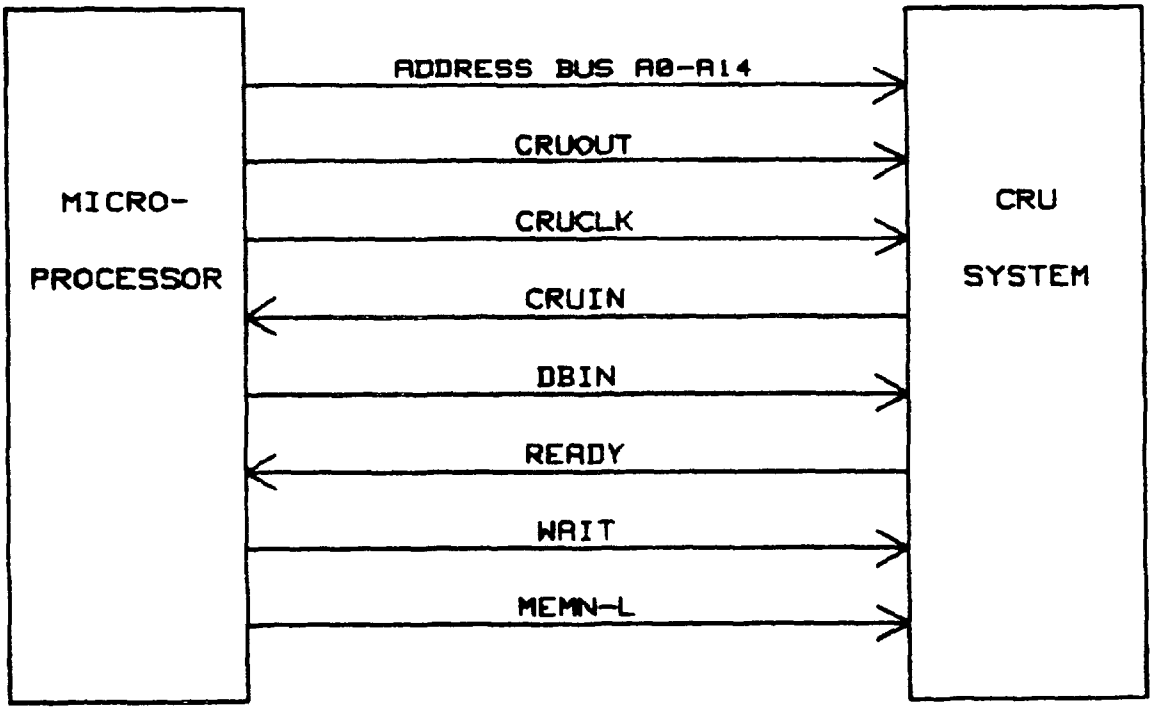
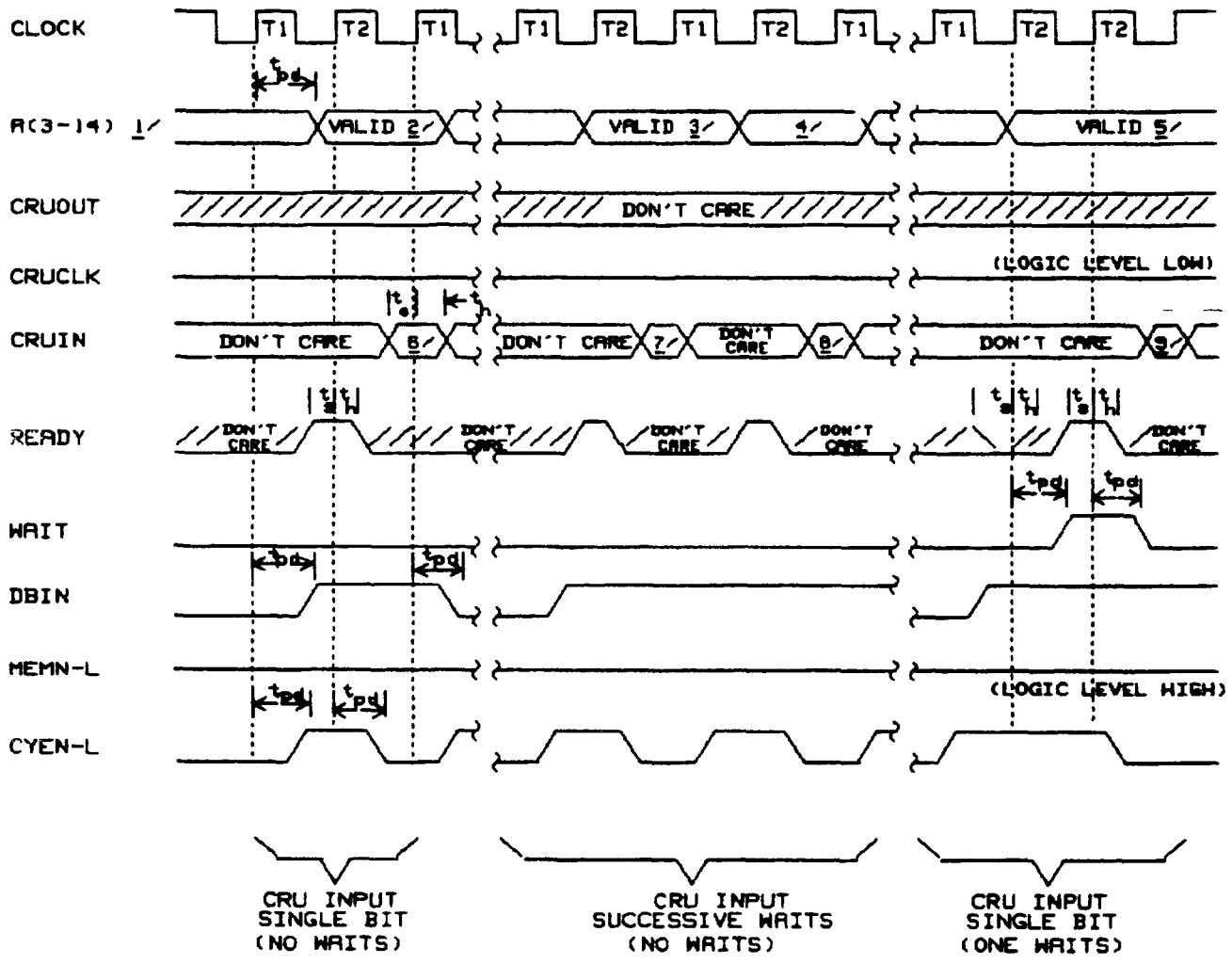


FIGURE 18 CRU interface

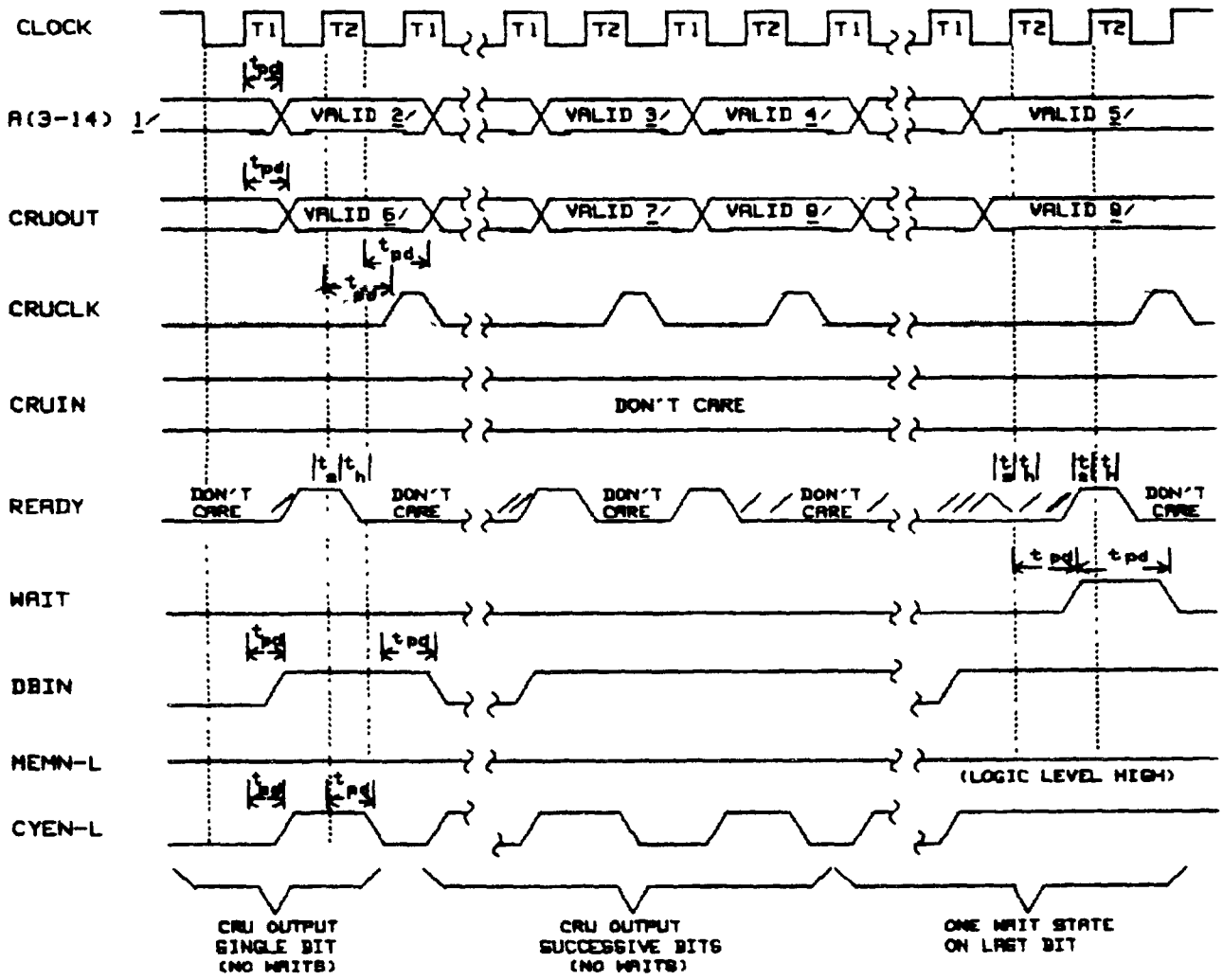
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1/ A(0-2) are set to zero for CRU operations
2/ CRU bit address M
3/ CRU bit address N
4/ CRU bit address N+1
5/ CRU bit address R
6/ Valid input bit M
7/ Valid input bit N
8/ Valid input bit N+1
9/ Valid input bit R

FIGURE 19 CRU timing (input operation)

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1/ A(0-2) are set to zero for CRU operations
2/ CRU bit address M
3/ CRU bit address N
4/ CRU bit address N+1
5/ CRU bit address R
6/ CRU data out address M
7/ CRU data out address N
8/ CRU data out address N+1
9/ CRU data out address R

FIGURE 20 CRU timing (output operation)

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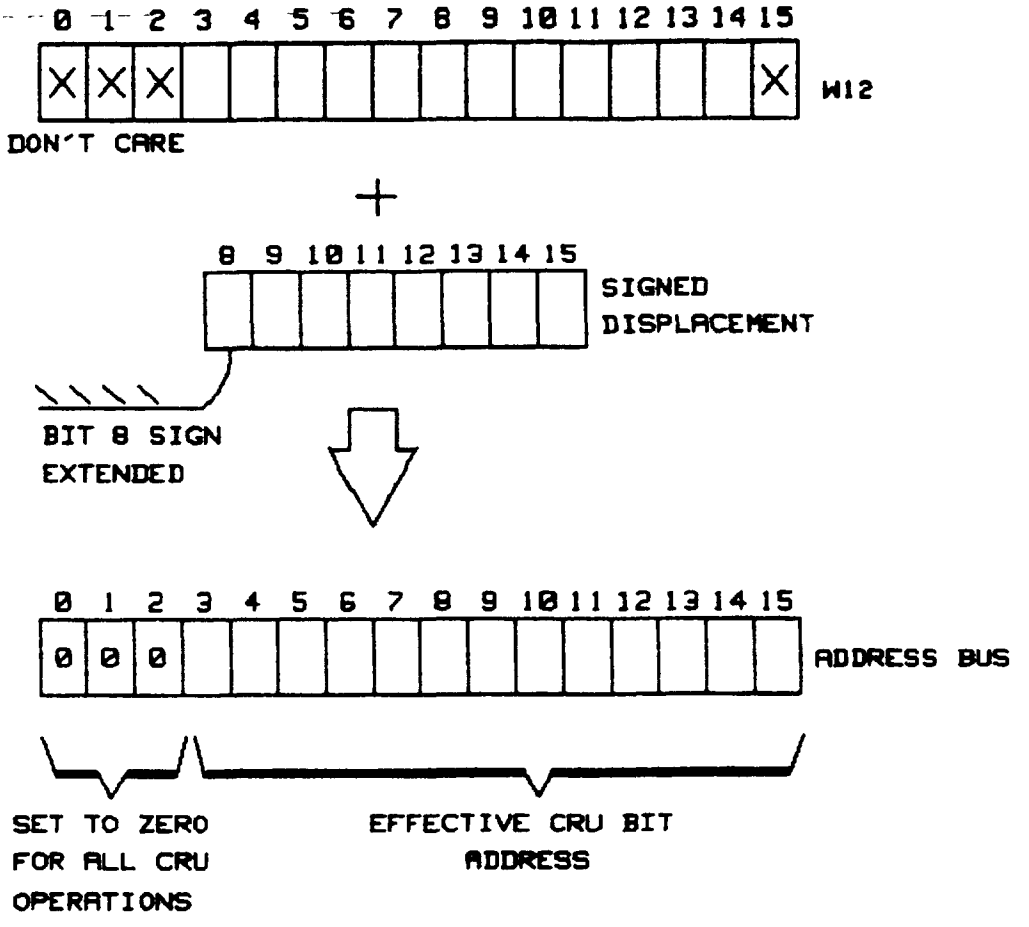
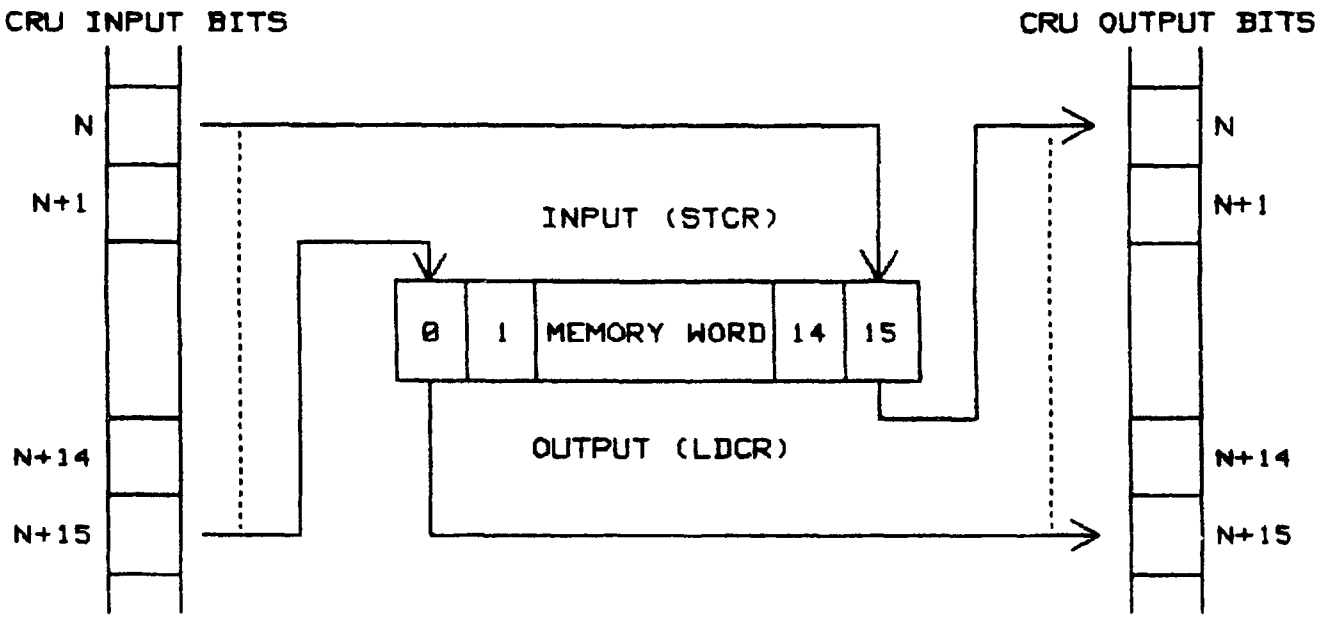


FIGURE 21 Single bit CRU address development

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N=BIT SPECIFIED BY CRU BASE REGISTER

FIGURE 22 LDCR/STCR data transfers

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receives an address that is sequentially greater than the address for the previous bit This addressing mechanism results in an order reversal of the bits, that is, bit 15 of the memory word (or bit 7) becomes the lowest addressed bit in the CRU field and bit 0 becomes the highest addressed bit in tre CRU field

A STCR instruction transfers data from the CRU to memory If the operation involves a byte or less transfer, the transferred data will be stored right-justified in the memory byte with leading bits set to zero If the operation involves from nine to sixteen bits, the transferred data is stored right-justified in the memory word with leading bits set to zero When the input from the CRU device is complete, the first bit from the CRU is the least-significant bit position in the memory word or byte

6 4 5 External Instructions The microprocessor has five external user defined instructions to be initiated under program control These instructions are CKON, CKOF, RSET, IDLE, and LREX IDLE causes the device to enter the idle state and remain until an interrupt, RST-L, or LOAD-L occurs When any of these five instructions are executed, a unique 3-bit code appears on the most-significant 3 bits of the ADDRESS BUS (A0 through A2) along with activation of DBIN and the occurrence of a CRUCLK pulse When in an idle state, the codes on the ADDRESS BUS and CRUCLK pulse occur repeatedly until the idle state is terminated The 3-bit codes are shown in Table III

6 4 5 1 LOAD-L Function The LOAD-L signal allows cold start ("bootstrap") ROM loaders and front panel functions to be implemented When active, LOAD-L causes a trap immediately following the instruction being executed Memory locations FFFC(16) and FFFE(16) are used to obtain the trap vector (WP and PC respectively) The old PC, WP and ST are loaded into the new workspace, status register bits ST57-ST15 are set to 0 (logic level LOW), then program execution resumes using the new PC and WP During a hold state (caused by activation of HOLD-L), the device will sample the LOAD-L input

During non-hold states, LOAD-L shall remain active for one instruction period (i e , LOAD-L should go active during IAQ and remain active until the next IAQ) until a memory read operation occurs at location FFFC(16) During hold states, LOAD-L shall remain active until a memory read operation occurs at location FFFC(16)

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TABLE III External Instruction Codes

EXTERNAL INSTRUCTION	A0	A1	A2
LREX	H	H	H
CKOF	H	H	L
CKON	H	L	H
RSET	L	H	H
IDLE	L	H	L

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6 5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
P1	1A22	
P2	1A2	
P3	1A21	
P4	1A24	
P5	1A23	
P6	1A5	
P7	1A4	
P8	1A27	Priority interrupt request lines (P1-P15). (Inputs)
P9	1A6	
P10	1A26	
P11	1A25	
P12	1A8	
P13	1A9	
P14	1A11	
P15	1A28	
EI-L	1A3	Enable interrupt. (Input)
CLK	1A22	Clock. (Input) Single phase clock input.
PRESET	2A13	Preset. (Input) Allows an asynchronous setup, independent of CLK.
LGH	2A30	(Input) Provides access to CLEAR and P0 (extra priority interrupt line).

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6.5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
READY	2A11	Ready. (Input) When active (high), READY indicates that the memory (for memory operations) or CRU device (for CRU operations) will be ready to read or write during the next clock cycle. When READY is not active (low), MHA enters a wait state and suspends internal operations until the memory system or CRU device activates READY.
HOLD-L	2A12	Hold. (Input) When active (low), HOLD-L indicates to MHA that an external controller (e.g., DMA device) desires to use the memory bus for direct memory data transfers. In response, the MHA enters the hold state after completion of its present cycle (memory or nonmemory). MHA then asserts HOLDA and allows its address bus, data bus, MEMN-L, WE-L, DBIN, IAQ and CYEN-L to be pulled to the high logic state. When HOLD-L is deactivated, MHA reassumes bus control and continues operation by resuming execution of the suspended instruction.
LOAD-L	2A22	Load. (Input) When active (low), LOAD-L causes MHA to store old PC,WP, and ST, set Status Register Bits 9-15 low, and execute a nonmaskable interrupt with unmapped memory addresses FFFC(16) & FFFE(16) containing the associated trap vectors (WP and PC). The load sequence is initiated after the instruction being executed is completed. LOAD-L will also terminate an idle state. If LOAD-L is active at the end of a reset function, the LOAD-L trap will occur after the reset function is completed. During hold states, LOAD-L will remain active until MHA leaves the hold state and the above conditions are met.

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6 5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
RST-L	2A25	Reset. (Input) When active (low logic level), RST-L causes MHA to reset itself, and inhibit WE-L and CRUCLK. When RST-L is released, MHA goes through a level-zero interrupt sequence by storing old PC, WP and ST, setting all status register bits low, acquiring the WP and PC trap vectors from memory locations 0000(16) and 0002(16), and then fetching the first instruction of the reset program environment if LOAD-L is not active. MHA continuously samples RST-L on low-to-high clock transitions. RST-L must be active for one low-to-high transition of the clock and satisfy the hold time requirements of this signal.
CLOCK	2A23	Clock. (Input)
CRUIN	1A30	CRU Data In. (Input) CRUIN receives input data from the external interface logic. When the MHA executes a STCR or TB instruction, it samples CRUIN for the level of the CRU bit specified by the address bus (A3-A14). (CRU= Communication Register Unit.)
DBIN	1A32	Data Bus In. (Output) When activated (high) by MHA during MEMN-L, DBIN indicates that the MHA has disabled its output buffers to allow the memory system to place memory read data on the bus. MHA will also activate DBIN during all CRU operations and during the execution of the five external instructions. In all other cases except when HOLDA is active, MHA will maintain DBIN at a low logic level.
MEMN-L	2A17	Memory Enable. (Output) When active (low), MEMN-L indicates that the address bus contains a valid memory address.
WE-L	2A9	Write Enable. (Output) When active (low), WE-L indicates that the MHA data bus is outputting data to be written into memory.

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6 5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
HOLDA	2A26	Hold Acknowledge. (Output) When active (high), HOLDA indicates that MHA is in a hold state and that its address bus, data bus, MEMN-L, WE-L, DBIN, IAQ, and CYEN-L are pulled to the high state. MHA will enter a hold state in response to the activation of HOLD-L (during the activation of an ILLOP or XOP instruction).
WAIT	2A28	Wait. (Output) When active (high), WAIT indicates the MHA has entered a wait state in response to a not READY condition from a memory system or a CRU device.
IAQ	2A24	Instruction Acquisition. (Output) IAQ is activated (high) by MHA during any MHA initiated instruction acquisition memory cycle. Consequently, IAQ may be used by an external device as an indication of when to sample the memory data bus to obtain instruction operations code data.
CYEN-L	2A7	End of Cycle. (Output) When active (low), CYEN-L indicates that the MHA will initiate a new microinstruction cycle on the next low-to-high transition of the clock.
CRUOUT	1A31	CRU Data Out. (Output) CRUOUT outputs serial data when the MHA executes a LDCR, SBZ, or SBO instruction. The data on CRUOUT should be sampled by the external interface logic when CRUCLK goes active.
CRUCLK	2A8	CRU Clock. (Output) When active (high), CRUCLK indicates to the external logic the presence of output data on CRUOUT or the presence of an encoded external instruction on A0-A2.

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6 5 Functional Description of I/O Pins.

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
A0	1A37	
A1	1A38	
A2	2A38	
A3	2A18	
A4	2A19	
A5	2A40	
A6	2A16	Address Bus. (Output) This open-collector bus provides the memory address to the memory system when MEMN-L is active and CRU I/O bit addresses to the I/O system when MEMN-L is inactive and DBIN is active.
A7	2A37	
A8	2A36	
A9	2A15	
A10	2A35	
A11	2A34	
A12	2A33	
A13	2A14	
A14	2A21	

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6 5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
D0	1A13	
D1	1A20	
D2	2A20	
D3	2A32	
D4	2A31	
D5	1A14	
D6	1A156	Data Bus. (Input/Output) D0-D15 comprise the bidirectional, open-collector data bus. This bus transfers memory data to (when writing) and from (when reading) the external memory system when MEMN-L is active.
D7	1A16	
D8	1A17	
D9	1A18	
D10	1A19	
D11	2A2	
D12	2A3	
D13	2A4	
D14	2A5	
D15	2A6	
INTR-L	1A29	Interrupt Request. (Input) When active (low), INTR-L indicates that an external interrupt is requesting service. If INTR-L is active, MHA loads the data on the interrupt code input lines IC0-IC3 into the internal interrupt-code storage register. The code is then compared to the interrupt mask bits of the status register. If the interrupt code is equal to or less than Status Register Bits 12-15 (equal or higher

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6 5 Functional Description of I/O Pins

FUNCTIONAL DESIGNATION	PIN NUMBER	FUNCTIONAL DESCRIPTION
INTR-L (con't)	1A29	priority than the previous enabled interrupt level), MHA initiates the interrupt sequence. If the comparison fails, MHA ignores the interrupt request. In that case, INTR-L should be held active. MHA will continue to sample IC0-IC3 until the program enables a sufficiently low interrupt level to accept the requesting interrupt.
IC0	1A34	Interrupt Codes. (Input) IC0 (MSB)-IC3 (LSB), indicating an interrupt identity code, are sampled by MHA when INTR-L is active (low). When IC0-IC3 are LLLL, the highest priority external interrupt is requesting service; when HHHH, the lowest priority external interrupt is requesting service.
IC1	1A33	
IC2	1A12	
IC3	1A7	
Q1	1A36	Output states for the clock shaping network. (See Figure 2)
Q2	1A40	
INJ	1A35	Injector Supply Current.
VCC	1A1	Vcc. 5.0 ± 0.5 Vdc
VCC	2A1	Vcc. 5.0 ± 0.5 Vdc
GND	1A10	Ground.
GND	1A39	Ground.
GND	2A10	Ground.
GND	2A39	Ground.

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6 6 Parallel outputs Paralleling of outputs is not permissible

6 7 Functional test vectors Functional test vectors are not printed herein because of their extreme length, and, even when presented as a hard copy, they require conversion to on-line format When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred) For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request

Preparing Activity:
Navy - EC

(Project 5963-N701)

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TABLE IV Isolation test

TEST COND	TEST DESCRIPTION	SET-UP CONDITIONS		25 + 5 deg C INITIAL LIMITS		UNITS	NOTES
		CONNECT A TO	CONNECT B TO	MIN	MAX		
A	ISOLATION FRAME TO I/O	ALL PINS	FRAME	10	INF	Mohm	1,2, 3,4, 5

Notes

1 Isolation test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
MEGOHM BRIDGE VOLTAGE	100 V	± 10%	10 uA	A
ISOLATION MEASUREMENT	NA	± 0.1 MOhm	NA	A

2 The megohm bridge used shall be item R06

3 In the set-up conditions of table IV, "A" shall correspond to the positive terminal of the megohm bridge, and "B" shall correspond to the negative terminal

4 Warning When using a megohm bridge, high voltage may be present at any of the output binding posts Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads

5 Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge

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TABLE V Continuity test

TEST COND	TEST DESCRIPTION	SET-UP CONDITIONS		25 + 5 deg C INITIAL LIMITS		UNITS	NOTES
		CONNECT A TO	CONNECT B TO	MIN	MAX		
B1	CONTINUITY, GND	1A10	2A10	0.0	1.0	Ohm	1,2,3
B2	CONTINUITY, GND	1A10	1A39				
B3	CONTINUITY, GND	1A10	2A39				
B4	CONTINUITY, VCC	1A1	2A1	0.0	1.0	Ohm	1,2,3

Notes

1 Continuity test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
CONTINUITY MEASUREMENT	NA	± 0.11 Ohm	NA	B1-B4

2 The ohmmeter used in shall be item R10

3 In the set-up conditions of table V, "A" shall correspond to the positive terminal of the ohmmeter, and "B" shall correspond to the negative terminal

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TABLE VI Power supply stress test

TEST COND	TEST DESCRIPTION	25 ± 5 deg C INITIAL LIMITS	NOTES
C	POWER SUPPLY STRESS TEST	Module shall pass the remaining tests.	1-4

Notes

1 Power supply stress test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY,(Vcc)	7.0 V	+0 mV,-70 mV	245 mA	C
Injector Current, (INJ)	750 mA	± 7.5mA	NA	C
POWER SUPPLY	3.6 V	± 36 mV	245 mA	C

2 There shall be no power supply overshoot on Vcc for this test

3 Precondition the module using the test tolerance table above by applying

a) 7 0 V to pins 1A1 and 2A1 (Vcc)

b) 3 6 V to pins 2A13 (PRESET), 1A22, 1A2, 1A21, 1A24, 1A23, 1A5, 1A4, 1A27, 1A6, 1A26, 1A25, 1A8, 1A9, 1A11, 1A28 (P1-P15)

c) 0 0 V to pins 1A10, 1A39, 2A10, 2A39 (GND), 2A12 (HOLD-L), 1A3 (EI-L)

d) 750 mA to pin 1A35 (INJ)

4 When performing the power supply stress test, no measurements are taken This condition shall be maintained for a minimum of 5 seconds and a maximum of 10 seconds

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TABLE VII Input breakdown current test.

TEST COND	SIGNAL NAME	PIN NO.	Vin (V)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
D1	P1	1A22	5.5	0.0	1.0	mA	1,2
D2	P2	1A2					
D3	P3	1A21					
D4	P4	1A24					
D5	P5	1A23					
D6	P6	1A5					
D7	P7	1A4					
D8	P8	1A27					
D9	P9	1A6					
D10	P10	1A26					
D11	P11	1A25					
D12	P12	1A8					
D13	P13	1A9					
D14	P14	1A11					
D15	P15	1A28					
D16	EI-L	1A3	5.5	0.0	1.0	mA	1,2
D17	CLK	2A22	5.5	0.0	0.1	mA	1,2
D18	PRESET	2A13	5.5	0.0	0.2	mA	1,2

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Notes

1 Input breakdown current test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY, (Vcc)	5.5 V	+55mV, -0mV	245 mA	D1-D18
FORCING VOLTAGE, (Vin)	5.5 V	+0mV, -55mV	5.0 mA	D1-D18
PATTERN VOLTAGE HIGH, (VH) LOW, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 NA NA	 D1-D18 D1-D18
INPUT CURRENT MEASUREMENT	NA NA NA	± 11 uA ± 1.1 uA ± 2.2 uA	NA NA NA	D1-D16 D17 D18

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ for preconditioning the module under test and make the indicated measurement at the pattern line shown in _____

c When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

d The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table VII and the current then measured The forcing voltage overshoot shall not exceed 5 5 V from ground reference level at any time

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TABLE VIII. Output short circuit current test

TEST COND	SIGNAL NAME	PIN No.	Vout (V)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
E1	DBIN	1A32	0.0	-9.1	-12.4	mA	1,2,3
E2	MEMN-L	2A17					
E3	WE-L	2A9					
E4	HOLDA	2A26					
E5	WAIT	2A28					
E6	IAQ	2A24					
E7	CYEN-L	2A7					
E8	CRUOUT	1A31					
E9	CRUCLK	2A8					
E10	A0	1A37					
E11	A1	1A38					
E12	A2	2A38					
E13	A3	2A18					
E14	A4	2A19					
E15	A5	2A40					
E16	A6	2A16					
E17	A7	2A37					
E18	A8	2A36					
E19	A9	2A15					
E20	A10	2A35					
E21	A11	2A34					
E22	A12	2A33					
E23	A13	2A14					
E24	A14	2A21					
E25	D0	1A13	0.0	-9.1	-12.4	mA	1,2,3

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TABLE VIII. Output short circuit current test

TEST COND	SIGNAL NAME	PIN No.	Vout (V)	25 ± 5 deg C INITIAL LIMITS MIN	MAX	UNITS	NOTES
E26	D1	1A20	0.0	-9.1	-12.4	mA	1,2,3
E27	D2	2A20					
E28	D3	2A32					
E29	D4	2A31					
E30	D5	1A14					
E31	D6	1A15					
E32	D7	1A16					
E33	D8	1A17					
E34	D9	1A18					
E35	D10	1A19					
E36	D11	2A2					
E37	D12	2A3					
E38	D13	2A4					
E39	D14	2A5					
E40	D15	2A6	0.0	-9.1	-12.4	mA	1,2,3
E41	INTR-L	1A29	0.0	-20	-55	mA	1,2,3
E42	IC0	1A34	0.0	-20	-100	mA	1,2,3
E43	IC1	1A33	0.0	-20	-55	mA	1,2,3
E44	IC2	1A12					
E45	IC3	1A7	0.0	-20	-55	mA	1,2,3
E46	Q1	1A36	0.0	-20	-100	mA	1,2,3
E47	Q2	1A40	0.0	-20	-100	mA	1,2,3

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Notes

1 Output short circuit current test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY, (Vcc)	5.5 V	± 55 mV	245 mA	E1-E47
FORCING VOLTAGE, (Vout)	0.0 V	+10 mV,-0 mV	NA	E1-E47
PATTERN VOLTAGE HIGH, (VH) LOW, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 NA NA	 E1-E47 E1-E47
OUTPUT CURRENT MEASUREMENT	NA NA NA	± 36.3 uA ± 385 uA ± 800 uA	NA NA NA	E1-E40 E41,E43-E45 E42,E46-E47

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ for preconditioning the module under test and make the indicated measurement at the pattern line shown in _____

c When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

d The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table VIII The forcing voltage undershoot shall not exceed 0 0 V from ground reference level at any time

3 When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 5 seconds

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TABLE IX Input clamp diode voltage test

TEST COND	SIGNAL NAME	PIN No.	Iin (mA)	25 + 5 deg C INITIAL LIMITS MIN	25 + 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
F1	P1	1A22	-12.0	-0.0	-1.5	V	1,2
F2	P2	1A2					
F3	P3	1A21					
F4	P4	1A24					
F5	P5	1A23					
F6	P6	1A5					
F7	P7	1A4					
F8	P8	1A27					
F9	P9	1A6					
F10	P10	1A26					
F11	P11	1A25					
F12	P12	1A8					
F13	P13	1A9					
F14	P14	1A11					
F15	P15	1A28					
F16	EI-L	1A3	-12.0				
F17	CLK	1A22	-18.0				
F18	PRESET	2A13	-18.0				
F19	READY	2A11	-12.0				
F20	HOLD-L	2A12					
F21	LOAD-L	2A27					
F22	RST-L	2A25					
F23	CLOCK	2A23					
F24	CRUIN	1A30	-12.0	0.0	-1.5	V	1,2

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TABLE IX Input clamp diode voltage test

TEST COND	SIGNAL NAME	PIN No.	Iin (mA)	25 + 5 deg C INITIAL LIMITS MIN	25 + 5 deg C INITIAL LIMITS MAX	UNITS	NOTE
F25	D0	1A13	-12.0	-0.0	-1.5	V	1,2
F26	D1	1A20					
F27	D2	2A20					
F28	D3	2A32					
F29	D4	2A31					
F30	D5	1A14					
F31	D6	1A15					
F32	D7	1A16					
F33	D8	1A17					
F34	D9	1A18					
F35	D10	1A19					
F36	D11	2A2					
F37	D12	2A3					
F38	D13	2A4					
F39	D14	2A5					
F40	D15	2A6	-12.0	0.0	-1.5	V	1,2

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Notes

1 Input clamp diode voltage test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY, (Vcc)	4.5 V	± 45 mV	245 mA	F1-F40
INJECTOR CURRENT, (INJ)	400 mA	± 4 mA	NA	F1-F40
FORCING CURRENT Iin Iin	 -12.0mA -18.0mA	 ± 1.2 mA ± 1.8 mA	 NA NA	 F1-F16, F19-F40 F17, F18
INPUT VOLTAGE MEASUREMENT	NA	± 16.5 mV	NA	F1-F40

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins, and the injector current forced into pin 1A35 The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ for preconditioning the module under test and make the indicated measurement at the pattern line shown in _____

c When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

d The forcing current shall be applied to the pin specified in the appropriate test conditions of table IX, and the voltage then measured

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TABLE X Power supply current test

TEST COND	PARAMETER	PIN NO.	25 + 5 deg C INITIAL LIMITS MIN	MAX	UNITS	NOTES
G1	INJECTOR SUPPLY VOLTAGE	1A35	0.88	1.63	V	1,2
G2	INJECTOR SUPPLY VOLTAGE	1A35	0.91	1.68	V	1,2
G3	VCC POWER SUPPLY CURRENT	1A1, 2A1	20	195	mA	1,2

Notes

1 Power supply current test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT/ VOLTAGE LIMIT	TEST CONDITION
POWER SUPPLY, (Vcc)	5.5 V	± 55 mV	245 mA	G1-G3
INJECTOR CURRENT, (INJ)	480 mA	± 4.8 mA	NA	G1
(INJ)	520 mA	± 5.2 mA	NA	G2
(INJ)	500 mA	± 5.0 mA	NA	G3
INJECTOR SUPPLY VOLTAGE MEASUREMENT	NA	8.25 mV	NA	G1
INJECTOR SUPPLY VOLTAGE MEASUREMENT	NA	8.47 mV	NA	G2
VCC POWER SUPPLY CURRENT MEASUREMENT	NA	1.925 mA	NA	G3

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2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply oltage applied to all Vcc pins and injector current applied to pin 1A35 (INJ) he DC return of the power supply shall be connected to all ground pins of the odule The power supply voltage overshoot shall not exceed 6 0 V from ground eference level at any time

b Apply 0 0 V to pins 2A12 (HOLD-L), 1A3 (EI-L) and 2A30 (LGH)

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TABLE XI. Input current, high level, test

TEST COND	SIGNAL NAME	PIN No.	Vih (V)	25 + 5 deg C INITIAL LIMITS MIN	MAX	UNITS	NOTES
H1	P1	1A22	2.4	0.0	80	uA	1,2
H2	P2	1A2					
H3	P3	1A21					
H4	P4	1A24					
H5	P5	1A23					
H6	P6	1A5	2.4	0.0	80	uA	1,2
H7	P7	1A4	2.4	0.0	40	uA	1,2
H8	P8	1A27	2.4	0.0	80	uA	1,2
H9	P9	1A6					
H10	P10	1A26					
H11	P11	1A25					
H12	P12	1A8					
H13	P13	1A9					
H14	P14	1A11	2.4	0.0	80	uA	1,2
H15	P15	1A28	2.4	0.0	40	uA	1,2
H16	EI-L	1A3	2.4	0.0	80	uA	1,2
H17	CLK	1A22	2.4	0.0	40	uA	1,2
H18	PRESET	2A13	2.4	0.0	80	uA	1,2
H19	LGH	2A30	2.4	-2.6	-3.4	mA	1,2
H20	READY	2A11	2.4	0.0	300	uA	1,2
H21	HOLD-L	2A12					
H22	LOAD-L	2A27					
H23	RST-L	2A25	2.4	0.0	300	uA	1,2
H24	CLOCK	2A23	2.4	0.0	1000	uA	1,2
H25	CRUIN	1A30	2.4	0.0	300	uA	1,2

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Notes

1 Input current, high level, test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY, (Vcc)	5.5 V	± 55 mV	245 mA	H1-H41
INJECTOR CURRENT, (INJ)	520 mA	± 5.2 mA	NA	H1-H41
FORCING VOLTAGE, (Vih)	2.4 V	+0 mV,-24 mV	NA	H1-H41
INPUT CURRENT MEASUREMENT	NA	± 880 nA	NA	H1-H6,H8-H14, H16,H18
	NA	± 440 nA	NA	H7,H15,H17
	NA	± 8.8 uA	NA	H19
	NA	± 3.3 uA	NA	H20-H23,H25

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

c The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table XI, all other inputs shall be tied to 0 0 V The forcing voltage overshoot shall not exceed 5 5 V from ground reference level at any time

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TABLE XII Input current, low level, test

TEST COND	SIGNAL NAME	PIN No.	Vil (V)	25 + 5 deg C INITIAL LIMITS MIN	MAX	UNITS	NOTES
J1	P1	1A22	0.4	0.0	-3.5	mA	1,2
J2	P2	1A2					
J3	P3	1A21					
J4	P4	1A24					
J5	P5	1A23					
J6	P6	1A5	0.4		-3.5	mA	1,2
J7	P7	1A4	0.4		-1.75	mA	1,2
J8	P8	1A27	0.4		-3.5	mA	1,2
J9	P9	1A6					
J10	P10	1A26					
J11	P11	1A25					
J12	P12	1A8					
J13	P13	1A9					
J14	P14	1A11	0.4		-3.5	mA	1,2
J15	P15	1A28	0.4		-1.75	mA	1,2
J16	EI-L	1A3	0.4		-3.5	mA	1,2
J17	CLK	1A22	0.4		-0.88	mA	1,2
J18	PRESET	2A13	0.4	0.0	-1.75	mA	1,2
J19	LGH	2A30	0.4	-5.0	-10.8	mA	1,2
J20	READY	2A11	0.4	0.0	-10	mA	1,2
J21	HOLD-L	2A12					
J22	LOAD-L	2A27					
J23	RST-L	2A25	0.4	0.0	-10	mA	1,2
J24	CLOCK	2A23	0.4	0.0	-20	mA	1,2
J25	CRUIN	1A30	0.4	0.0	-10	mA	1,2

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Notes

1 Input current, low level, test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY	5.5 V	± 55 mV	245 mA	J1-J41
INJECTOR CURRENT	520 mA	± 5.2 mA	NA	J1-J41
FORCING VOLTAGE, (Vil)	0.4 V	± 4 mV	NA	J1-J41
INPUT CURRENT MEASUREMENT	NA	38.5 uA	NA	J1-J6,J8-J14,J16
	NA	19.25 uA	NA	J7,J15,J18
	NA	9.68 uA	NA	J17
	NA	63.8 uA	NA	J19
	NA	110 uA	NA	J20-J23,J25
	NA	220 uA	NA	J24

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

c The forcing voltage shall be applied to the pin specified in the appropriate test conditions of table XII All other inputs shall be tied to 3 0 V The forcing voltage overshoot shall not exceed 5 5 V from ground reference level at any time

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TABLE XIII. Output voltage, high level, test.

TEST COND	SIGNAL NAME	PIN No.	Ioh (mA)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
K1	DBIN	1A32	-0.25	5.22	5.55	V	1,2
K2	MEMN-L	2A17					
K3	WE-L	2A9					
K4	HOLDA	2A26					
K5	WAIT	2A28					
K6	IAQ	2A24					
K7	CYEN-L	2A7					
K8	CRUOUT	1A31					
K9	CRUCLK	2A8					
K10	A0	1A37					
K11	A1	1A38					
K12	A2	2A38					
K13	A3	2A18					
K14	A4	2A19					
K15	A5	2A40					
K16	A6	2A16					
K17	A7	2A37					
K18	A8	2A36					
K19	A9	2A15					
K20	A10	2A35					
K21	A11	2A34					
K22	A12	2A33					
K23	A13	2A14					
K24	A14	2A21	-0.25	5.22	5.55	V	1,2
K25	D0	1A13	-1.0	5.05	5.55	V	1,2

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TABLE XIII Output voltage, high level, test

TEST COND	SIGNAL NAME	PIN No.	Ioh (mA)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
K26	D1	1A20	-1.0	5.05	5.55	V	1,2
K27	D2	2A20					
K28	D3	2A32					
K29	D4	2A31					
K30	D5	1A14					
K31	D6	1A15					
K32	D7	1A16					
K33	D8	1A17					
K34	D9	1A18					
K35	D10	1A19					
K36	D11	2A2					
K37	D12	2A3					
K38	D13	2A4					
K39	D14	2A5					
K40	D15	2A6	-1.0	5.05	5.55	V	1,2
K41	INTR-L	1A29	-0.56	2.4	4.5	V	1,2
K42	IC0	1A34	-0.13				
K43	IC1	1A33	-0.56				
K44	IC2	1A12	-0.56				
K45	IC3	1A7	-0.56				
K46	Q1	1A36	-0.38				
K47	Q2	1A40	-0.38	2.4	4.5	V	1,2

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Notes

1 Output voltage, high level, test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY	4.5 V	± 45 mV	245 mA	K1-K47
INJECTOR CURRENT	320 mA	± 5.2 mA	NA	K1-K47
FORCING CURRENT, Ioh	-0.25mA	± 2.5 uA	NA	K1-K24
Ioh	-1.0 mA	± 10 uA	NA	K25-K40
Ioh	-0.56mA	± 5.6 uA	NA	K41,K43-K45
Ioh	-0.13mA	± 1.3 uA	NA	K42
Ioh	-0.38mA	± 3.8 uA	NA	K46,K47
PATTERN VOLTAGE				
HIGH, (VH)	2.0 V	+20,-0 mV	NA	K1-K47
LOW, (VL)	0.8 V	+0,-8 mV	NA	K1-K47
OUTPUT VOLTAGE	NA	± 3.63 mV	NA	K1-K24
	NA	± 5.5 mV	NA	K25-K40
	NA	± 23.1 mV	NA	K41-K47
MEASUREMENT				

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins, and the injector current applied to pin 1A35 A The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ for preconditioning the module under test and make the indicated measurement at the pattern line shown in _____

c Apply 3 6 V ± 1% to pins 2A27 (LOAD-L) and 1A3 (EI-L) Apply 0 0 V to pin 1A30 (CRUIN)

d When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

e The forcing current shall be applied to the pin specified in the appropriate test conditions of table XII, and the voltage then measured

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TABLE XIV Output voltage, low level, test

TEST COND	SIGNAL NAME	PIN No.	Ioh (mA)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
L1	DBIN	1A32	10.7	0.0	0.4	V	1,2
L2	MEMN-L	2A17					
L3	WE-L	2A9					
L4	HOLDA	2A26					
L5	WAIT	2A28					
L6	IAQ	2A24					
L7	CYEN-L	2A7					
L8	CRUOUT	1A31					
L9	CRUCLK	2A8					
L10	A0	1A37					
L11	A1	1A38					
L12	A2	2A38					
L13	A3	2A18					
L14	A4	2A19					
L15	A5	2A40					
L16	A6	2A16					
L17	A7	2A37					
L18	A8	2A36					
L19	A9	2A15					
L20	A10	2A35					
L21	A11	2A34					
L22	A12	2A33					
L23	A13	2A14					
L24	A14	2A21					
L25	D0	1A13	10.7	0.0	0.4	V	1,2

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TABLE XIV Output voltage, low level, test

TEST COND	SIGNAL NAME	PIN No.	Ioh (mA)	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
L26	D1	1A20	10.7	0.0	0.4	V	1,2
L27	D2	2A20					
L28	D3	2A32					
L29	D4	2A31					
L30	D5	1A14					
L31	D6	1A15					
L32	D7	1A16					
L33	D8	1A17					
L34	D9	1A18					
L35	D10	1A19					
L36	D11	2A2					
L37	D12	2A3					
L38	D13	2A4					
L39	D14	2A5					
L40	D15	2A6	10.7				
L41	INTR-L	1A29	16.0				
L42	IC0	1A34	4.0				
L43	IC1	1A33	16.0				
L44	IC2	1A12	16.0				
L45	IC3	1A7	16.0				
L46	Q1	1A36	3.2				
L47	Q2	1A40	3.6	0.0	0.4	V	1,2

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Notes

1 Output voltage, low level, test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
POWER SUPPLY	4.5 V	± 45 mV	245 mA	L1-L47
INJECTOR CURRENT	480 mA	± 4.8 mA	NA	L1-L47
FORCING CURRENT, Ioh	10.7 mA	107 uA	NA	L1-L40
Ioh	16.0 mA	160 uA	NA	L41,L43-L45
Ioh	4.0 mA	40 uA	NA	L42
Ioh	3.2 mA	32 uA	NA	L46
Ioh	3.6 mA	36 uA	NA	L47
PATTERN VOLTAGE				
HIGH, (VH)	2.0 V	+20,-0 mV	NA	L1-L47
LOW, (VL)	0.8 V	+0,-8 mV	NA	L1-L47
OUTPUT VOLTAGE MEASUREMENT	NA	± 4.4 mV	NA	L1-L47

2 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins, and the injector current applied to pin 1A35, the DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ for preconditioning the module under test and make the indicated measurement at the pattern line shown in _____

c When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test

d The forcing current shall be applied to the pin specified in the appropriate test conditions of table XIV, and the voltage then measured

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TABLE XV Functional test

TEST COND	TEST DESCRIPTION	NOTES
M1	FUNCTIONAL TEST, Priority Interrupt Minimum Power Supply Voltage (4.5 V) Maximum Clock Frequency	1,2, 3
M2	FUNCTIONAL TEST, Clock Shaping Maximum Power Supply Voltage (5.5 V) Maximum Clock Frequency	1,2, 3

Notes

1 The following procedure is required to perform this test

a Apply power to the module The module shall have the power supply voltage applied to all Vcc pins The DC return of the power supply shall be connected to all ground pins of the module The power supply voltage overshoot shall not exceed 6 0 V from ground reference level at any time

b Use the pattern indicated in _____ to verify the functional performance of the module using a capacitive load of 50 pF and 100 pF (see Fig 23)

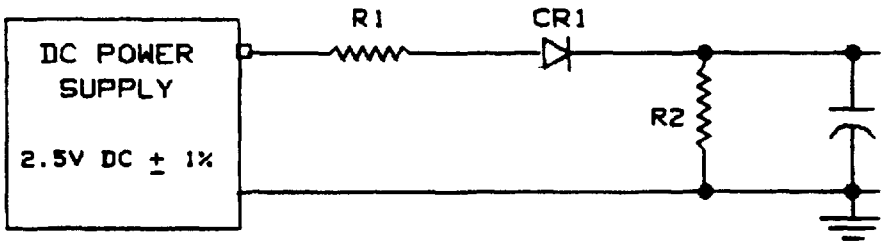
2 The following input voltages shall be used for the static functional test.

VIH = 2 0V±1%
VIL = 0 8V±1% for M1
VIL = 0 7V±1% for M2

3 The following output voltages shal be used for the static functional test

H ≥ 2.4V
L ≤ 0 4V

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NOTES
1 CR1 = 1N4150 or equivalent
2 CL = 50 pF ±5% (for pins 1A7, 1A12, 1A29, 1A33, 1A34, 1A36, 1
CL = 100 pF ±5% (for all other pins)
3 R2 = Select value for each output load as specified in table
4 R1 = Select value for a current flow of IOL when point IOL is found in the following table

Outputs 3/	R2 1/	IOL 2/
Q1	6.34Kohms	320 mA
Q2	6.34Kohms	3.60 mA
IC0	17.8Kohms	4.0 mA
INTR-L, IC1, IC2, IC3	4.32Kohms	16.0 mA
CPU outputs	---	10.7 mA

1/ All resistors are metal film, ±1%, 1/4W
2/ IOL tolerance, ±1%
3/ CPU outputs are A0-A14, MEMN-L, WE-L, HOLDA, WAIT, IAQ, CRUOUT, CRUCLK, D0-D15

FIGURE 23 Logic load circuit

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TABLE XVI Dynamic test (min and max injector current)

TEST COND.	PARAMETER	INPUT TABLE	NOTES	TEST FIGURE
N1	Dynamic functional test, minimum injector current	XVII	1,2	24
N2	Dynamic functional test, maximum injector current	XVII	1,2	24
N3	Dynamic functional test, minimum injector current	XVIII	1,2	24
N4	Dynamic functional test, maximum injector current	XVIII	1,2	24

Notes

1 Apply the vectors (see Appx A) for each test condition using the input setup and hold times, power supply voltage, and injector current as specified in Tables XVII through XIX and figure 25 Verify the output levels are VOL $\leq$0 8V and VOH $\geq$2 0V at a point in time not less than the maximum propagation delay time shown in table XIX for test conditions N1 and N2

2 Verify the output levels are VOL $\leq$0 9V and VOH $\geq$2 0V at a point in time not less than the maximum propagation delay time shown in table XIX for test conditions N3 and N4

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Table XVII CPU input conditions for table XVI, tests N1,N2

PARAMETER	PIN(s)	INPUT CONDITIONS	UNIT	FIG
INJ	1A35	1/,2/	mA	24,25
Vcc	1A1,2A1	4.5±1%	V	
PRP	2A23	1000±1%	nS	
PWA	2A23	333±1%		
tTLH,tTHL	All CPU inputs	15±5		
tsu1 HOLD-L	2A12	332±10		
tsu2 READY	2A11	292±10		
tsu3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	287±10		
tsu4 CRUIN	1A30	272±10		
tsu5 INTR-L	1A29	247±10		
tsu6 IC0-IC3	1A7,1A12,1A33,1A34	247±10		
tsu7 RST-L	2A25	332±10		
tsu8 LOAD-L	2A27	292±10		
th1 HOLD-L	2A12	160±10		
th2 READY	2A11	170±10		
th3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	145±10		
th4 CRUIN	1A30	180±10		
th5 INTR-L	1A29	160±10		
th6 IC0-IC3	1A7,1A12,1A33,1A34	160±10		
th7 RST-L	2A25	160±10		
th8 LOAD-L	2A27	180±10	nS	
VIH	All	2.0+0.1,-0.0	V	
VIL	All	0.6+0.0,-0.1	V	24,25

1/ INJ = 480mA±1% for test conditions N1,T1

2/ INJ = 520mA±1% for test conditions N2,T2

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Table XVIII CPU input conditions for table XVI, tests N3,N4

PARAMETER	PIN(s)	INPUT CONDITIONS	UNIT	FIG
INJ	1A35	1/,2/	mA	24,25
Vcc	1A1,2A1	4.5±1%0,-0.1	V	
PRP	2A23	333±1%	nS	
PWA	2A23	111±1%		
tTLH,tTHL	All CPU inputs	15±5		
tsu1 HOLD-L	2A12	5+2,-0		
tsu2 READY	2A11	35+2,-0		
tsu3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	5+2,-0		
tsu4 CRUIN	1A30	20+2,-0		
tsu5 INTR-L	1A29	5+2,-0		
tsu6 IC0-IC3	1A7,1A12,1A33,1A34	10+2,-0		
tsu7 RST-L	2A25	0+2,-0		
tsu8 LOAD-L	2A27	35+2,-0		
th1 HOLD-L	2A12	25+2,-0		
th2 READY	2A11	30+2,-0		
th3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	35+2,-0		
th4 CRUIN	1A30	25+2,-0		
th5 INTR-L	1A29	25+2,-0		
th6 IC0-IC3	1A7,1A12,1A33,1A34	25+2,-0		
th7 RST-L	2A25	30+2,-0		
th8 LOAD-L	2A27	25+2,-0	nS	
VIH	All	3.0+0.1,-0.0	V	
VIL	All	0+0.0,-0.1	V	24,25

1/ INJ = 480mA±1% for test condition N3
2/ INJ = 520mA±1% for test condition N4

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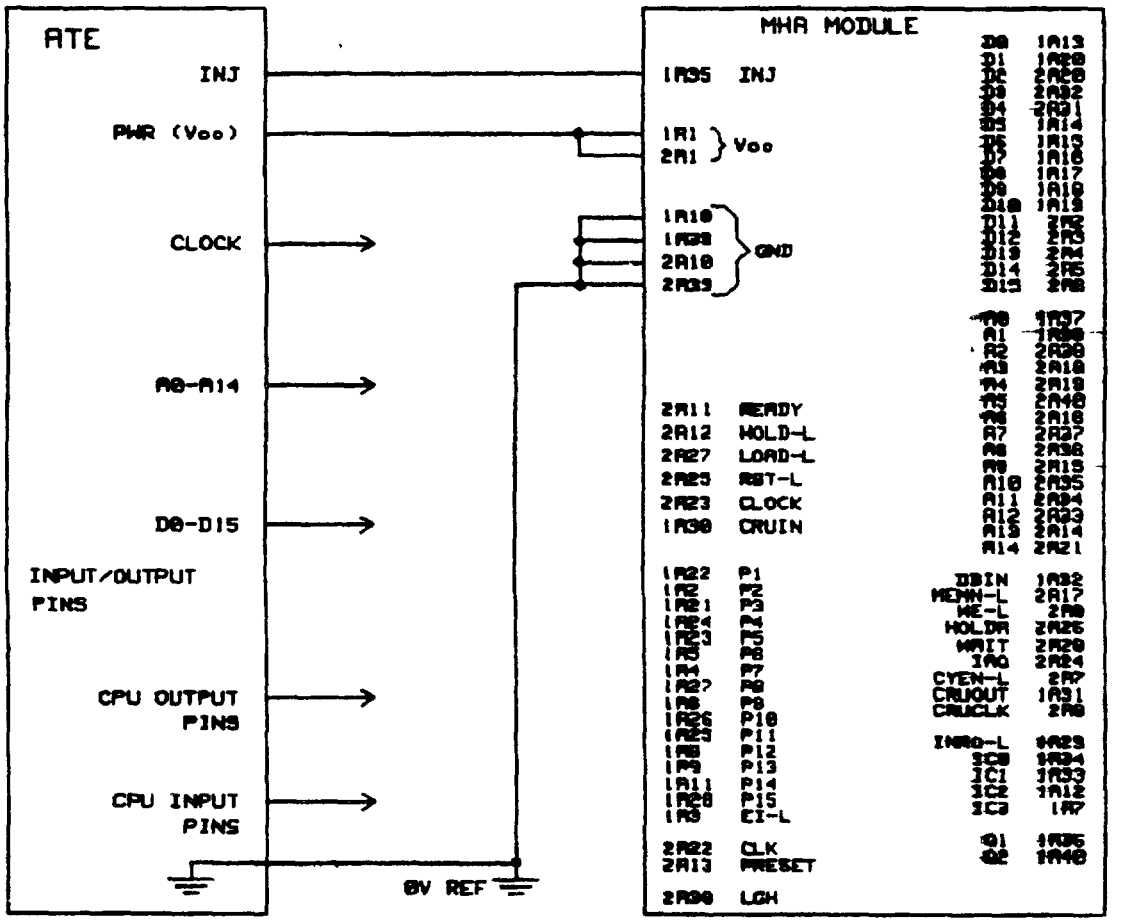
Table XIX CPU output conditions for Table XVI, tests N1-N4 (CL=100pF)

PARAMETER	PIN(s)	OUTPUT CONDITIONS MIN	OUTPUT CONDITIONS MAX	UNIT	FIG
tpd1 D0-D15	1A13-1A20,2A2-2A6, 2A20,2A31,2A32	-	140	nS	24,25
tpd2 A0-A14	1A37,1A38,2A14-2A19,2A21,2A33-2A38-2A40	-	140		
tpd3 IAQ	2A24	-	140		
tpd4 WE-L	2A9	-	180		
tpd5 CRUOUT	1A31	-	175		
tpd6 DBIN	1A32	-	155		
tpd7 MEMN-L	2A17	-	140		
tpd8 WAIT	2A28	-	130		
tpd9 CRUCLK	2A8	-	185		
tpd10 CYEN-L	2A7	-	175		
tpd11 HOLDA	2A26	-	150		
When leaving a hold state:					
tpd1 D0-D15	1A13-1A20,2A2-2A6, 2A20,2A31,2A32	-	200		
tpd2 A0-A14	1A37,1A38,2A14-2A19 2A21,2A33-2A38,2A40	-	200		
tpd3 IAQ	2A24	-	200		
tpd6 DBIN	1A32	-	200		
tpd7 MEMN-L	2A17	-	200	nS	
Output switch voltage	All outputs	1.5	1.5	V	24,25

1/ For test conditions N1 and N2, the outputs shall be checked with the compare window set at 900 nS and continuing for 10 nS

2/ For test conditions N3 and N4, check the propagation delay times by starting output comparison at the maximum propagation delay time limits and continuing for 10 nS

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NOTES

1 Load each output as specified on figure 23

2 ATE must be capable of sinking 16 mA when connected to any data bus pin (D0-D15)

3 INTR-L, IC0-IC3 inputs are connected to the priority interrupt outputs and therefore item U04 must be capable of sinking 80 mA when attached to either of these inputs and forcing a logic low level of 0 4V The inputs shall never be held low in a static condition for more than 0 5 second

FIGURE 24 Dynamic functional (minimum and maximum injector current) test circuit for tables XVI, XVII, XVIII and XIX

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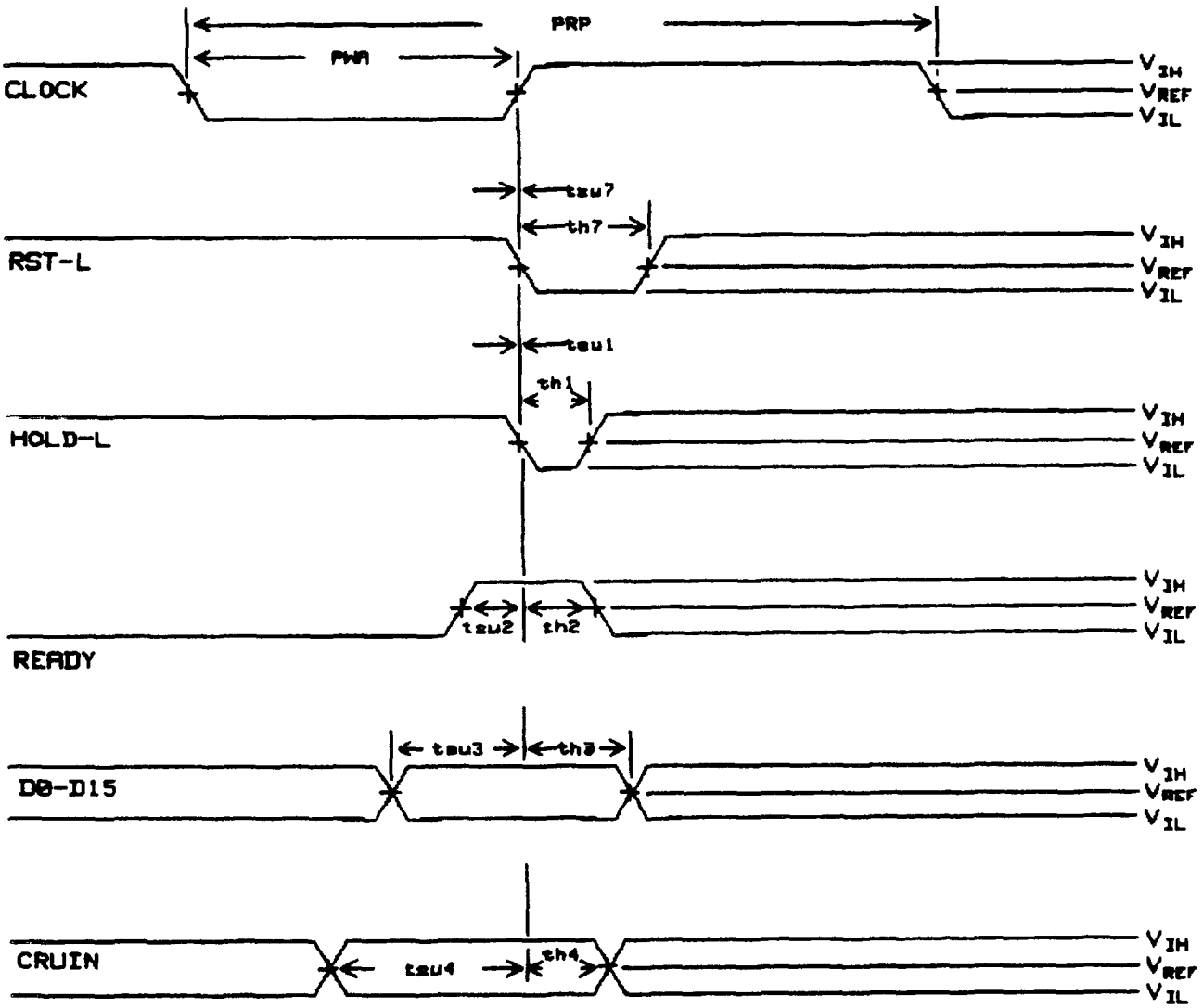


FIGURE 25 Dynamic functional waveforms for tables XVI, XVII, XVIII and XIX

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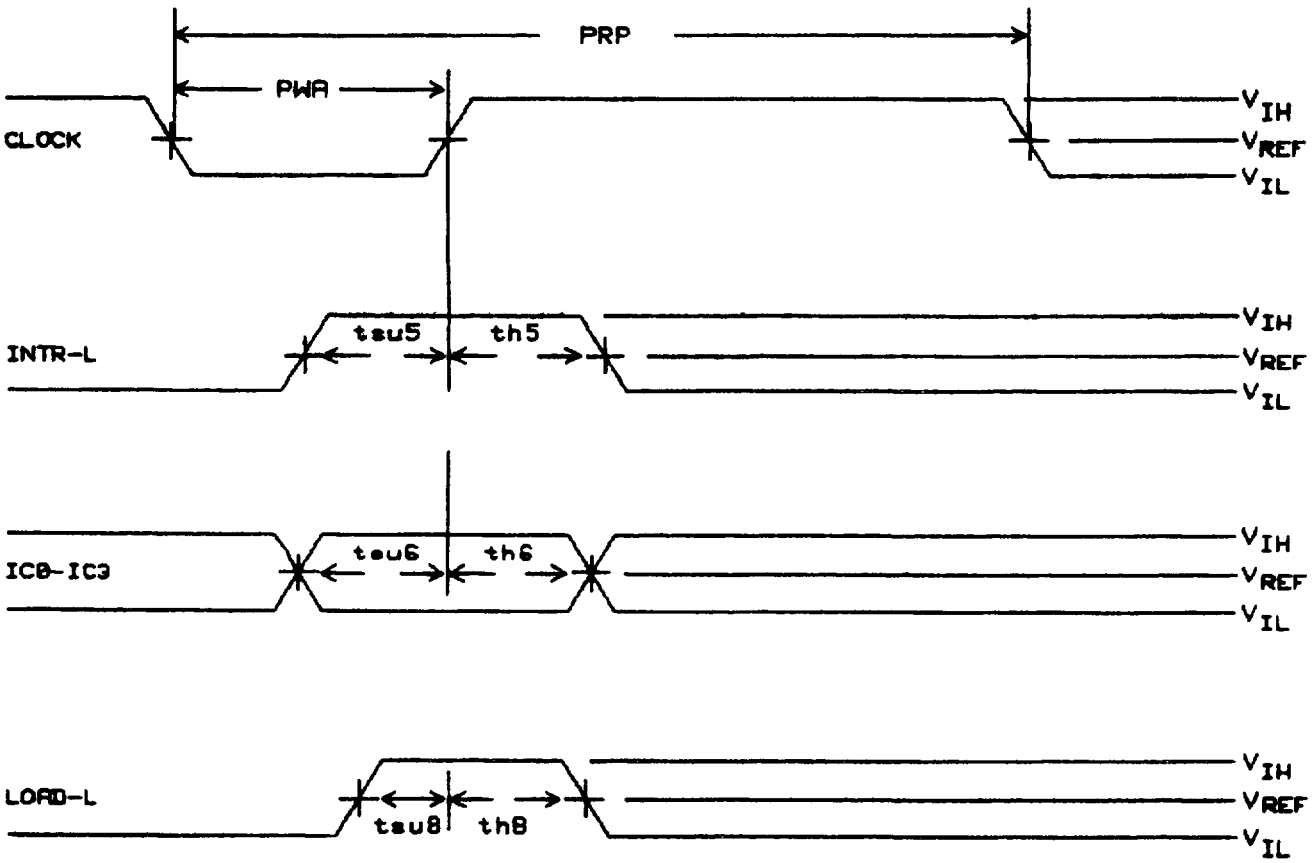


FIGURE 25 Dynamic functional waveforms for tables XVI, XVII, XVIII and XIX (con't)

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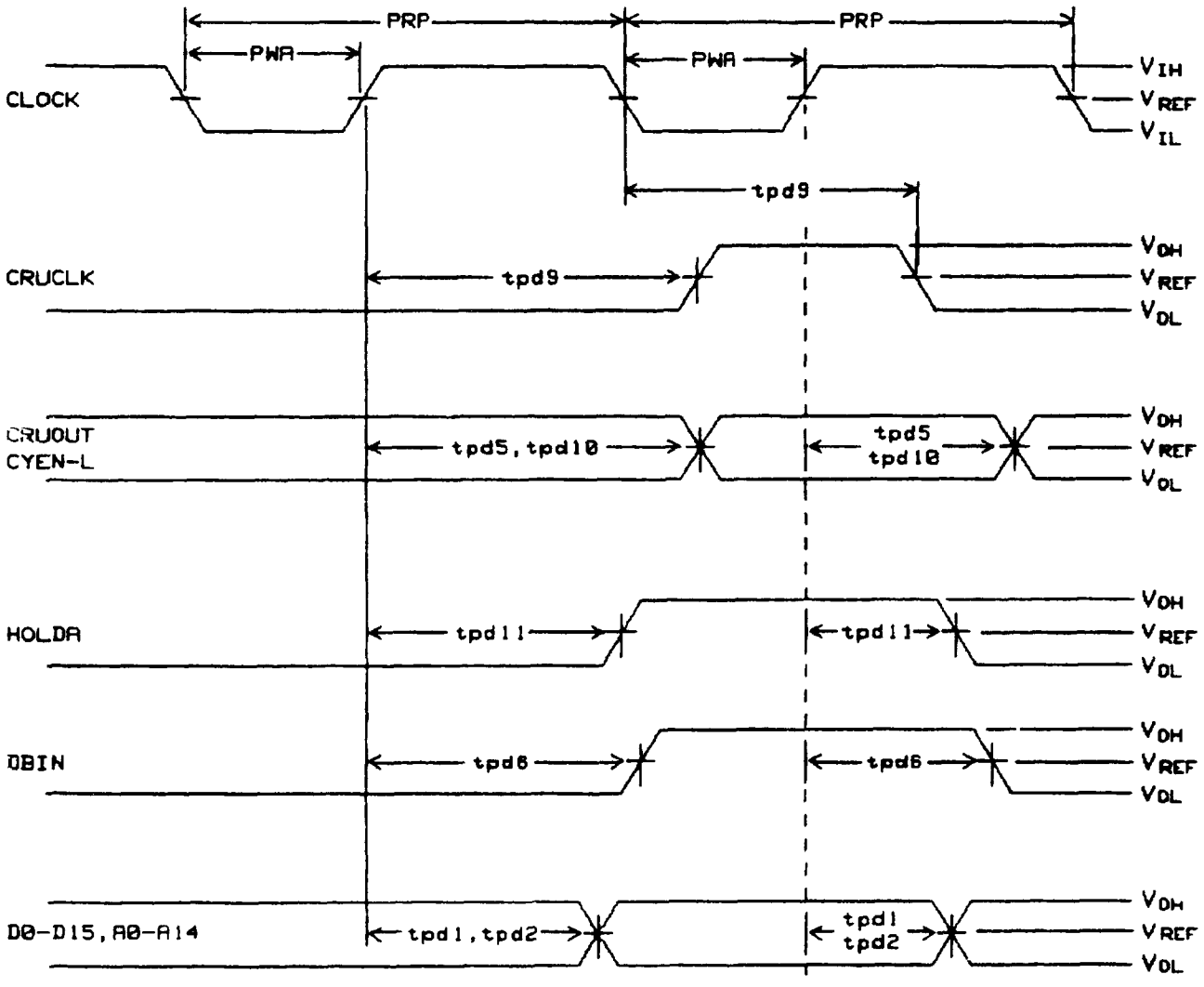


FIGURE 25 Dynamic functional waveforms for tables XVI, XVII, XVIII and XIX (con't)

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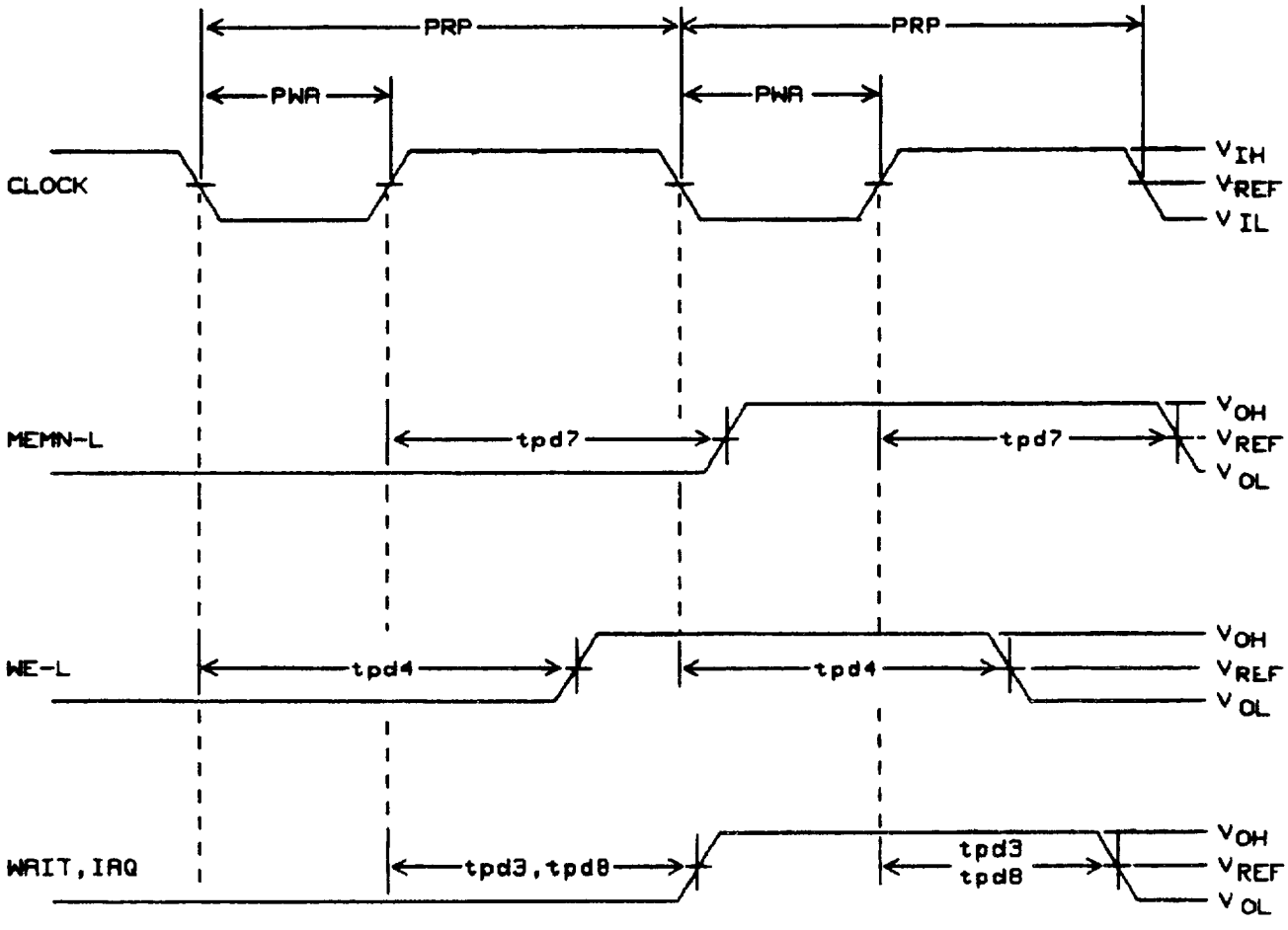


FIGURE25 Dynamic functional waveforms for tables XVI, XVII, XVIII and XIX (con't)

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NOTES

1 For input pulse and output pulse parameters, refer to tables XVII, XVIII and XIX

2 Input transition times and output transition times are measured from the 10% level to the 90% level

3 In addition to the switching characteristics shown, the following signal relationships exist

a During write operations
(1) ADDR lines are valid prior to WE-L active (low)
(2) WE-L goes inactive (high) prior to ADDR lines going valid
(3) Data lines are valid prior to WE-L active (low)
(4) WE-L goes inactive (high) prior to data lines going invalid
(5) WE-L does not go active (low) before MEMN-L goes active (low)
(6) WE-L goes inactive (high) before MEMN-L goes inactive (high)

b During read operations
(1) DBIN does not go inactive (low) before the maximum data hold time, th3, runs out
(2) MEMN-L does not go inactive (high) before the maximum data hold time, th3, runs out

c During CRU output operations
(1) CRUCLK goes inactive (low) prior to ADDR lines going invalid
(2) CRUCLK goes inactive (low) prior to CRUOUT going invalid

FIGURE 25 Dynamic functional waveforms for tables XVI,XVII,XVIII and XIX (con't)

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Table XX Transition time (tTHL) test (CL= 100pF)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg + 5 degC AMBIENT MAX	UNIT	FIGS
P1	DBIN	1A32	5	40	nS	26,27
P2	MEMN-L	2A17				
P3	WE-L	2A9				
P4	HOLDA	2A26				
P5	WAIT	2A28				
P6	IAQ	2A24				
P7	CYEN-L	2A7				
P8	CRUOUT	1A31				
P9	CRUCLK	2A8				
P10	A0	1A37				
P11	A1	1A38				
P12	A2	2A38				
P13	A3	2A18				
P14	A4	2A19				
P15	A5	2A40				
P16	A6	2A16				
P17	A7	2A37				
P18	A8	2A36				
P19	A9	2A15				
P20	A10	2A35				
P21	A11	2A34				
P22	A12	2A33				
P23	A13	2A14				
P24	A14	1A13	5	40	nS	26,27

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Table XX Transition time (tTHL) test (Continued)

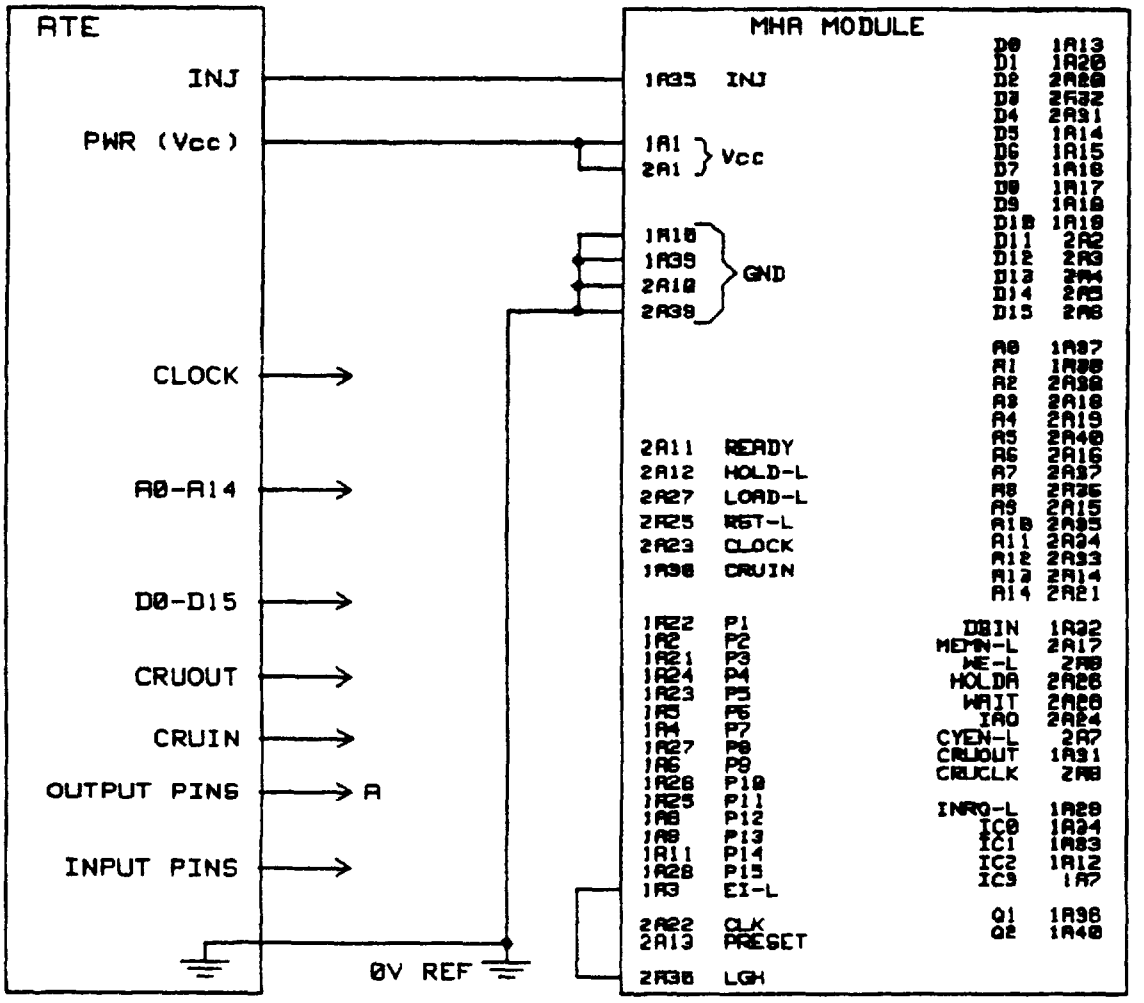
TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	MAX	UNIT	TEST FIG
P25	D0 Transition time, CL = 100 pF	1A13	5	40	nS	26,27
P26	D1	1A20				
P27	D2	2A20				
P28	D3	2A32				
P29	D4	2A31				
P30	D5	1A14				
P31	D6	1A15				
P32	D7	1A16				
P33	D8	1A17				
P34	D9	1A18				
P35	D10	1A19				
P36	D11	2A2				
P37	D12	2A3				
P38	D13	2A4				
P39	D14	2A5				
P40	D15	2A6	5	40		26,27
P41	INTR-L	1A29	4	50		26,31
P42	IC0	1A34		20		26,29
P43	IC1	1A33		50		26,29
P44	IC2	1A12		50		26,30
P45	IC3	1A7		50		26,30
P46	Q1	1A36		20		26,34
P47	Q2	1A40	4	20	nS	26,34

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Table XXI Transition time (tTLH) test

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT		UNIT	TEST FIG
			MIN	MAX		
Q1	INTR-L	1A29	10	80	nS	26,31
Q2	IC0	1A34	3	45		26,29
Q3	IC1	1A33	10	80		26,29
Q4	IC2	1A12	10	80		26,30
Q5	IC3	1A7	10	80		26,30
Q6	Q1	1A36	3	45		26,34
Q7	Q2	1A40	3	45	nS	26,34

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NOTES
1 INJ = 480 mA ±1%
2 VCC = 5 0 V ±1%
3 Load each output under test as specifiedon figure 34

FIGURE 26 Transition time (tTHL, tTLH) test circuit for tables XX and XXI

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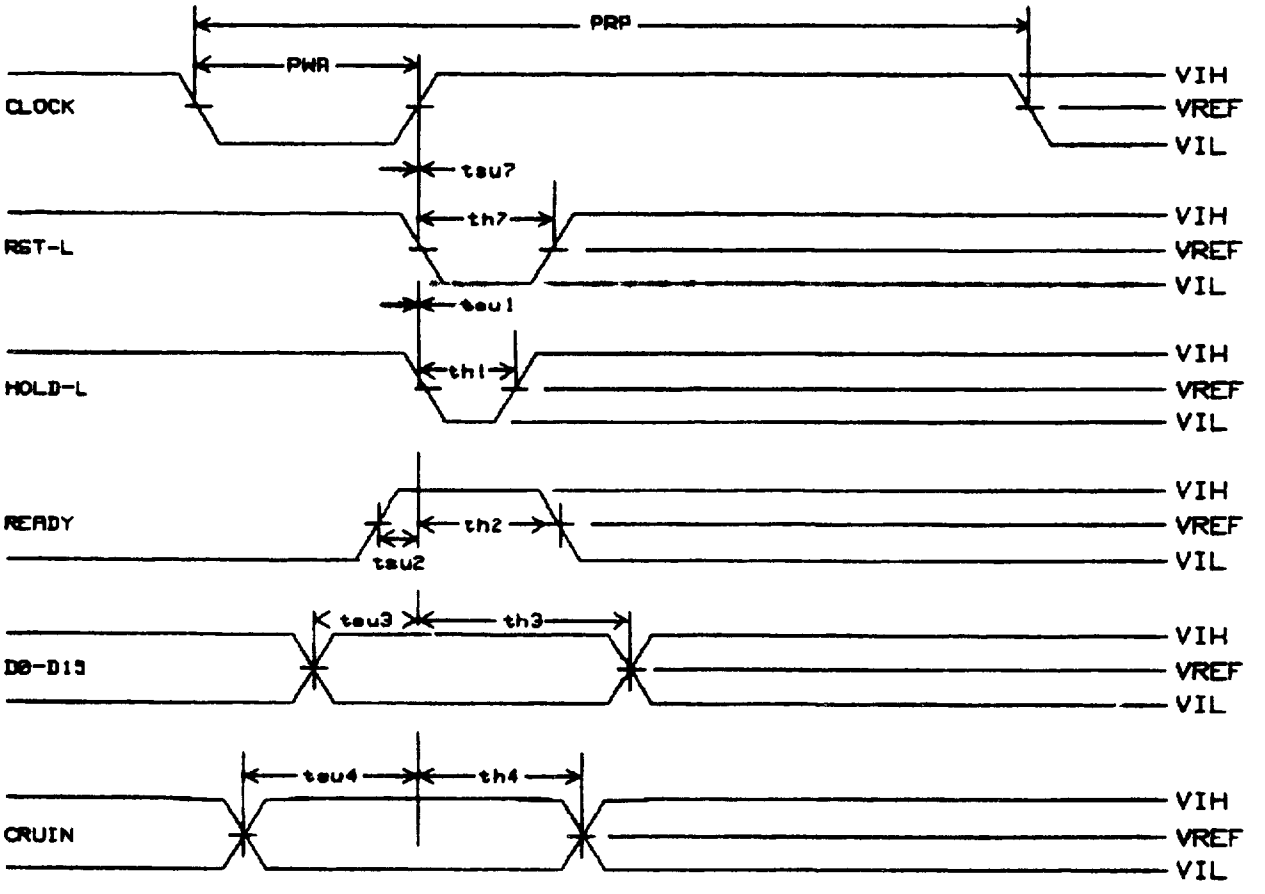


FIGURE 27 Transition time (tTHL) waveforms for CPU outputs, table XX

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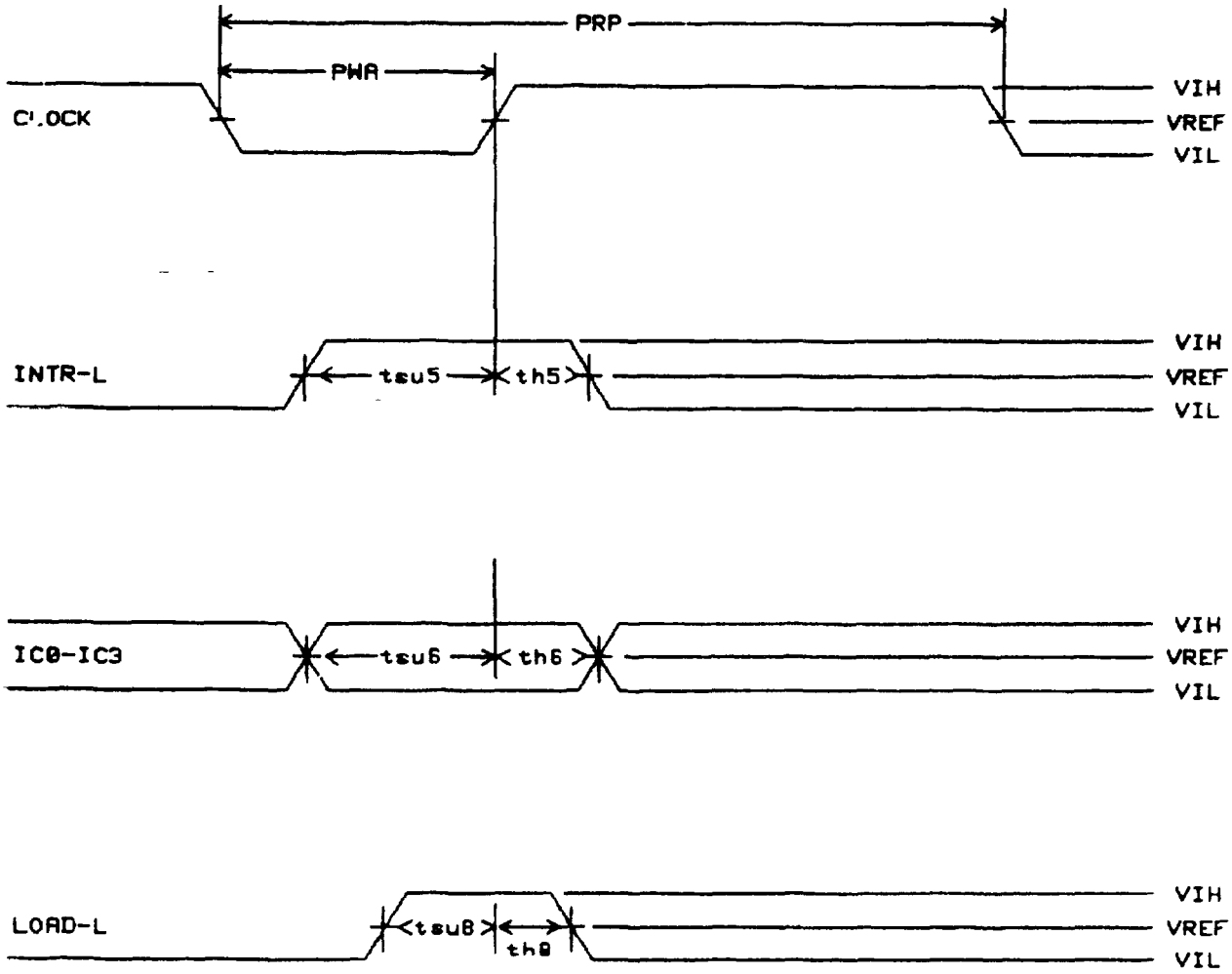


FIGURE 27 Transition time (tTHL) waveforms for CPU outputs, table XX (con't)

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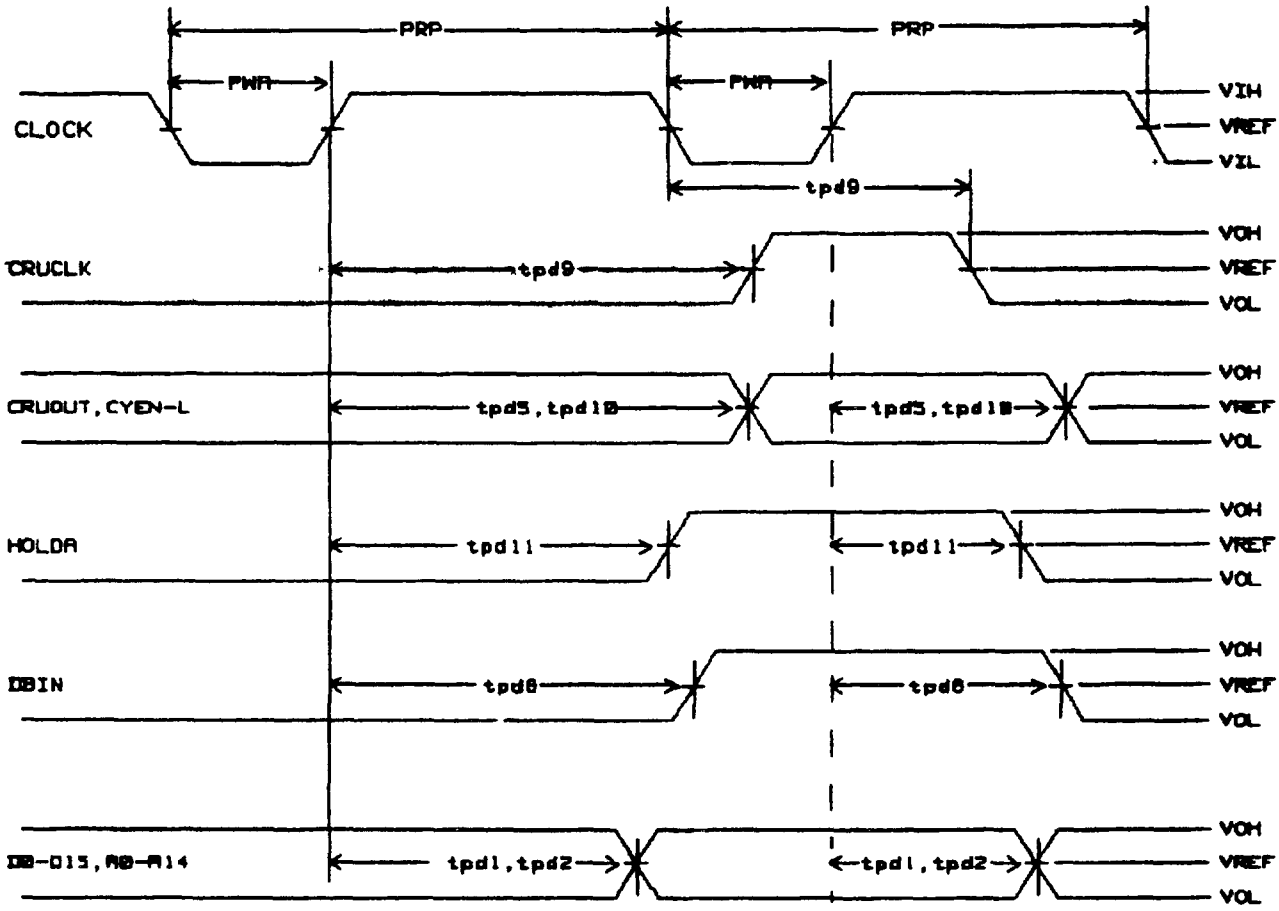


FIGURE 27 Transition time (tTHL) waveforms for CPU outputs, table XX (con't)

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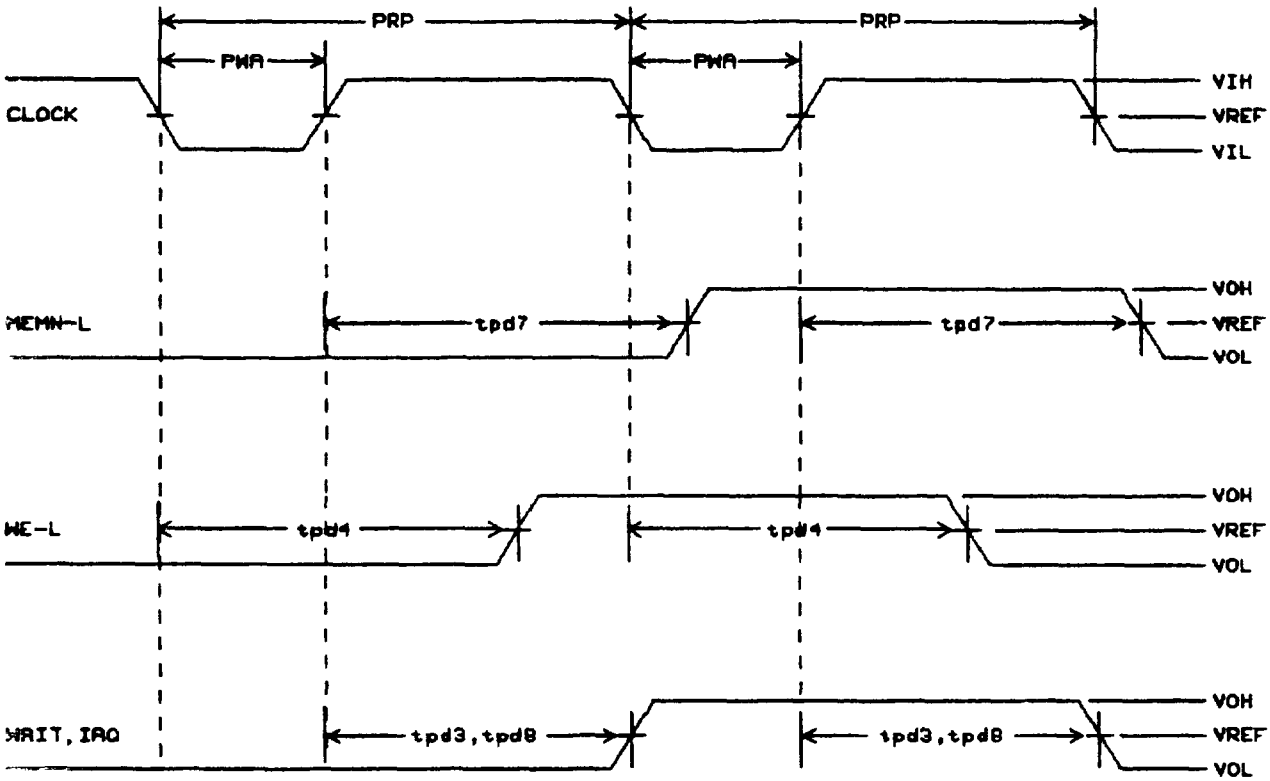


FIGURE 27 Transition time (tTHL) waveforms for CPU outputs, table XX (con't)

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Check the source to verify that this is the current version before use.

NOTES

1 For input pulse and output pulse parameters, refer to tables XVII through XIX

2 Input transition times and output transition times are measured from the 10% level to the 90% level

3 In addition to the switching characteristics shown, the following signal relationships exist

a During write operations
(1) ADDR lines are valid prior to WE-L active (low)
(2) WE-L goes inactive (high) prior to ADDR lines going valid
(3) Data lines are valid prior to WE-L active (low)
(4) WE-L goes inactive (high) prior to datalines going invalid
(5) WE-L does not go active (low) before MEMN-L goes active (low)
(6) WE-L goes inactive (high) before MEMN-L goes inactive (high)

b During read operations
(1) DBIN does not go inactive (low) before the maximum data hold time, th3, runs out
(2) MEMN-L does not go inactive (high) before the maximum data hold time, th3, runs out

c During CRU output operations
(1) CRUCLK goes inactive (low) prior to ADDR lines going invalid
(2) CRUCLK goes inctive (low) prior to CRUOUT going invalid

FIGURE 27 Transition time (tTHL) waveforms for CPU outputs, table XX (con't)

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Table XXII Propagation delay time (tPLH) test

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg + 5 degC AMBIENT MAX	UNIT	TEST FIG
R1	EI-L to INTR-L	1A29	10	58	nS	28,31
R2	EI-L to INTR-L			43		
R3	LGH to INTR-L			61		
R4	P1 to INTR-L					
R5	P2 to INTR-L					
R6	P3 to INTR-L					
R7	P4 to INTR-L					
R8	P5 to INTR-L					
R9	P6 to INTR-L					
R10	P7 to INTR-L					
R11	P8 to INTR-L					
R12	P9 to INTR-L					
R13	P10 to INTR-L					
R14	P11 to INTR-L					
R15	P12 to INTR-L					
R16	P13 to INTR-L					
R17	P14 to INTR-L					
R18	P15 to INTR-L	1A29		61		28,31
R19	EI-L to IC0	1A34		56		28,29
R20	LGH to IC0			51		28,31
R21	P1 to IC0					
R22	P2 to IC0					
R23	P3 to IC0					
R24	P4 to IC0					
R25	P5 to IC0	1A34	10	51	nS	28,31

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Table XXII Propagation delay time (tPLH) test (Continued)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg ± 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg ± 5 degC AMBIENT MAX	UNIT	TEST FIG
R26	P6 to IC0	1A34	10	51	nS	28,31
R27	P7 to IC0	1A34		51		28,31
R28	EI-L to IC1	1A33		46		28,32
R29	EI-L to IC1					
R30	LGH to IC1					
R31	P1 to IC1					
R32	P2 to IC1					
R33	P3 to IC1			46		28,32
R34	P4 to IC1			56		28,29
R35	P5 to IC1			56		
R36	P6 to IC1			56		
R37	P7 to IC1			56		28,29
R38	P8 to IC1			46		28,32
R39	P9 to IC1			46		
R40	P10 to IC1			46		
R41	P11 to IC1	1A33		46		28,32
R42	EI-L to IC2	1A12		61		
R43	EI-L to IC2			46		
R44	LGH to IC2			46		
R45	P1 to IC2			46		28,32
R46	P2 to IC2			50		28,30
R47	P2 to IC2			56		
R48	P3 to IC2			50		
R49	P3 to IC2			56		28,30
R50	P4 to IC2	1A12	10	46	nS	28,32

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Table XXII Propagation delay time (tPLH) test (Continued)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg + 5 degC AMBIENT MAX	UNIT	TEST FIG
R51	P5 to IC2	1A12	10	46	nS	28,32
R52	P6 to IC2			56		28,30
R53	P7 to IC2			56		28,30
R54	P8 to IC2			46		28,32
R55	P9 to IC2			46		28,32
R56	P10 to IC2			50		28,32
R57	P11 to IC2			50		28,30
R58	P12 to IC2			46		28,32
R59	P13 to IC2	1A12		46		28,32
R60	EI-L to IC3	1A7		61		28,33
R61	EI-L to IC3			46		28,33
R62	LGH to IC3			46		28,33
R63	P1 to IC3			50		28,30
R64	P1 to IC3			56		28,30
R65	P2 to IC3			46		28,33
R66	P3 to IC3			50		28,30
R67	P3 to IC3			56		28,30
R68	P4 to IC3			46		28,33
R69	P5 to IC3			50		
R70	P5 to IC3			56		
R71	P6 to IC3			46		
R72	P7 to IC3			56		
R73	P8 to IC3			46		28,33
R74	P9 to IC3			50		28,30
R75	P10 to IC3	1A	10	46	nS	28,33

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Table XXII Propagation delay time (tPLH) test. (Continued)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg + 5 degC AMBIENT MAX	UNIT	TEST FIG
R76	P11 to IC3	1A7	10	50	nS	28,30
R77	P12 to IC3	1A7		46		28,33
R78	P13 to IC3	1A7		50		28,30
R79	P14 to IC3	1A7		46		28,33
R80	CLK to Q1	1A36		29		28,34
R81	CLK to Q2	1A40		29		28,34
R82	PRESET to Q1	1A36		29		28,35
R83	PRESET to Q2	1A40	10	29	nS	28,35

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Table XXIII Propagation delay time (tPHL) test

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT		UNIT	TEST FIG
			MIN	MAX		
S1	EI-L to INTR-L	1A29	10	68	nS	28,31
S2	EI-L to INTR-L			38		
S3	LGH to INTR-L			48		
S4	P1 to INTR-L					
S5	P2 to INTR-L					
S6	P3 to INTR-L					
S7	P4 to INTR-L					
S8	P5 to INTR-L					
S9	P6 to INTR-L					
S10	P7 to INTR-L					
S11	P8 to INTR-L					
S12	P9 to INTR-L					
S13	P10 to INTR-L					
S14	P11 to INTR-L					
S15	P12 to INTR-L					
S16	P13 to INTR-L					
S17	P14 to INTR-L					
S18	P15 to INTR-L	1A29		48		28,31
S19	EI-L to IC0	1A34		41		28,29
S20	LGH to IC0			36		28,31
S21	P1 to IC0					
S22	P2 to IC0					
S23	P3 to IC0					
S24	P4 to IC0					
S25	P5 to IC0	1A34	10	36	nS	28,31

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Table XXIII Propagation delay time (tPHL) test (Continued)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg + 5 degC AMBIENT MAX	UNIT	TEST FIG
S26	P6 to IC0	1A34	10	36	nS	28,31
S27	P7 to IC0	1A34		36		28,31
S28	EI-L to IC1	1A33		68		28,32
S29	EI-L to IC1			38		
S30	LGH to IC1			37		
S31	P1 to IC1			37		
S32	P2 to IC1			37		
S33	P3 to IC1			37		28,32
S34	P4 to IC1			63		28,29
S35	P5 to IC1			63		
S36	P6 to IC1			63		
S37	P7 to IC1			63		28,29
S38	P8 to IC1			37		28,32
S39	P9 to IC1			37		
S40	P10 to IC1			37		
S41	P11 to IC1	1A33		37		28,32
S42	EI-L to IC2	1A12		68		
S43	EI-L to IC2			38		
S44	LGH to IC2			37		
S45	P1 to IC2			37		28,32
S46	P2 to IC2			42		28,30
S47	P2 to IC2			63		
S48	P3 to IC2			42		
S49	P3 to IC2			63		28,30
S50	P4 to IC2	1A12	10	37	nS	28,32

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Table XXIII Propagation delay time (tPHL) test (Continued)

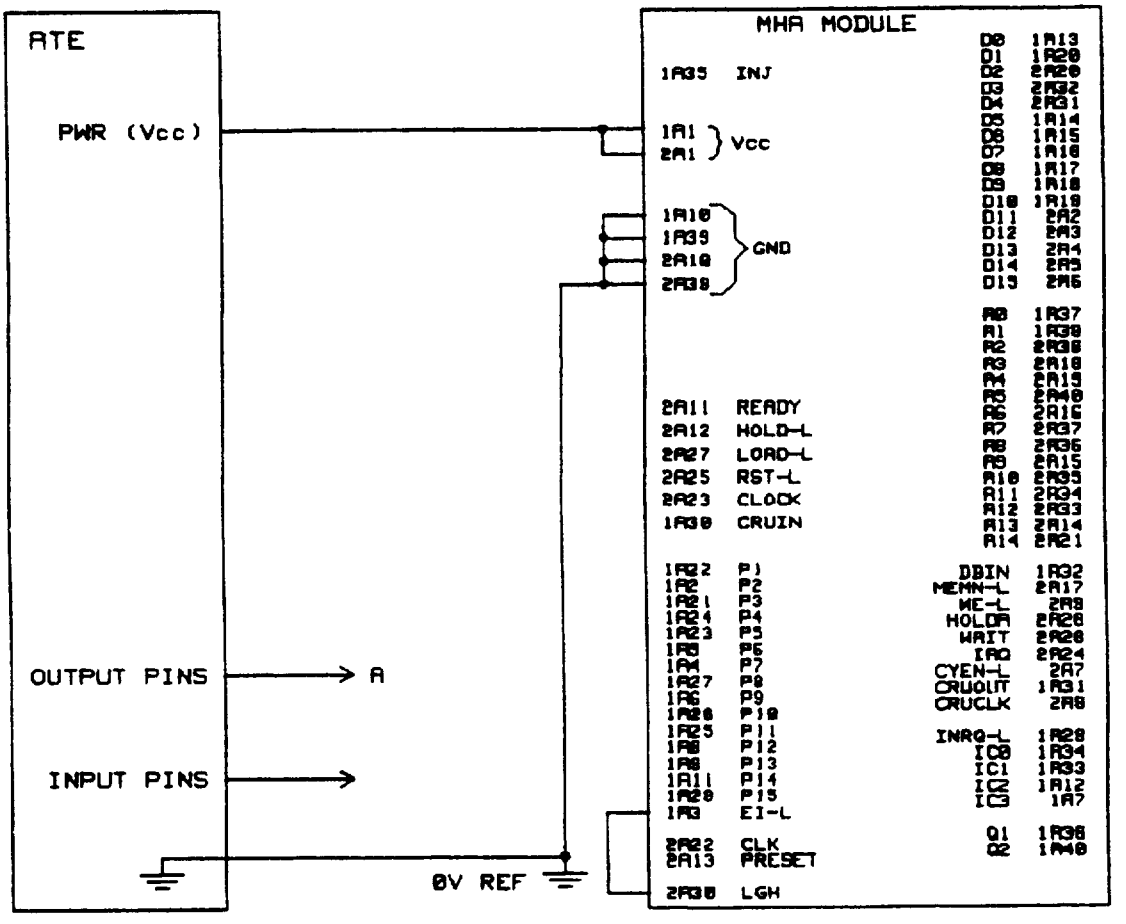
TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg ± 5 degC AMBIENT MIN	INITIAL LIMITS 25 deg ± 5 degC AMBIENT MAX	UNIT	TEST FIG
S51	P5 to IC2	1A12	10	37	nS	28,32
S52	P6 to IC2			63		28,30
S53	P7 to IC2			63		28,30
S54	P8 to IC2			37		28,32
S55	P9 to IC2			37		28,32
S56	P10 to IC2			42		28,32
S57	P11 to IC2			42		28,30
S58	P12 to IC2			37		28,32
S59	P13 to IC2	1A12		37		28,32
S60	EI-L to IC3	1A7		68		28,33
S61	EI-L to IC3			38		28,33
S62	LGH to IC3			37		28,33
S63	P1 to IC3			42		28,30
S64	P1 to IC3			63		28,30
S65	P2 to IC3			37		28,33
S66	P3 to IC3			42		28,30
S67	P3 to IC3			63		28,30
S68	P4 to IC3			37		28,33
S69	P5 to IC3			42		
S70	P5 to IC3			63		
S71	P6 to IC3			37		
S72	P7 to IC3			63		
S73	P8 to IC3			37		28,33
S74	P9 to IC3			42		28,30
S75	P10 to IC3	1A7	10	37	nS	28,33

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Table XXIII Propagation delay time (tPHL) test (Continued)

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg + 5 degC AMBIENT MIN	MAX	UNIT	TEST FIG
S76	P11 to IC3	1A7	10	42	nS	28,30
S77	P12 to IC3	1A7		37		28,33
S78	P13 to IC3	1A7		42		28,30
S79	P14 to IC3	1A7		37		28,33
S80	CLK to Q1	1A36		44		28,34
S81	CLK to Q2	1A40		44		28,34
S82	CLEAR to Q1	1A36		44		28,35
S83	CLEAR to Q2	1A40	10	44	nS	28,35

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NOTES
1 Load each output under test as specified on figure 23
2 VCC = 5 0 V ±1%

FIGURE 28 Propagation delay time (tPLH, tPHL) test circuit for tables XXII and XXIII

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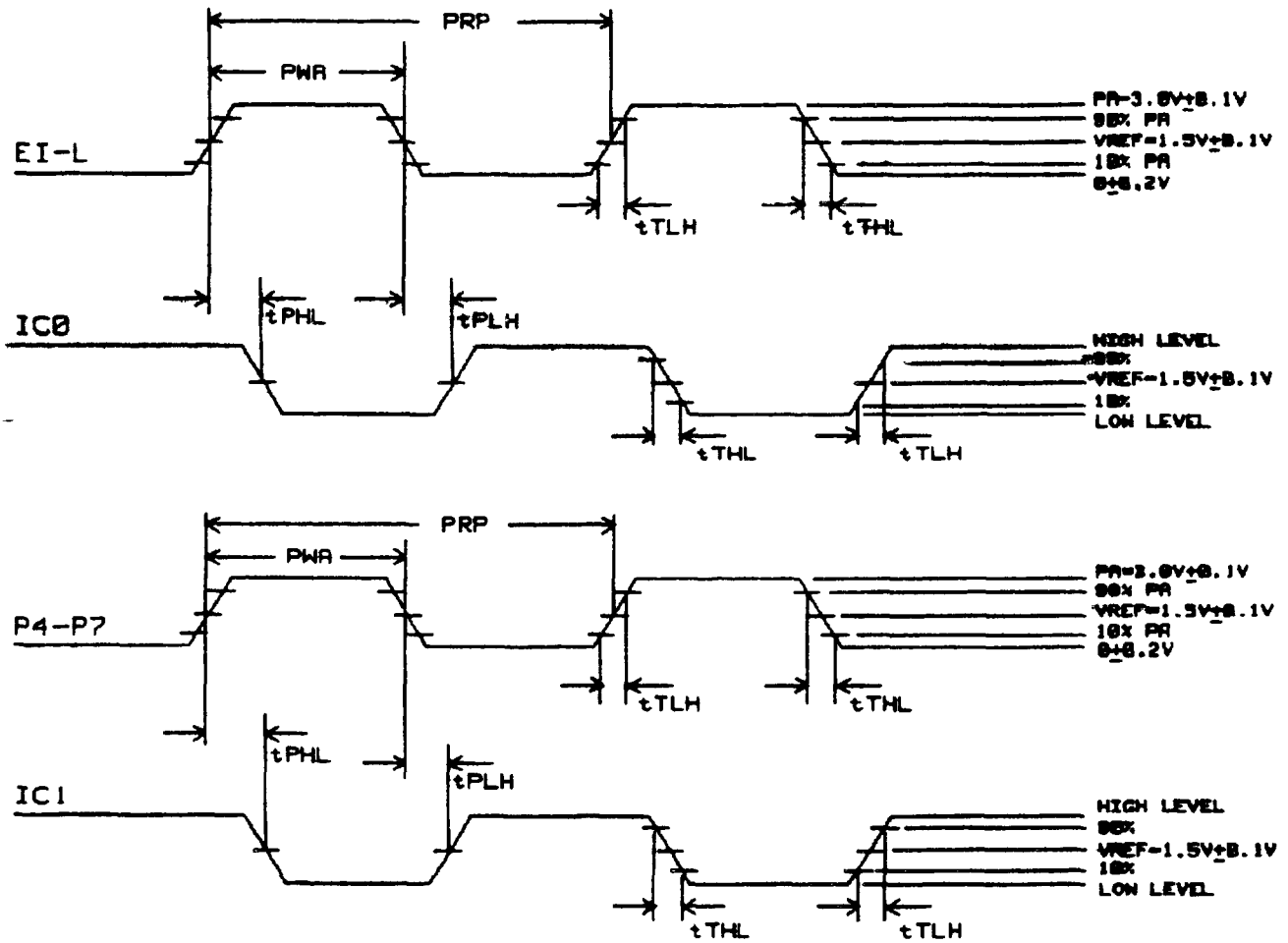


FIGURE 29 Transition time (tTLH, tTHL) and propagation delay time (tPLH, tPHL) test circuit waveforms for tables XX, XXI, XXII and XXIII

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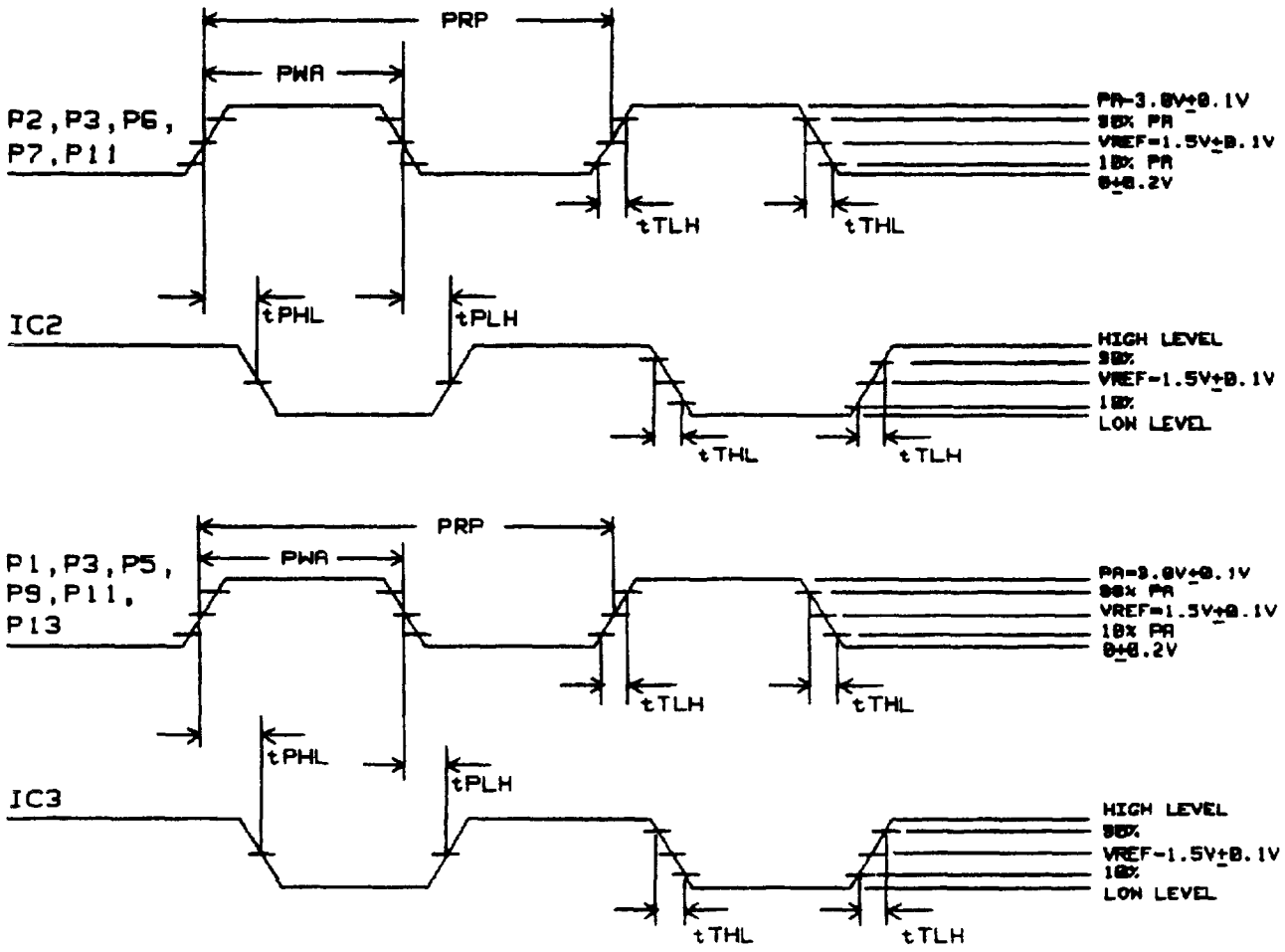


FIGURE 30 Transition time (tTLH, tTHL) and propagation delay time (tPLH, tPHL) test circuit waveforms for tables XX, XXI, XXII, and XXIII

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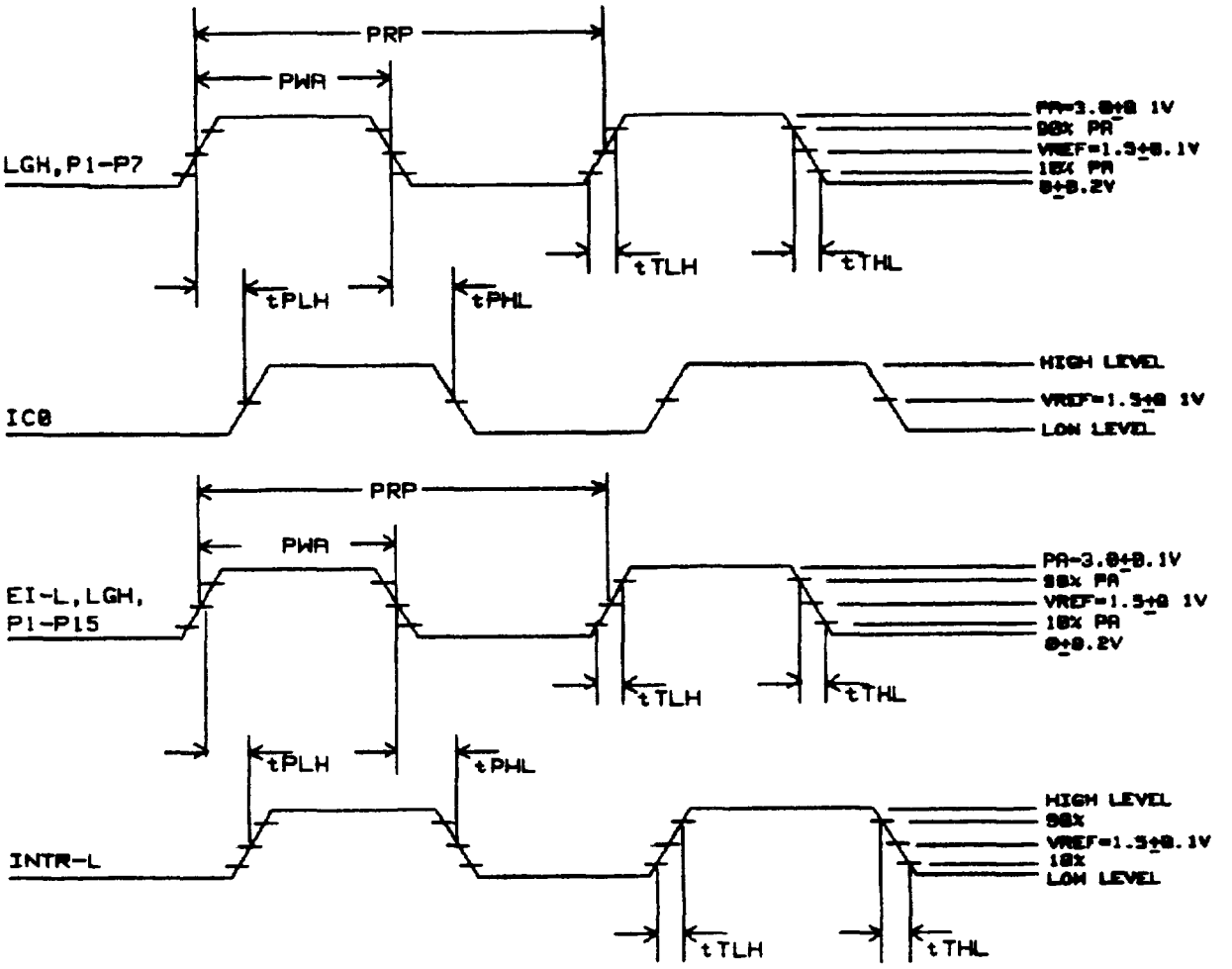


FIGURE 31 Transition time (tTLH, tTHL) and propagation delay time (tPLH, tPHL) test circuit waveform for tables XX, XXI, XXII, and XXIII

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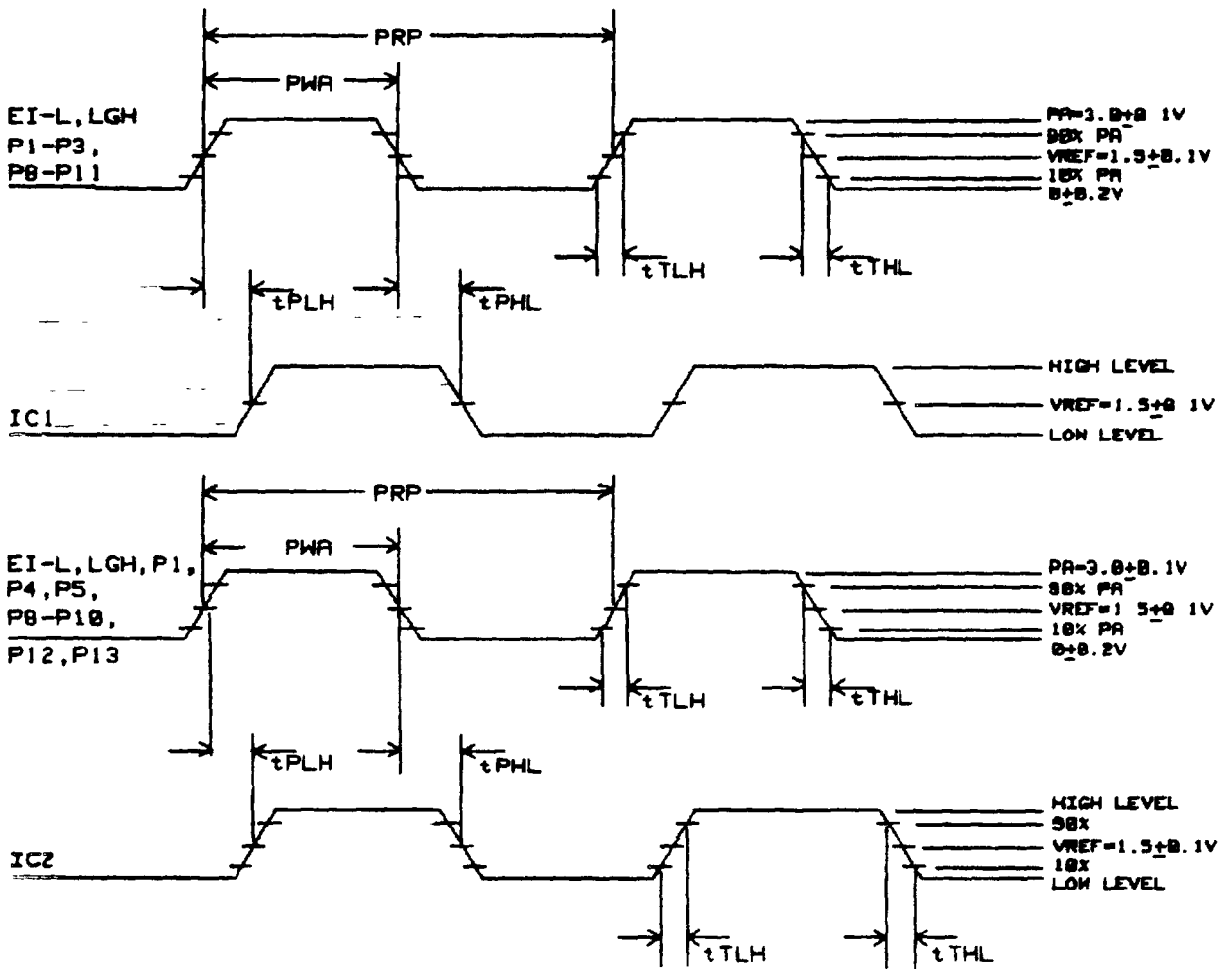
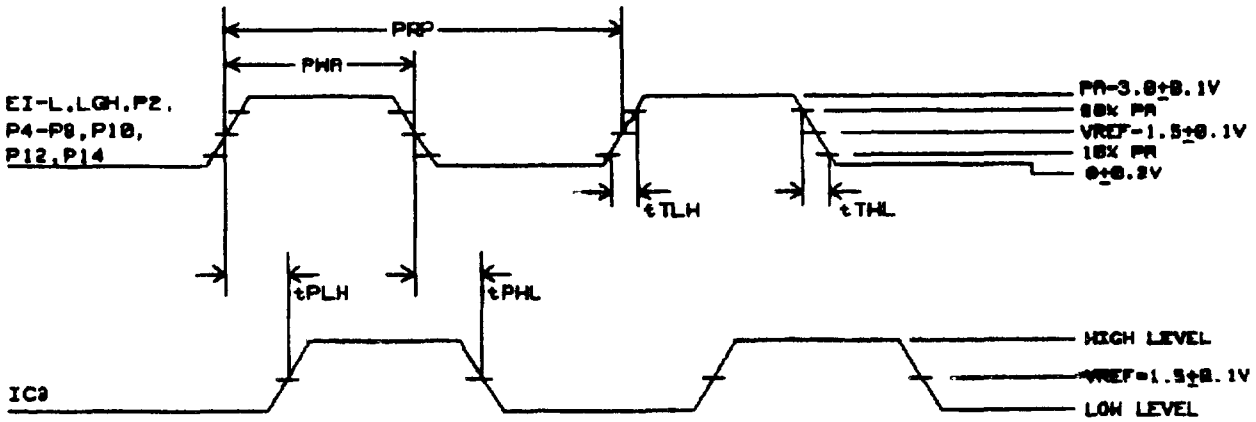


FIGURE 32 Transition time (tTLH, tTHL) and propagation delay time (tPLH, tPHL) test circuit waveforms for tables XX, XXI, XXII, and XXIII

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NOTES

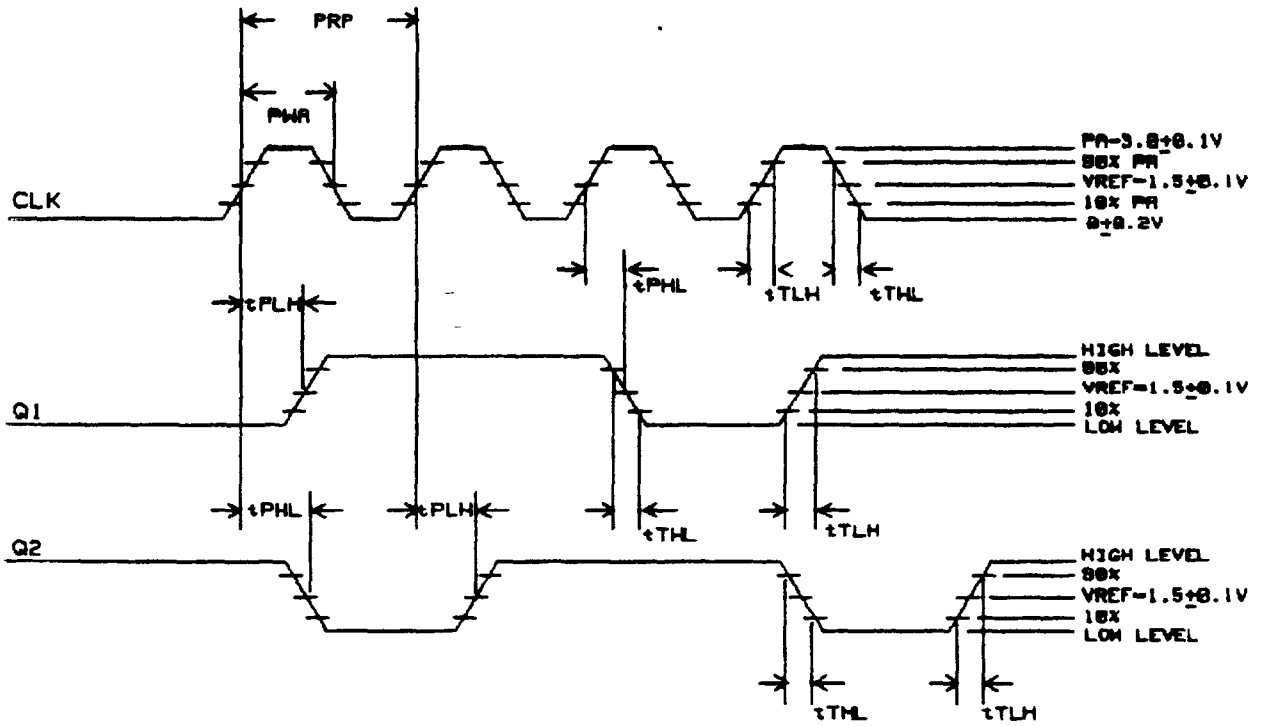
1 Input pulse parameters
tTLH = tTHL = 10 ±2 nS
PWA = 500 ± 50 nS
PRP = 1000 ± 100 nS

2 Before making the measurements specified in tables XX through XXIII, verify that the output pulse high level is ≥ 2 4V and the low level is ≤ 0 4V

3 All voltages are referenced to ground

FIGURE 33 Transition time (tTLH, tTHL) and propagation delay time (tPLH, tPHL) test circuit waveforms for tables XX, XXI, XXII, and XXIII

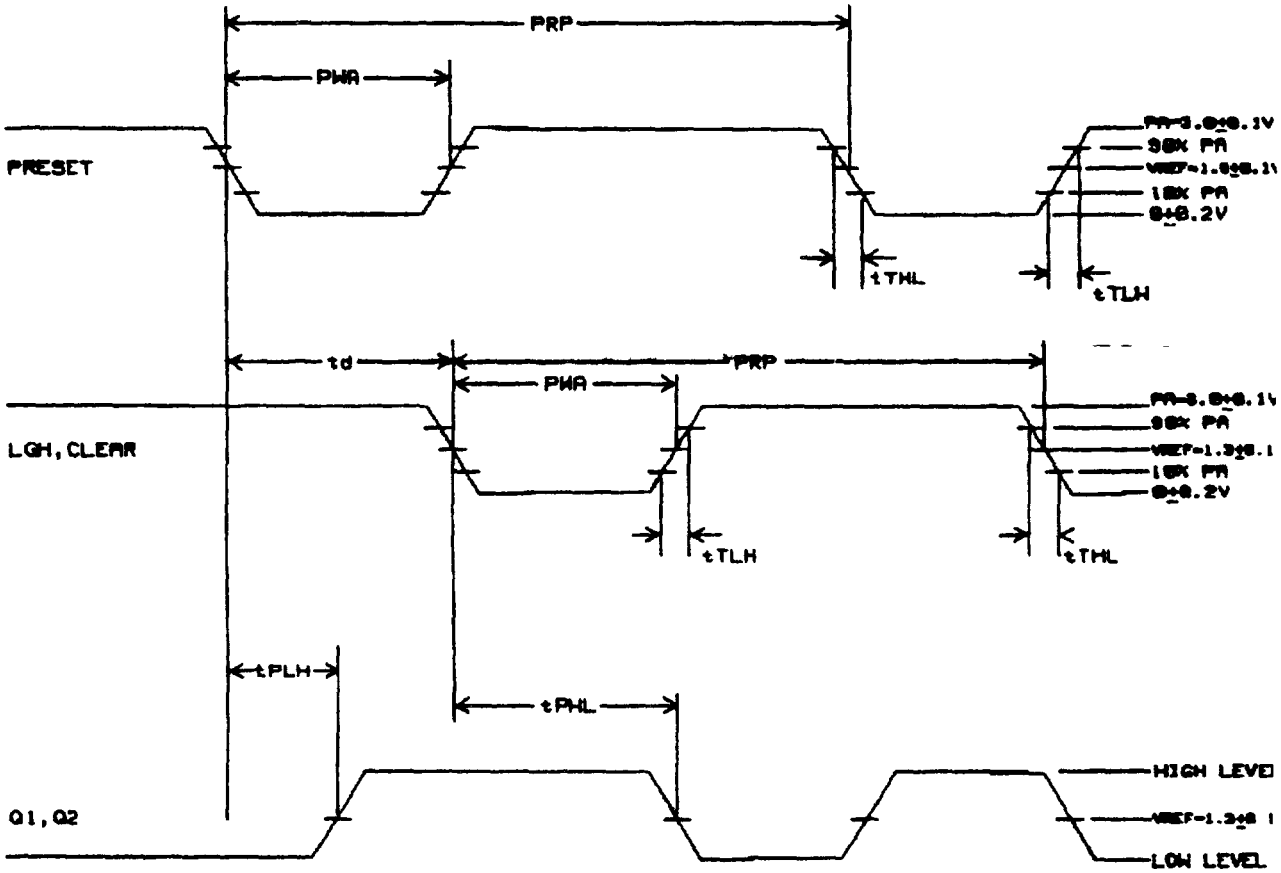
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NOTES
1 Input pulse parameters
tTLH = tTHL = 10 ± 2 nS
PWA = 400 ± 40 nS
PRP = 1000 ± 100 nS
2 Before making the measurements specified in tables XX to XXIII, verify that the output pulse high level is ≥ 2 4V and the low level is ≤ 0 4V
3 All voltages are referenced to ground

FIGURE 34 Transition time (tTLH,tTHL) and propagation delay time (tPLH,tPHL) test circuit waveforms for tables XX, XXI, XXII, and XXIII

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NOTES

1 Input pulse parameters
tTLH = tTHL = 10 ± 2 nS
PWA = 100 +2, -0 nS
PRP = 400 ± 40 nS
td = 200 ± 20 nS

2 Before making the measurements specified in tables XX to XXIII, verify that the output pulse high level is ≥ 2 4V and the low level is ≤ 0 4V

3 All voltages are referenced to ground

FIGURE 35 Propagation delay time (tPLH, tPHL) test circuit waveforms for tables XX, XXI, XXII, and XXIII

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Table XXIV Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH and tPHL) check 1/

CONDITION	TEST DESCRIPTION	INPUT TABLE	OUTPUT TABLE	WAVEFORM	TEST FIGURE
T1	Dynamic functional test, minimum injector current and propagation delay time check	XXV	XXVI	37	36
T2	Dynamic functional test, maximum injector current and propagation delay time check	XXV	XXVI	37	36

1/ Apply the vectors (see Appendix A) with the input parameters of table XXV and checking the output parameters of table XXVI

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Table XXV CPU input conditions for table XXIV

PARAMETER	PIN(s)	INPUT CONDITIONS	UNIT	FIG
INJ	1A35	1/,2/	mA	36,37
Vcc	1A1,2A1	4.5±1%	V	
PRP	2A23	333±1%	nS	
PWA	2A23	111±1%		
tTLH,tTHL	All CPU inputs	15±5		
tsu1 HOLD-L	2A12	5+2,-0		
tsu2 READY	2A11	35+2,-0		
tsu3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	5+2,-0		
tsu4 CRUIN	1A30	20+2,-0		
tsu5 INTR-L	1A29	5+2,-0		
tsu6 IC0-IC3	1A7,1A12,1A33,1A34	10+2,-0		
tsu7 RST-L	2A25	0+2,-0		
tsu8 LOAD-L	2A27	35+2,-0		
th1 HOLD-L	2A12	25+2,-0		
th2 READY	2A11	30+2,-0		
th3 D0-D15	1A13-1A20,2A2-2A6,2A20,2A31, 2A32	35+2,-0		
th4 CRUIN	1A30	25+2,-0		
th5 INTR-L	1A29	25+2,-0		
th6 IC0-IC3	1A7,1A12,1A33,1A34	25+2,-0		
th7 RST-L	2A25	30+2,-0		
th8 LOAD-L	2A27	25+2,-0	nS	
VIH	All	3.0+0.1,-0.0	V	
VIL	All	0.0+0.0,-0.1	V	36,37

1/ INJ = 480mA±1% for test condition T1
2/ INJ = 520mA±1% for test condition T2

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Table XXVI CPU output conditions for Table XXIV

PARAMETER	PIN(s)	OUTPUT CONDITIONS MIN	OUTPUT CONDITIONS MAX	UNIT	FIG
tpd1 D0-D15, CL= 100 pF	1A13-1A20,2A2-2A6, 2A20,2A31,2A32	-	140	nS	36,37
tpd2 A0-A14, CL= 100 pF	1A37,1A38,2A14-2A19,2A21,2A33-2A38-2A40	-	140		
tpd3 IAQ, CL= 100 pF	2A24	-	140		
tpd4 WE-L, CL= 100 pF	2A9	-	180		
tpd5 CRUOUT, CL= 100 pF	1A31	-	175		
tpd6 DBIN, CL= 100 pF	1A32	-	155		
tpd7 MEMN-L, CL= 100 pF	2A17	-	140		
tpd8 WRIT, CL= 100 pF	2A28	-	130		
tpd9 CRUCLK, CL= 100 pF	2A8	-	185		
tpd10 CYEN-L, CL= 100 pF	2A7	-	175		
tpd11 HOLDA, CL= 100 pF	2A26	-	150		
When leaving a hold state:					
tpd1 D0-D15, CL= 100 pF	1A13-1A20,2A2-2A6, 2A20,2A31,2A32	-	200		
tpd2 A0-A14, CL= 100 pF	1A37,1A38,2A14-2A19 2A21,2A33-2A38,2A40	-	200		
tpd3 IAQ, CL= 100 pF	2A24	-	200		
tpd6 DBIN, CL= 100 pF	1A32	-	200		
tpd7 MEMN-L, CL= 100 pF	2A17	-	200	nS	
Output switch voltage	All outputs	1.5	1.5	V	36,37

1/ Check the propagation delay times by starting output comparison at the maximum propagation delay time limits and continuing for 10 nS

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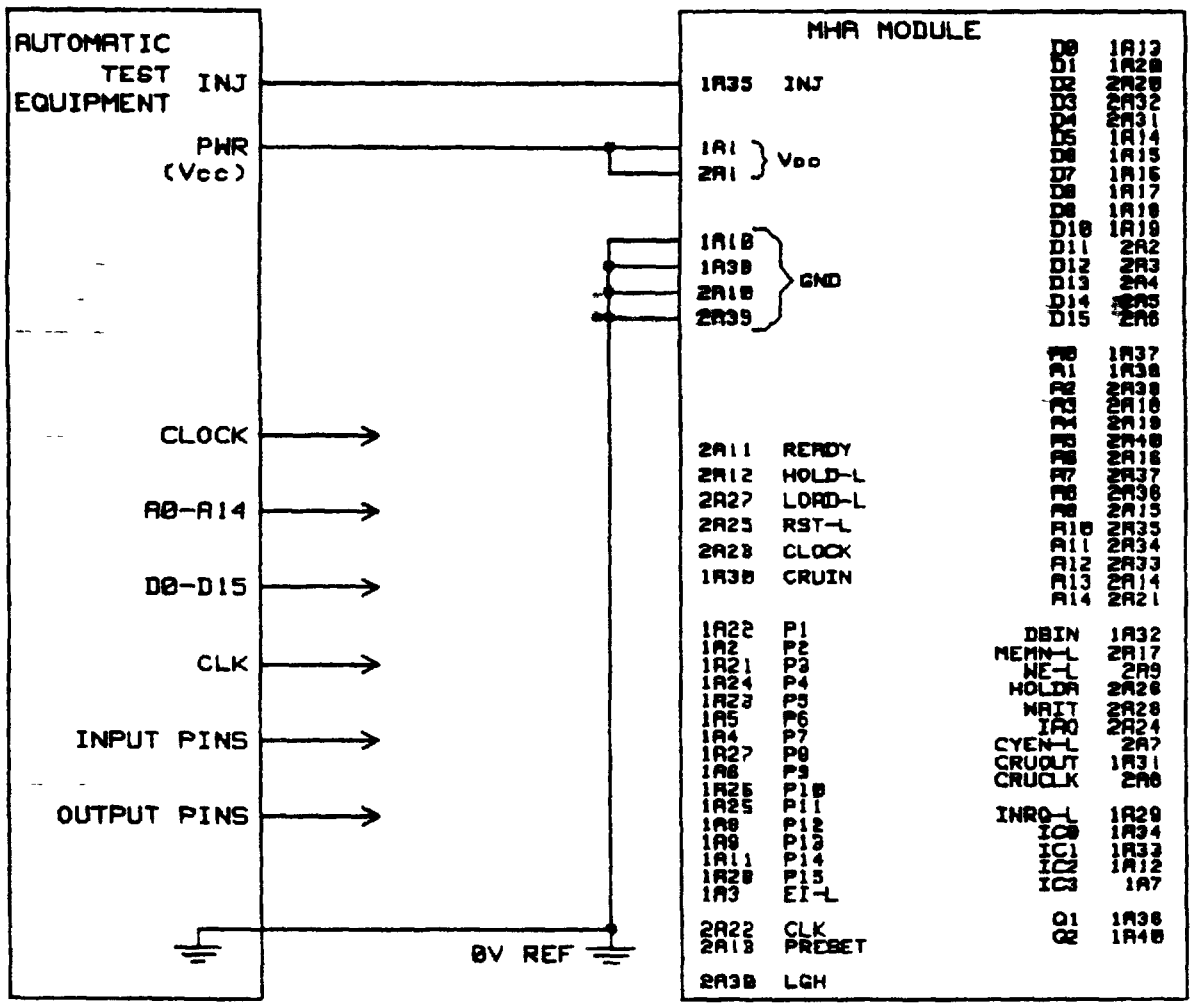


FIGURE 36 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH and tPHL) check circuit for tables XXIV, XXV, and XXVI

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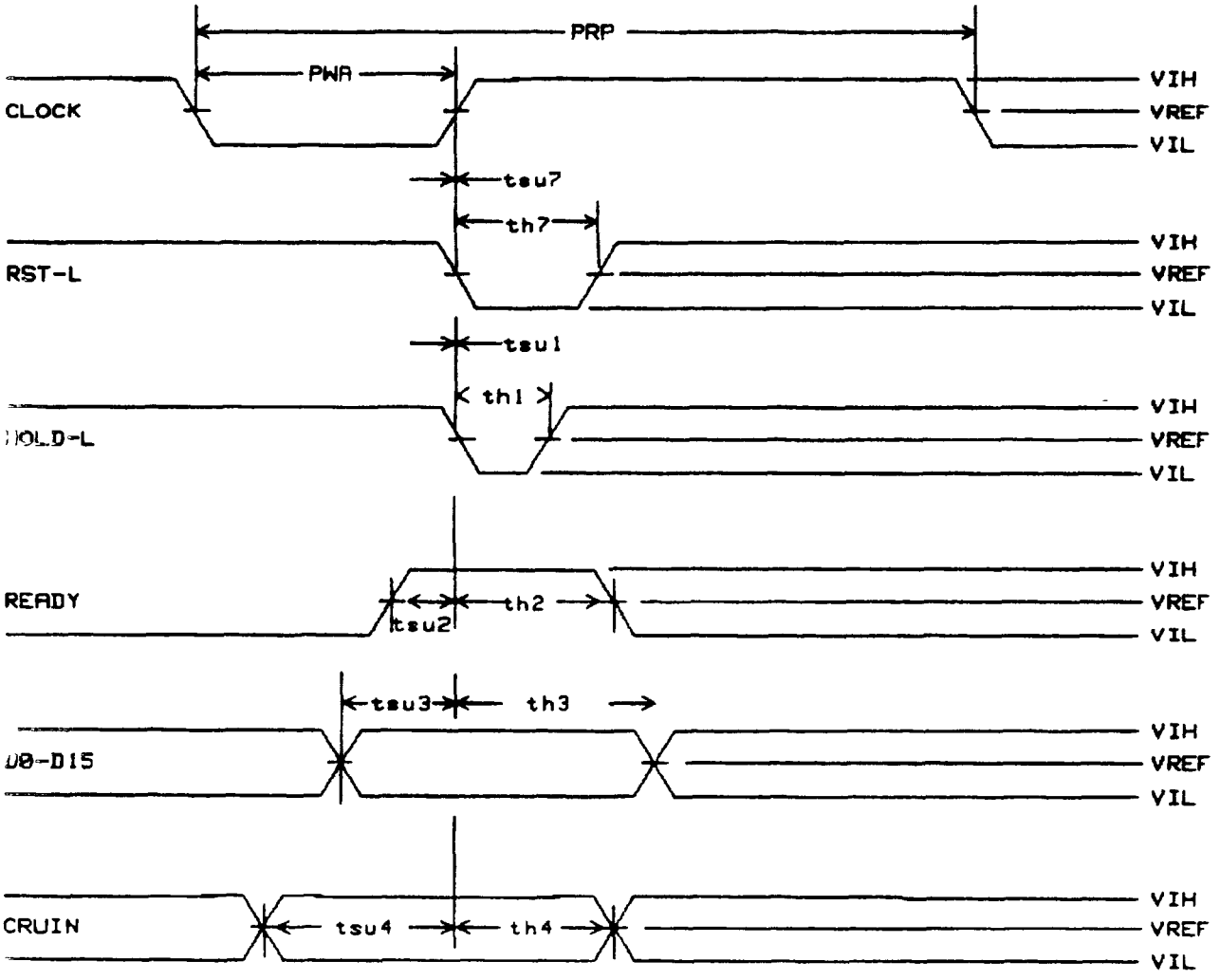


FIGURE 37 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH, tPHL) check waveforms for tables XXIV to XXVI

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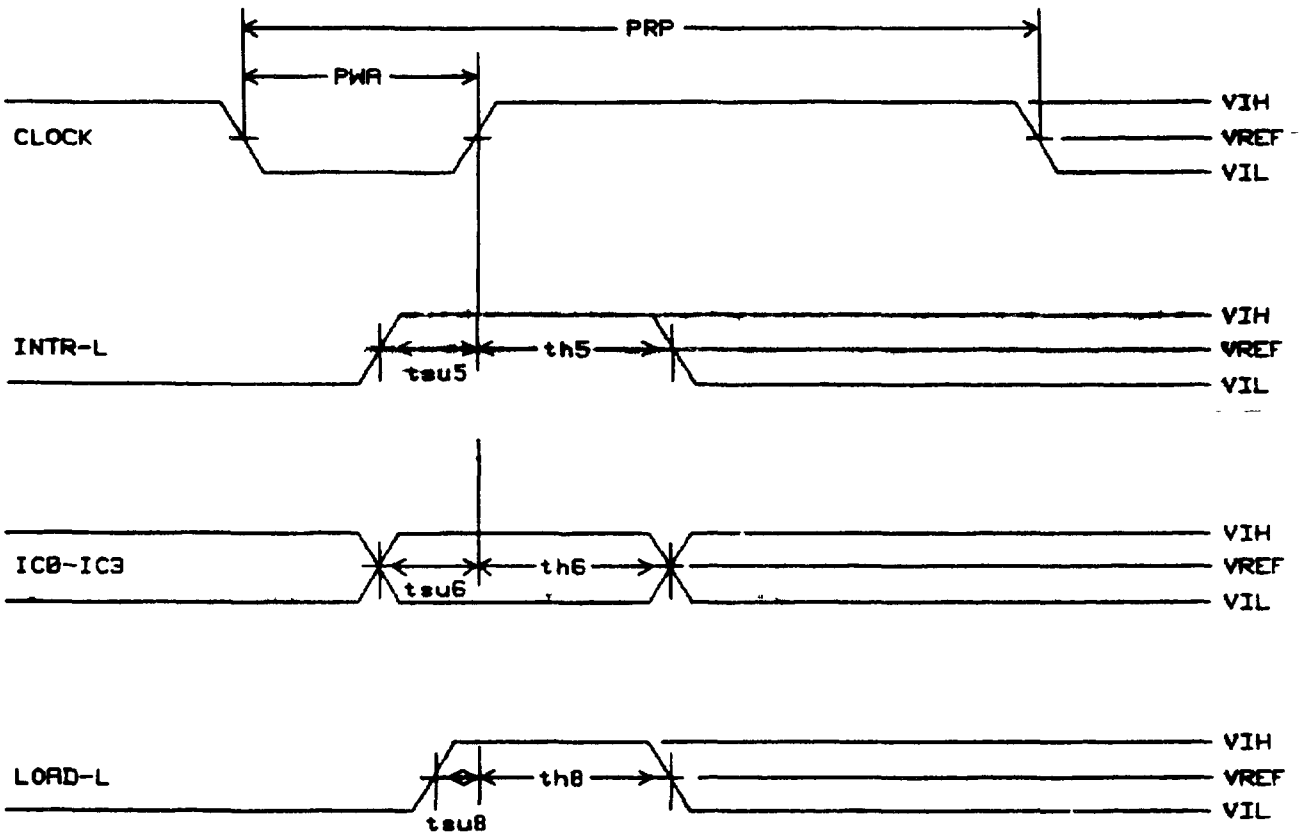


FIGURE 37 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH, tPHL) check waveforms for tables XXIV to XXVI (con't)

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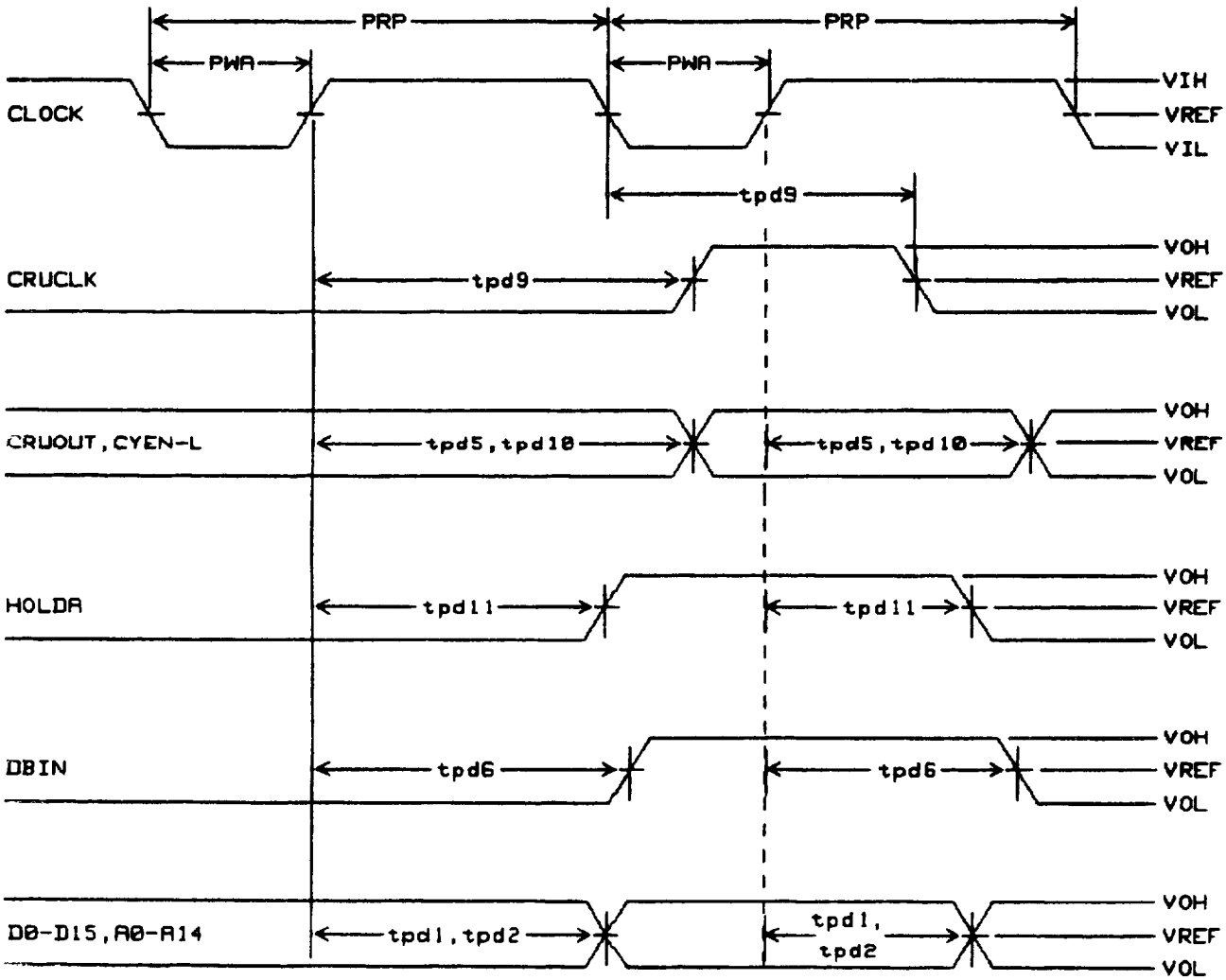


FIGURE 37 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH, tPHL) check waveforms for tables XXIV to XXVI (con't)

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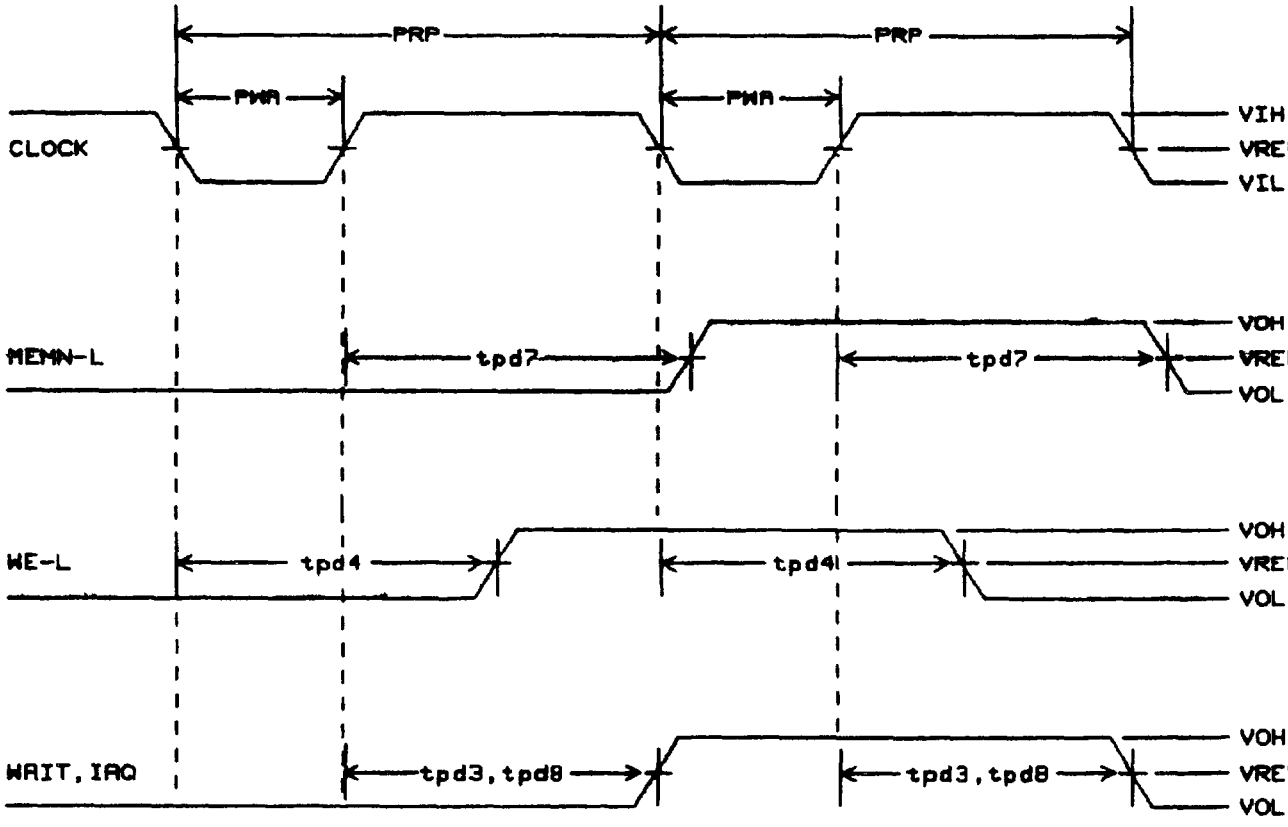


FIGURE 37 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH, tPHL) check waveforms for tables XXIV to XXVI (con't)

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NOTES
1 For input pulse and output pulse parameters, refer to tables XXIV to XXVI
2 Input transition times and output transition times are measured from the 10% level to the 90% level
3 In addition to the switching characteristics shown, the following signal relationships exist

a During write operations
(1) ADDR lines are valid prior to WE-L active (low)
(2) WE-L goes inactive (high) prior to ADDR lines going valid
(3) Data lines are valid prior to WE-L active (low)
(4) WE-L goes inactive (high) prior to data lines going invalid
(5) WE-L does not go active (low) before MEMN-L goes active (low)
(6) WE-L goes inactive (high) before MEMN-L goes inactive (high)

b During read operations
(1) DBIN does not go inactive (low) before the maximum data hold time, th3, runs out
(2) MEMN-L does not go inactive (high) before the maximum data hold time, th3, runs out

c During CRU output operations
(1) CRUCLK goes inactive (low) prior to ADDR lines going invalid
(2) CRUCLK goes inactive (low) prior to CRUOUT going invalid

FIGURE 37 Dynamic functional (minimum and maximum injector current) test and propagation delay time (tPLH, tPHL) check waveforms for tables XXIV to XXVI (con't)

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Table XXVII Maximum frequency (clock shaping) check test

TEST COND.	PARAMETER	CONNECT A TO	INITIAL LIMITS 25 deg ± 5 degC AMBIENT		UNIT	TEST FIG
			MIN	MAX		
U1	Q1, PRP2	1A36	201	207	nS	38,39
U2	Q1, PWA2	1A36	52	69	nS	38,39
U3	Q2, PRP2	1A40	201	207	nS	38,39
U4	Q2, PWA2	1A40	52	69	nS	38,39

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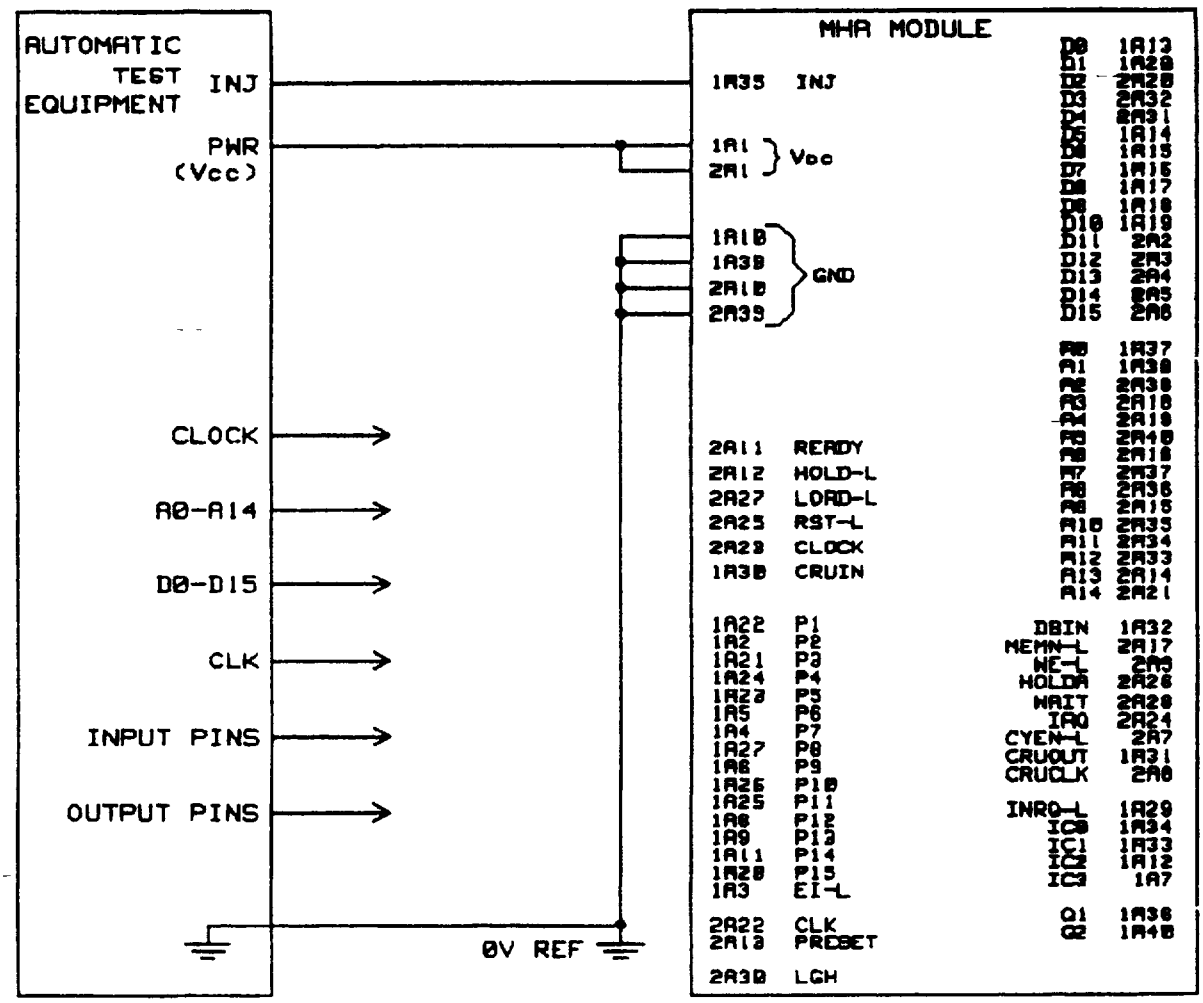
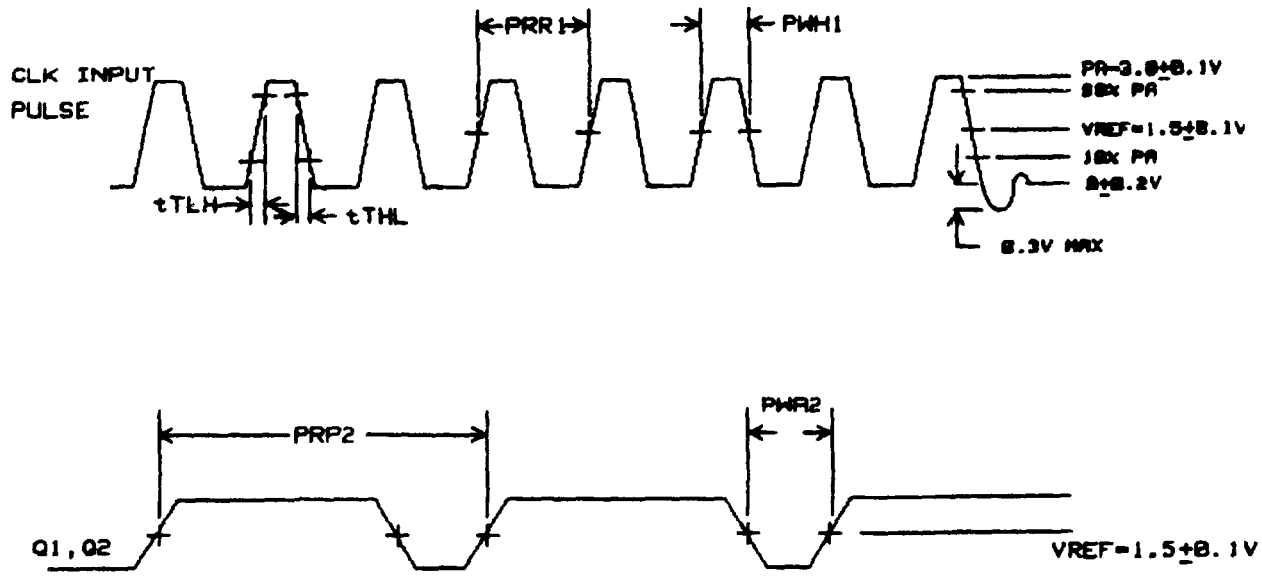


FIGURE 3B Maximum frequency (clock shaping) check test circuit for table XXVII

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NOTE Input pulse parameters
tTLH = tTHL = 10 ± 2 nS
PRR1 = 68 +2,-0 nS
PWH1 = 25 +2,-0 nS

FIGURE 39 Maximum frequency (clock shaping) check test circuit waveforms for table XXVII

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TABLE XXVIII Power supply filter capacitance test

TEST COND	SIGNAL NAME	PIN NUMBER	25 ± 5 deg C INITIAL LIMITS MIN	MAX	UNITS	NOTES
V1	VCC POWER SUPPLY CAPACITANCE	1A1,2A1	3.22	5.98	uF	1,2
V2	INJ POWER SUPPLY CAPACITANCE	1A35	200	1500	pF	3,4

Notes

1 VCC Power supply filter capacitance test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
CAPACITANCE BRIDGE FREQUENCY	1 kHz	± 100 Hz	NA	V1
CAPACITANCE BRIDGE VOLTAGE	0.3 Vp	± 30 mVp	NA	V1
CAPACITANCE MEASUREMENT	NA	± 30.36 nF	NA	V1

2 The capacitance bridge used in this test shall be item C01 The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXVIII

3 INJ power supply capacitance test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
CAPACITANCE METER FREQUENCY	1 MHz	± 100 KHz	NA	V2
CAPACITANCE BRIDGE VOLTAGE	0.3 Vp	± 30 mVp	NA	V2
CAPACITANCE MEASUREMENT	NA	± 14.3 pF	NA	V2

4 The capacitance meter used in this test shall be item C05 The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXVIII

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TABLE XXIX. Input capacitance test.

TEST COND	SIGNAL NAME	PIN NUMBER	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
W1	P1	1A22	2.0	20	pF	1,2
W2	P2	1A2				
W3	P3	1A21				
W4	P4	1A24				
W5	P5	1A23				
W6	P6	1A5				
W7	P7	1A4				
W8	P8	1A27				
W9	P9	1A6				
W10	P10	1A26				
W11	P11	1A25				
W12	P12	1A8				
W13	P13	1A9				
W14	P14	1A11				
W15	P15	1A28				
W16	EI-L	1A3	2.0	20	pF	1,2
W17	CLK	1A22	2.0	40	pF	1,2
W18	PRESET	2A13				
W19	LGH	2A30				
W20	READY	2A11				
W21	HOLD-L	2A12				
W22	LOAD-L	2A27				
W23	RST-L	2A25				
W24	CLOCK	2A23				
W25	CRUIN	1A30	2.0	40	pF	1,2

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TABLE XXIX Input capacitance test.

TEST COND	SIGNAL NAME	PIN NUMBER	25 ± 5 deg C INITIAL LIMITS MIN	25 ± 5 deg C INITIAL LIMITS MAX	UNITS	NOTES
W26	D0	1A13	2.0	40	pF	1,2
W27	D1	1A20				
W28	D2	2A20				
W29	D3	2A32				
W30	D4	2A31				
W31	D5	1A14				
W32	D6	1A15				
W33	D7	1A16				
W34	D8	1A17				
W35	D9	1A18				
W36	D10	1A19				
W37	D11	2A2				
W38	D12	2A3				
W39	D13	2A4				
W40	D14	2A5				
W41	D15	2A6	2.0	40	pF	1,2

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Notes

1 Input capacitance test tolerance table

CONDITION	VALUE	ACCURACY	CURRENT LIMIT	TEST CONDITION
CAPACITANCE METER FREQUENCY	1 MHz	± 100 KHz	NA	W1-W41
CAPACITANCE METER VOLTAGE	0.3 Vp	± 30 mVp	NA	W1-W41
CAPACITANCE MEASUREMENT	NA NA	± 0.198 pF ± 0.418 pF	NA NA	W1-W16 W17-W41

2 The capacitance meter used in this test shall be item C05 The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIX

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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